合肥大學
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Metaheuristic Optimization: State of the Art

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Outline

1. Introduction
2. Metaheuristics
3. Discrete Optimization
4. (Multi-)Permutation Problems
5. Summary
6. Advertisement

Introduction

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- And many, many literature references.
- Therefore, if this lecture interests you in this field, you will have many starting points for your own research.

Metaheuristics

- So let's begin.

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- So what is optimization?

What is Optimization?

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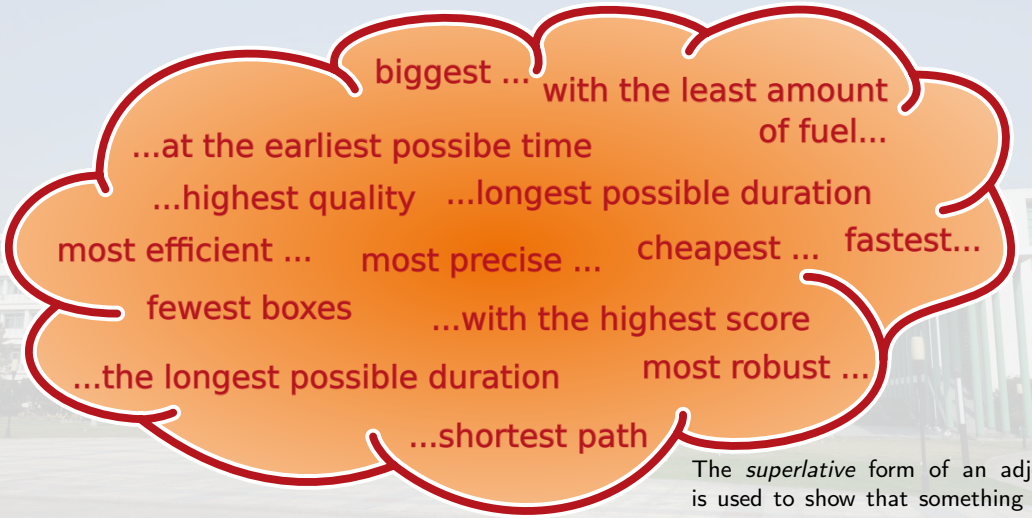
  1. Optimization means searching for superlatives.
  2. Optimization is the art of solving hard problems.
  3. Optimization allows us to find acceptable solutions for problems where our tools from high school maths *and* traditional algorithms from computer science are no longer enough, problems where exact solutions cannot always be reached.

Views on Optimization, Part 1

- Let's start with the first view: Superlatives.

Optimization is the search for superlatives

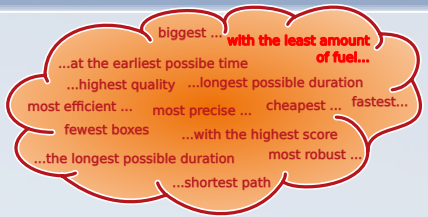
- Optimization means finding superlatives.



The *superlative* form of an adjective is used to show that something has a quality to the greatest or least degree.

Optimization is the search for superlatives

- Optimization means finding superlatives.
- Pick up and deliver packages from different places to customers using the least amount of fuel[195,240–242].



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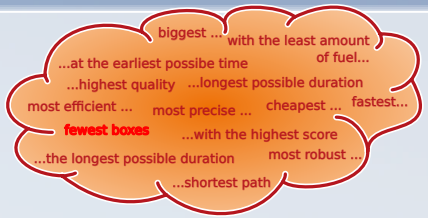

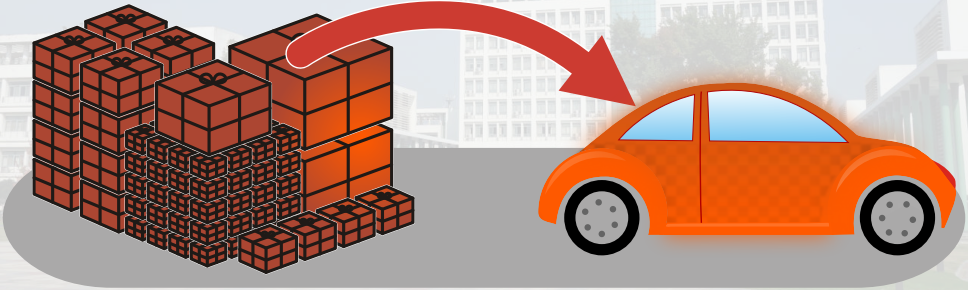

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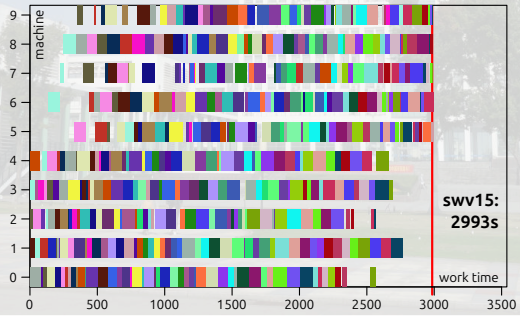
- Assign tasks to machines such that we can finish our work the at earliest possible time.



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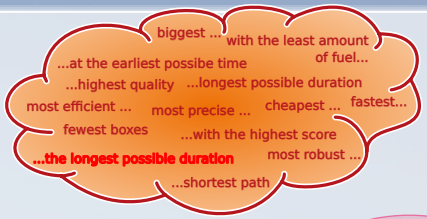

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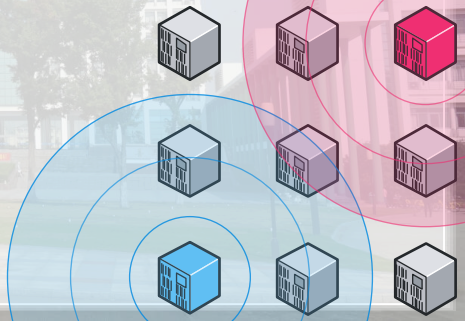
- Find a strategy to manage the power of the nodes in this sensor network so that full coverage is guaranteed for the longest possible duration.



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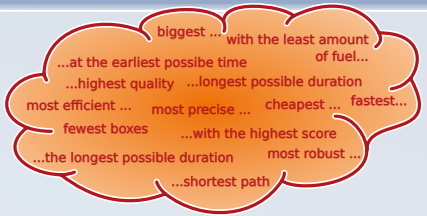

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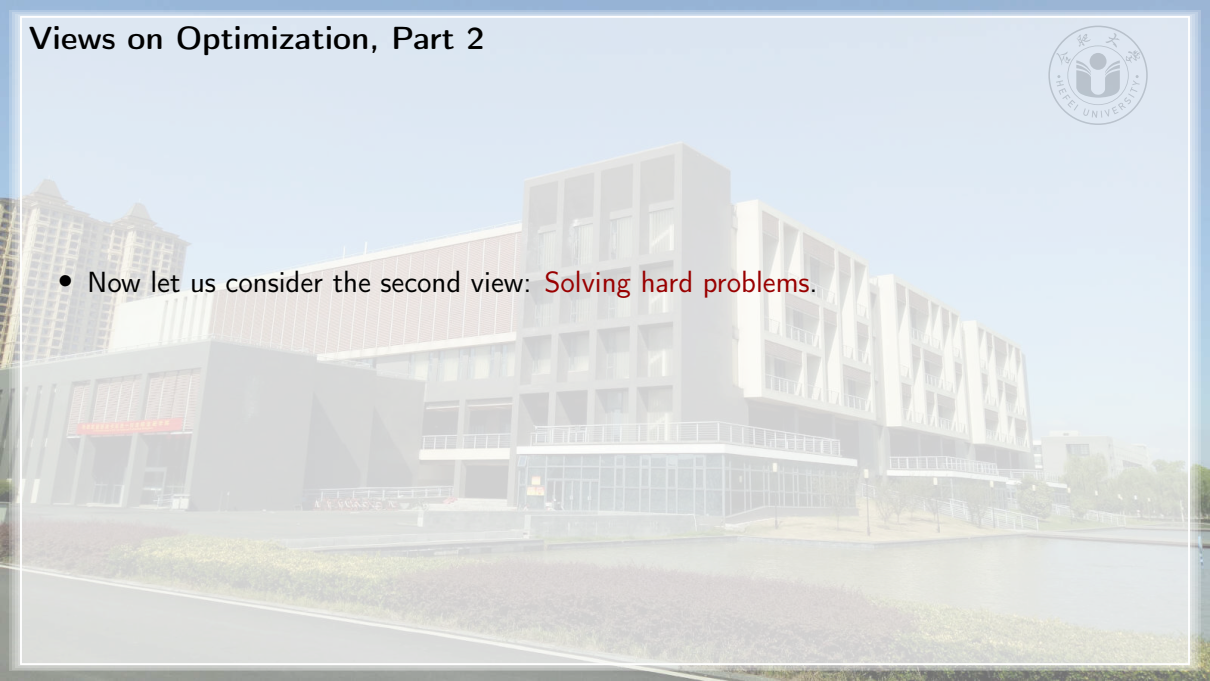
- Find a strategy to manage the power of the nodes in this sensor network so that full coverage is guaranteed for the longest possible duration.
- And so on.

Views on Optimization, Part 2



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Views on Optimization, Part 2

- Now let us consider the second view: Solving hard problems.
- Later, we learn how to quantify *hard*.
- For now, let's first think about how to define *problem*?

What are Optimization Problems?

- There are two ways to define optimization problems.

What are Optimization Problems?

- The economic view.

Optimization Problem

An optimization problem is a situation which requires deciding for one choice from a set of possible alternatives in order to reach a predefined or required benefit at minimal costs.

What are Optimization Problems?

- The mathematical view.

Optimization Problem

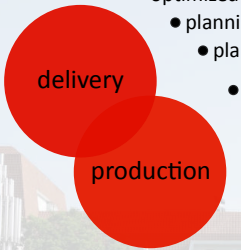
An optimization problem is a situation which requires deciding for one choice from a set of possible alternatives in order to reach a predefined or required benefit at minimal costs.

Solving an optimization problem requires finding an input element $x^\star$ within a set $\mathbb{X}$ of allowed elements for which a mathematical function $f\colon \mathbb{X} \mapsto \mathbb{R}$ takes on the smallest possible value.

Optimization in Business

- optimized logistics (business-to-customer)
  - planning and scheduling of maintenance visits
    - planning and scheduling of supply visits

Optimization in Business



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  - optimized assignment of jobs/orders to machines
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  - optimization of factory layouts and intra-factory logistics

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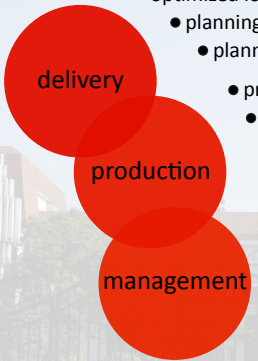

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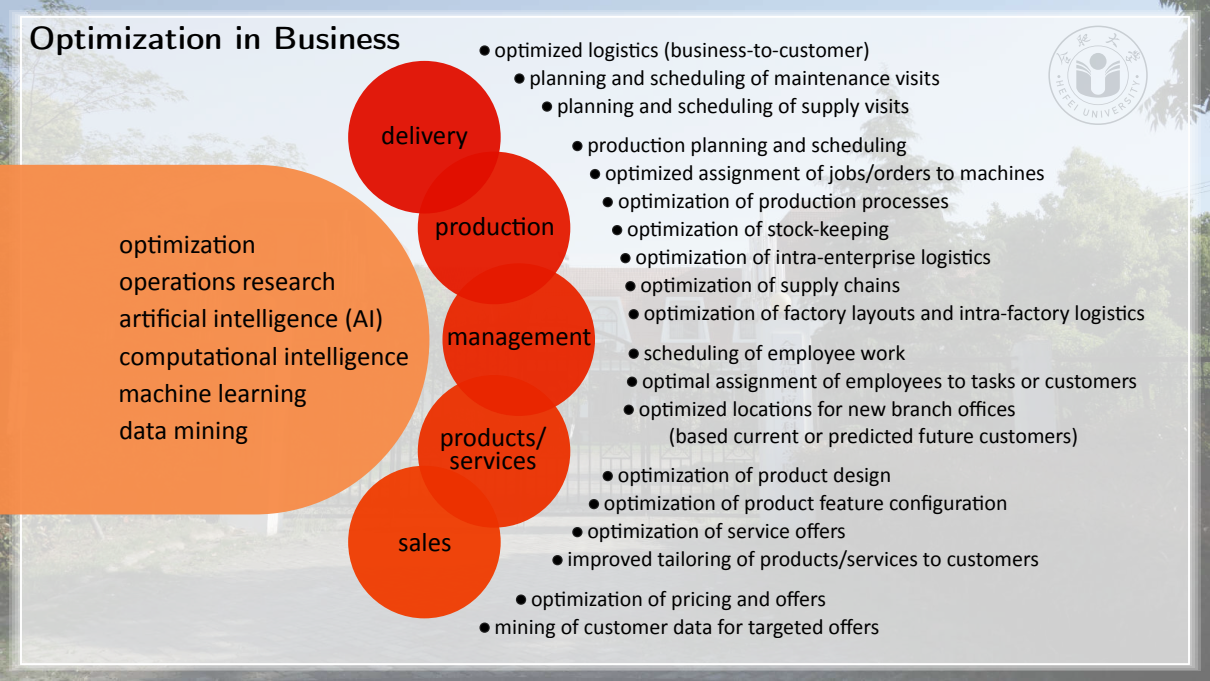

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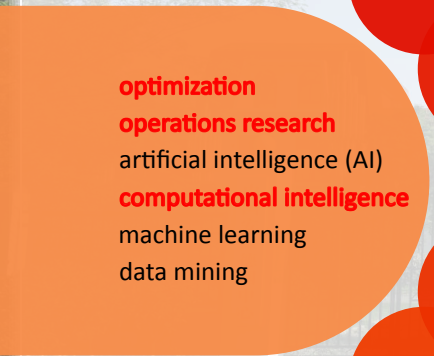

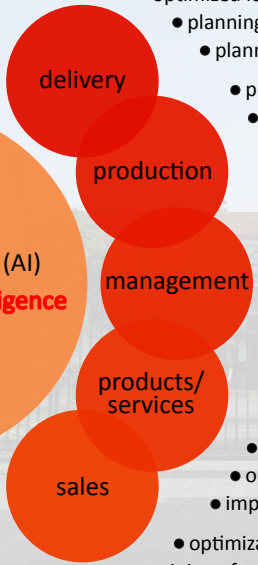

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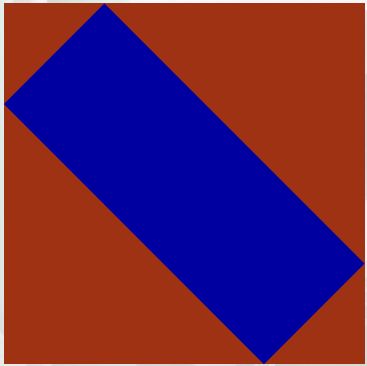
- Let's start with high school Maths, move to undergraduate computer science, and then approach this field of problems.

A Problem that we can solve with high school Maths equations

- A rectangle should be placed inside a square with side length 13cm as sketched.



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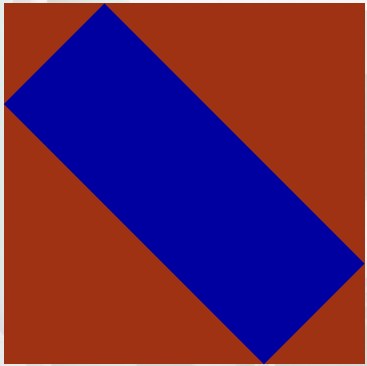

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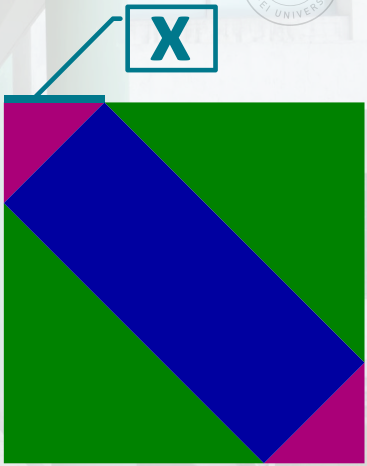

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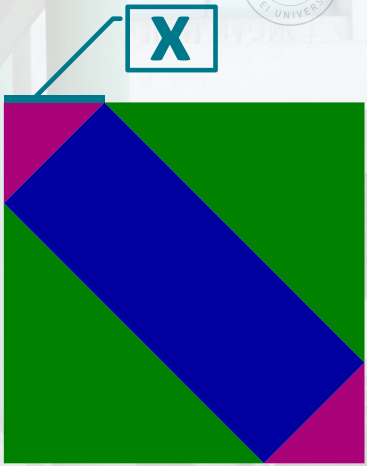

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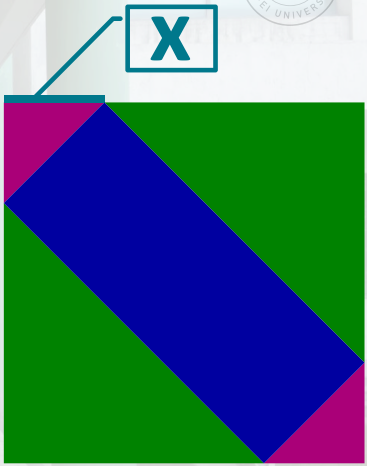

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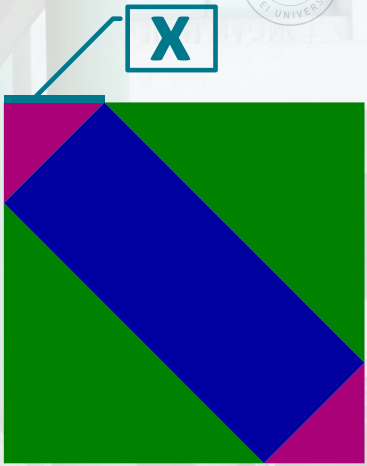

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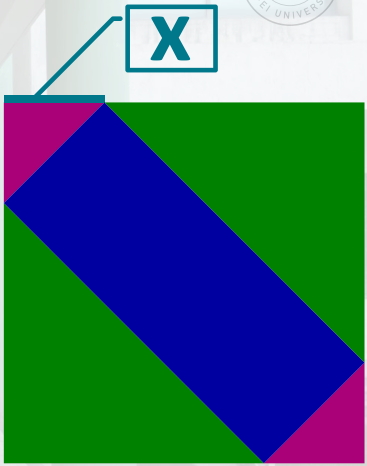

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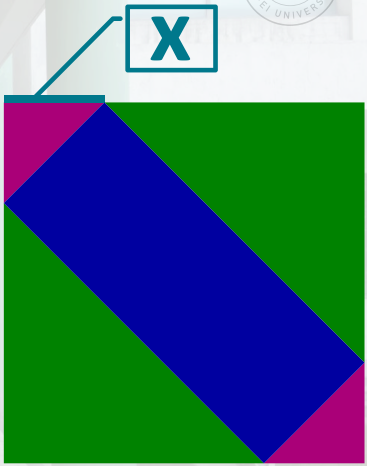
- We get $\hat{x} = 6.5\text{cm}$ and, hence, $\widehat{A_{\blacksquare}} = 84.5\text{cm}^2$.
- Maxima and minima are superlatives = optima.
- Sometimes we can simply compute them and do not need fancy algorithms.



Problems that algorithms can solve efficiently

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- I want to go to the goal: 食堂.

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- I am at the starting point 合肥大学南二区北大门.
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- How do I get there the fastest?

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- I am at the starting point 合肥大学南二区北大门.
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- How do I get there the fastest?
- We know the campus map.

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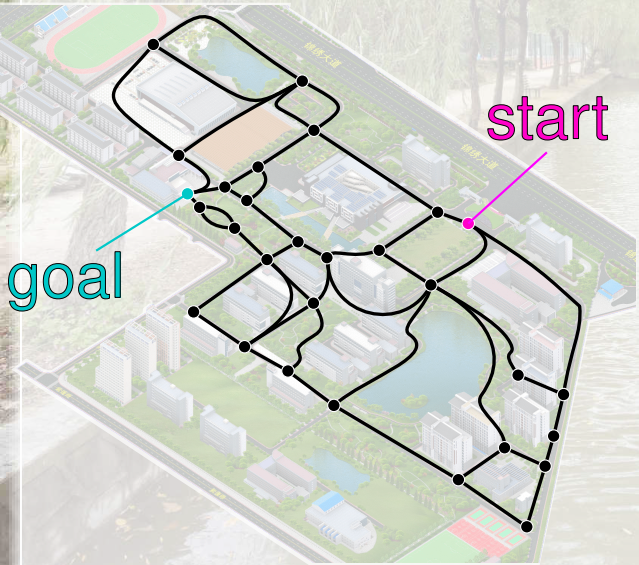
- We know all the intersections where I could make turns.

Problems that algorithms can solve efficiently



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- There is no mathematical equation that we can use to solve this.

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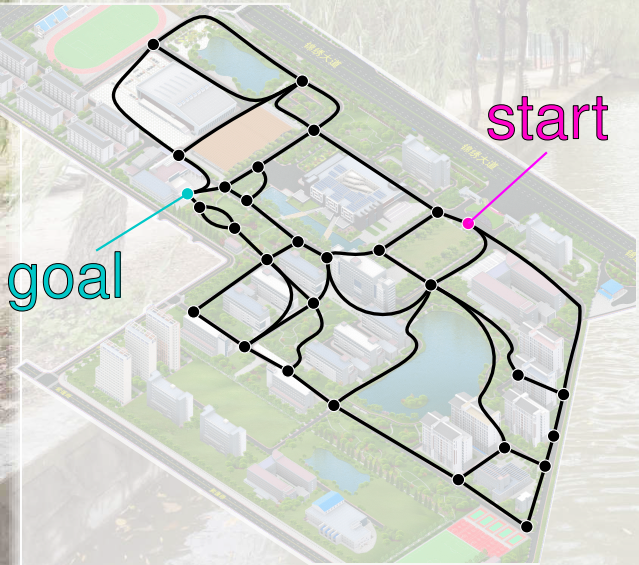

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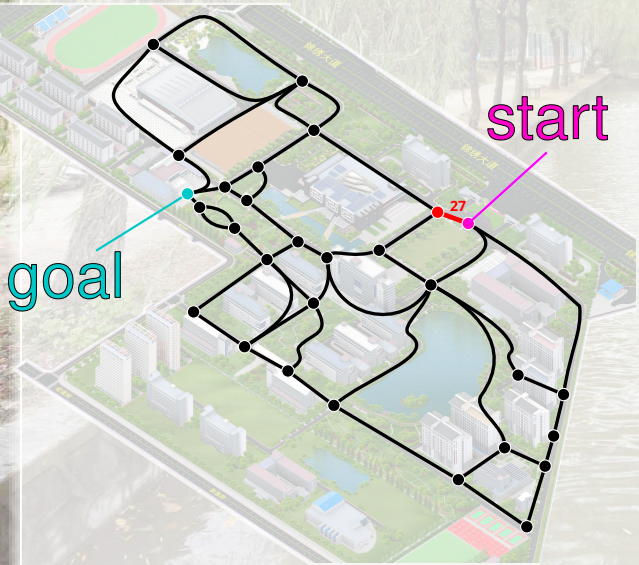
- We know all the intersections where I could make turns.
- There is no mathematical equation that we can use to solve this.
- But we can use the A* Algorithm[110,190], which iteratively constructs a solution by adding a heuristic to a cost function.

Problems that algorithms can solve efficiently



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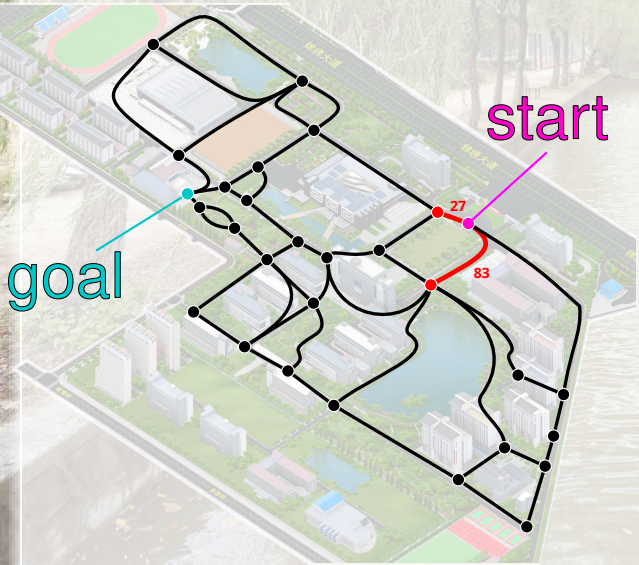
- For example, I could walk for $27s$ to this intersection.

Problems that algorithms can solve efficiently



- But we can use the A* Algorithm[110,190], which iteratively constructs a solution by adding a heuristic to a cost function.
- For example, I could walk for $27s$ to this intersection.
- Or for $83s$ to that one.

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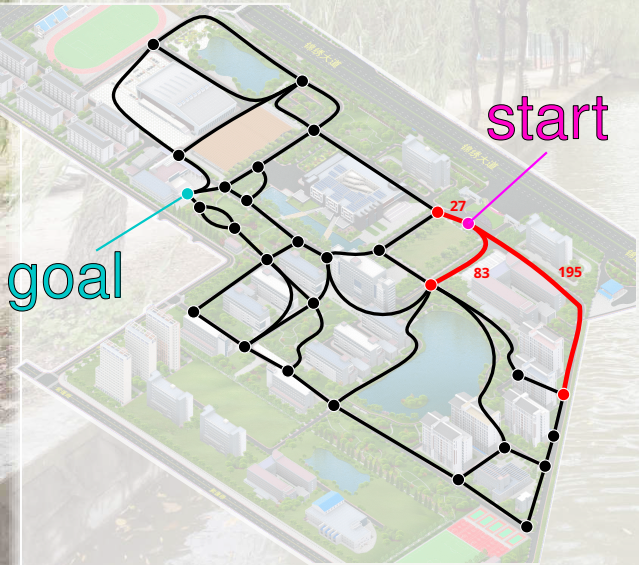

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- For example, I could walk for $27s$ to this intersection.
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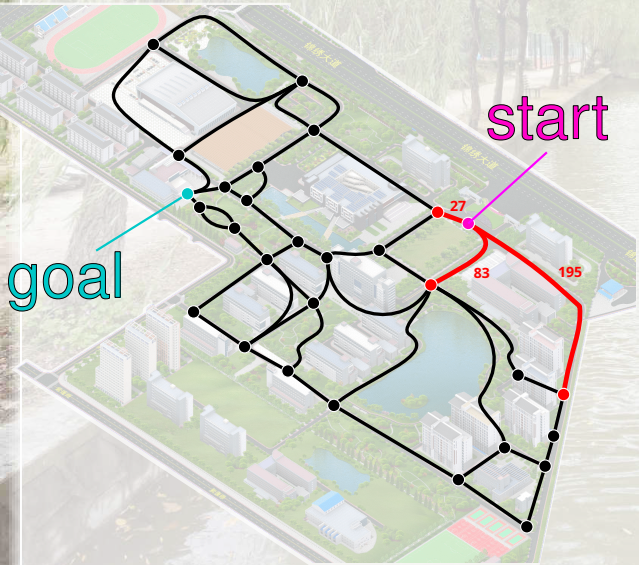

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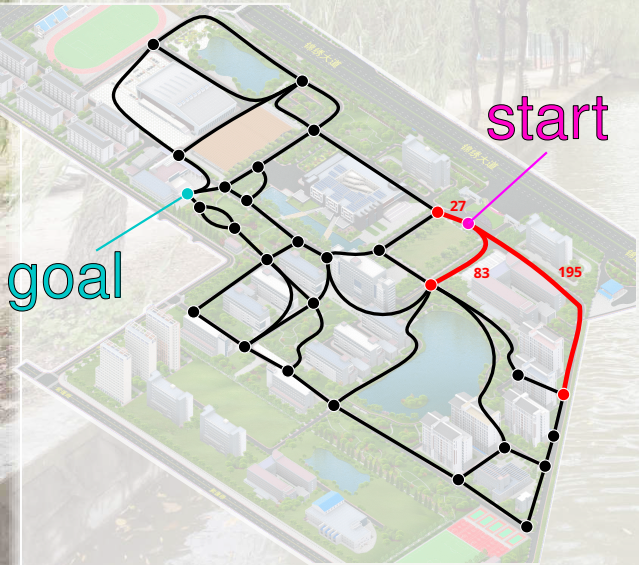

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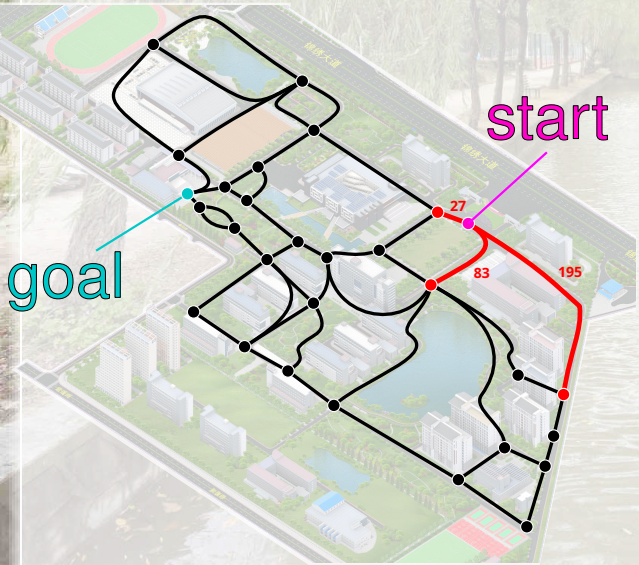

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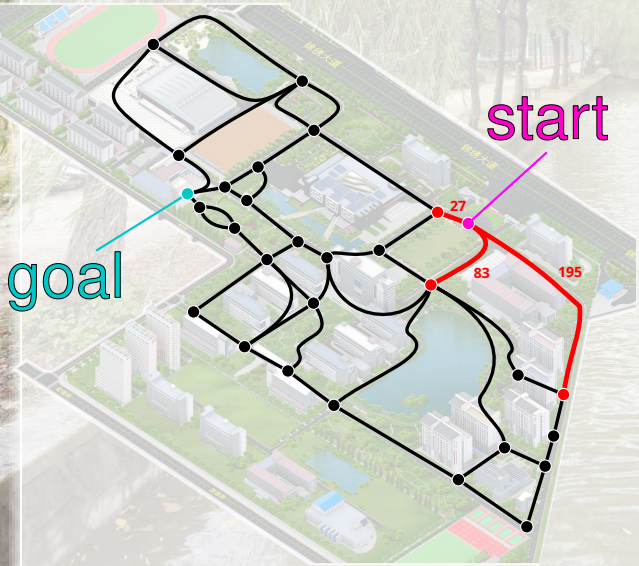

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- **Adding** the two distances for each node gives us a good impression of how useful going there would be.

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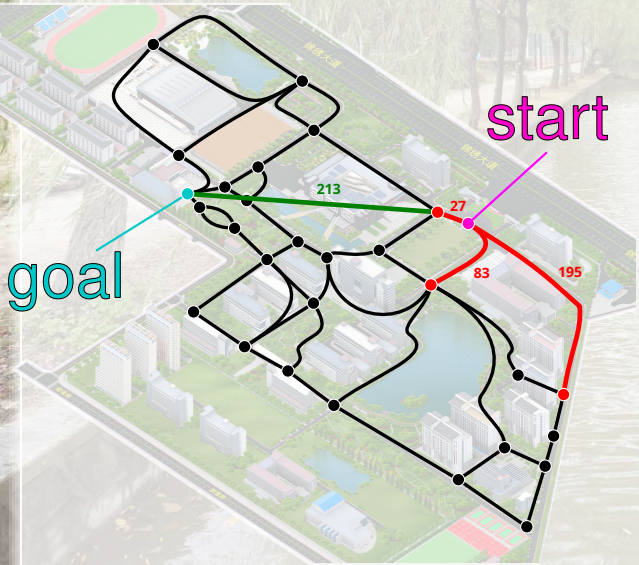

- But we can use the A* Algorithm[110,190], which iteratively constructs a solution by adding a heuristic to a cost function.
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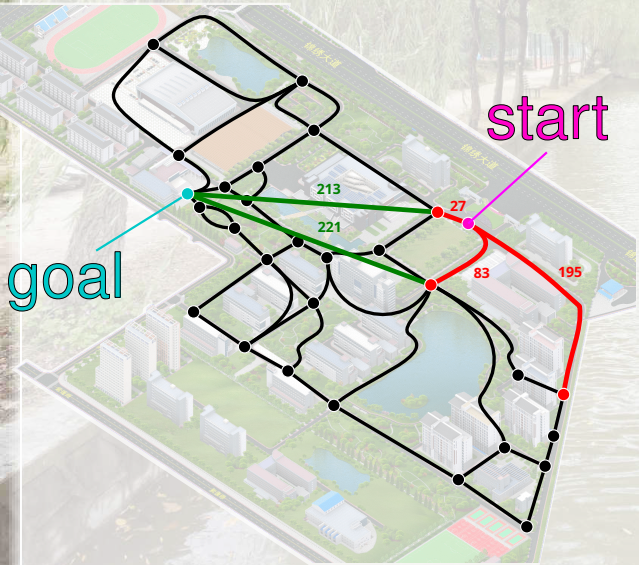

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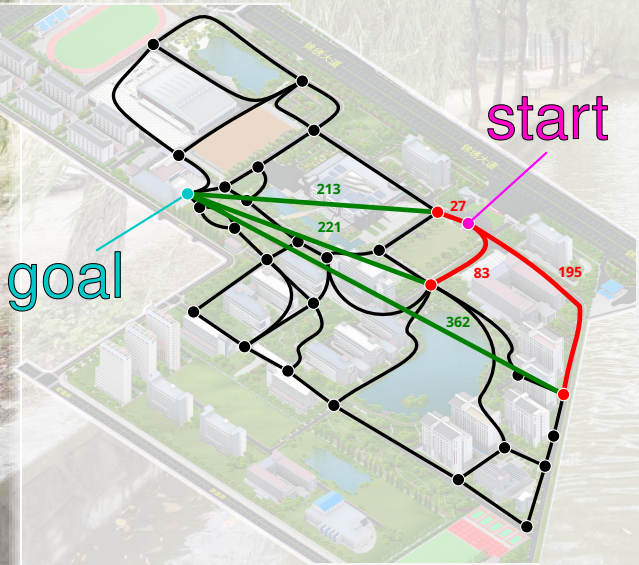

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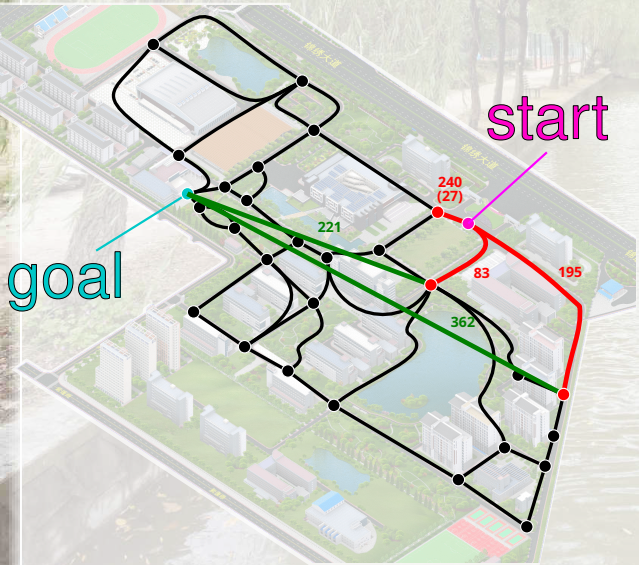

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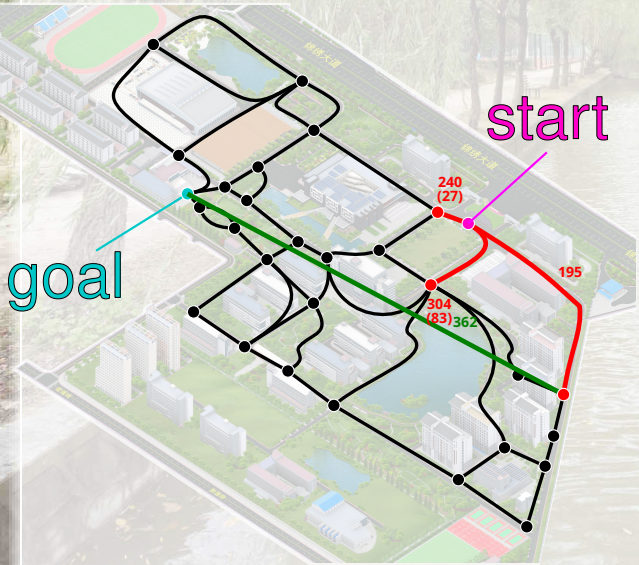

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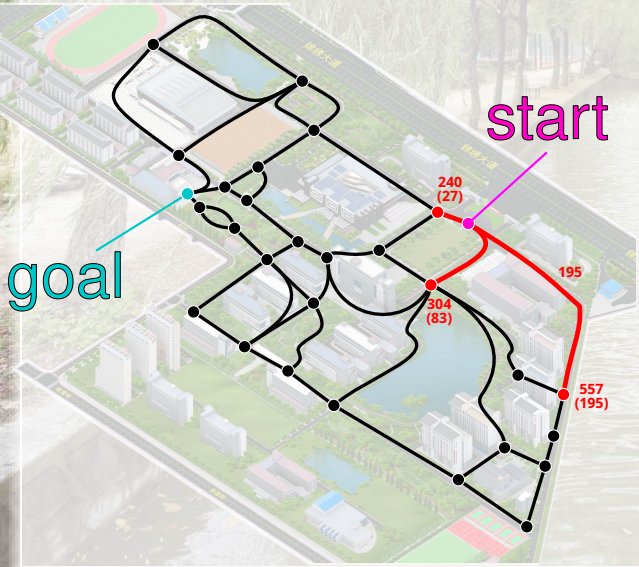

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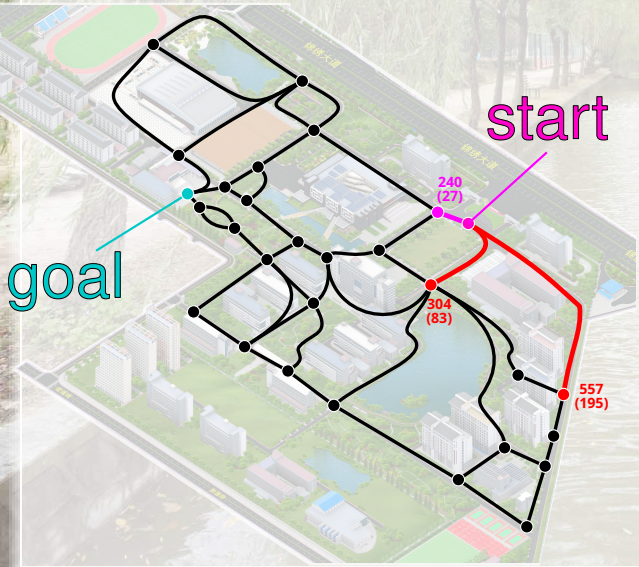

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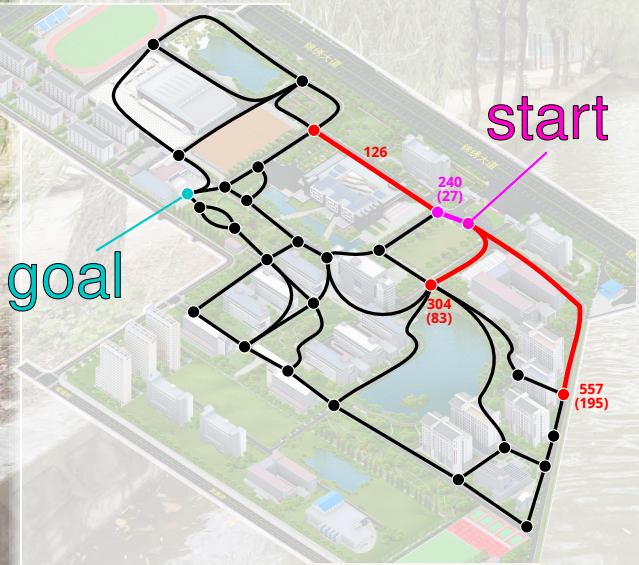

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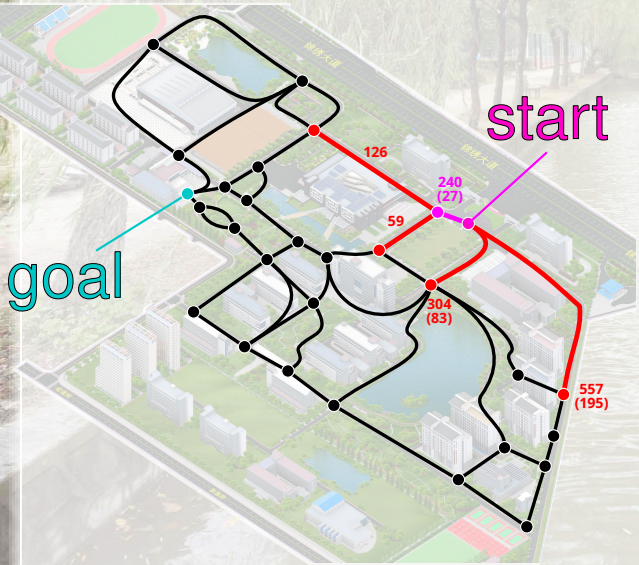

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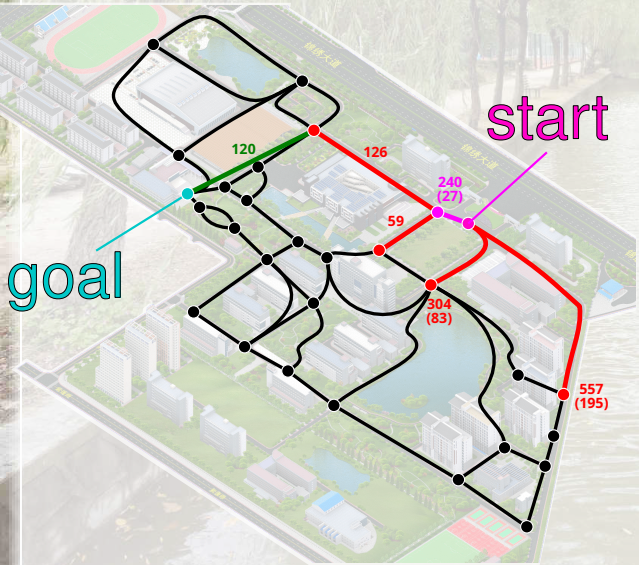

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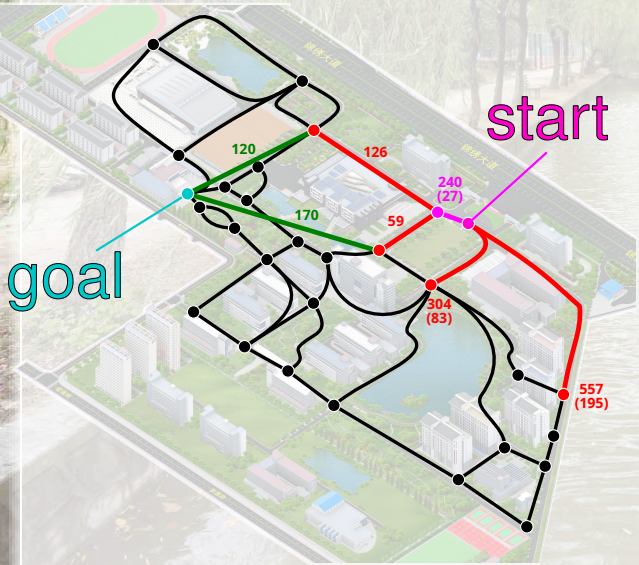

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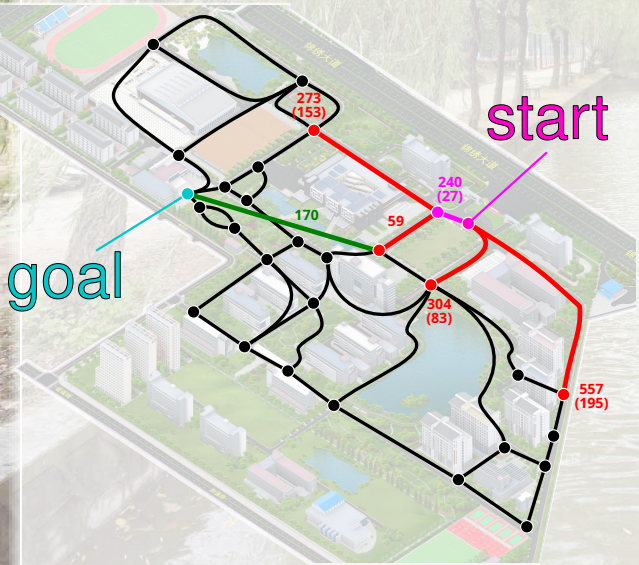

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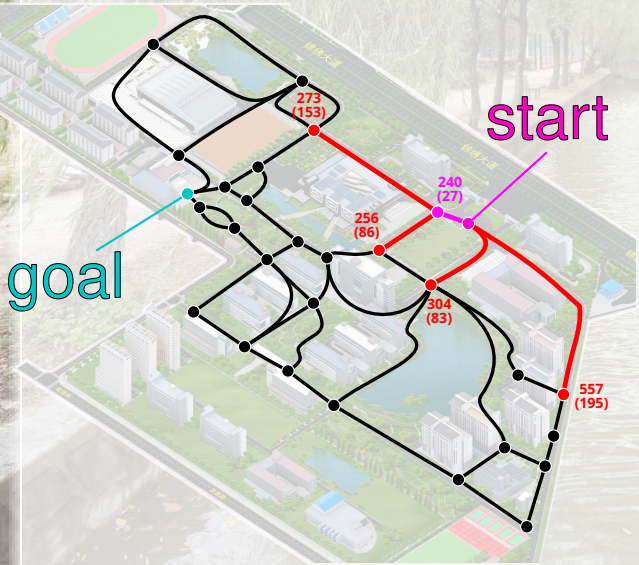

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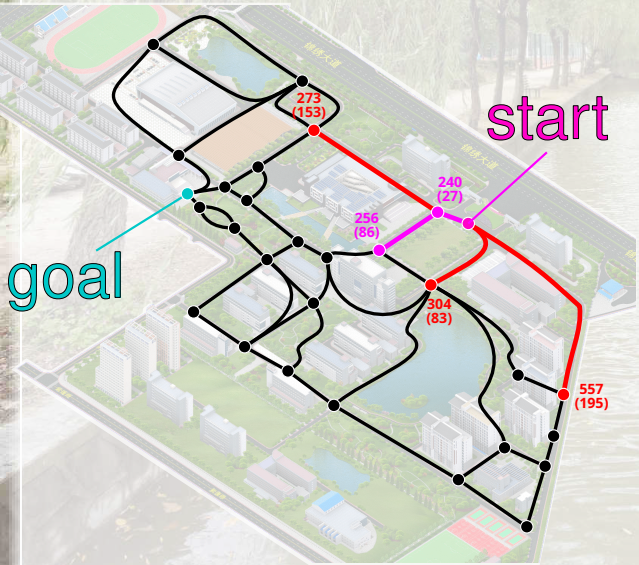

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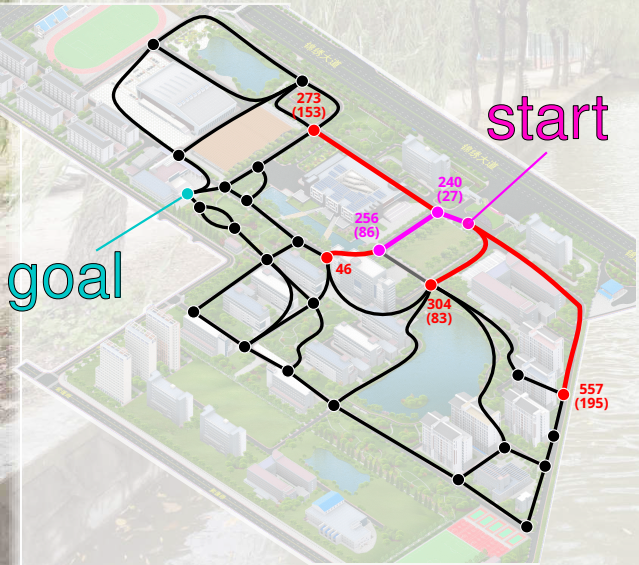

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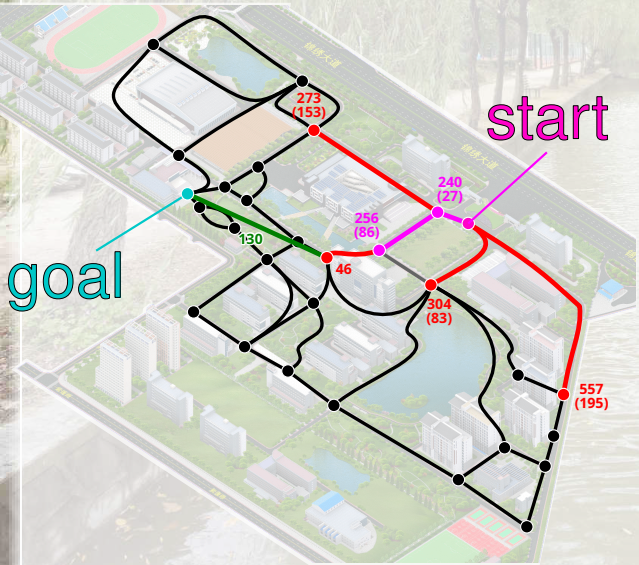

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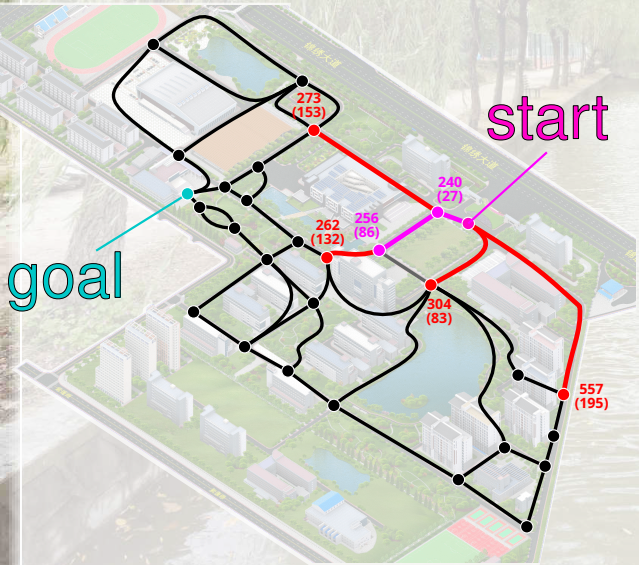

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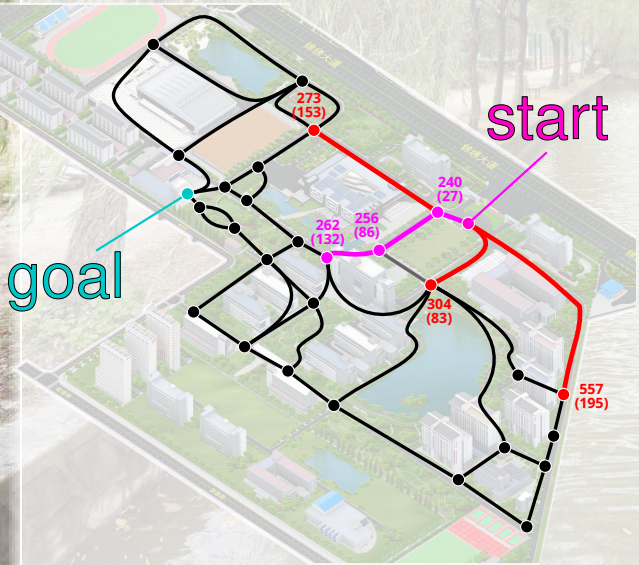

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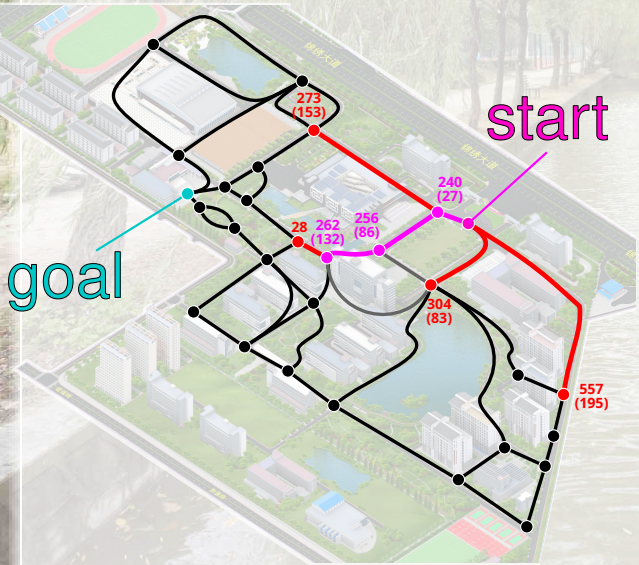

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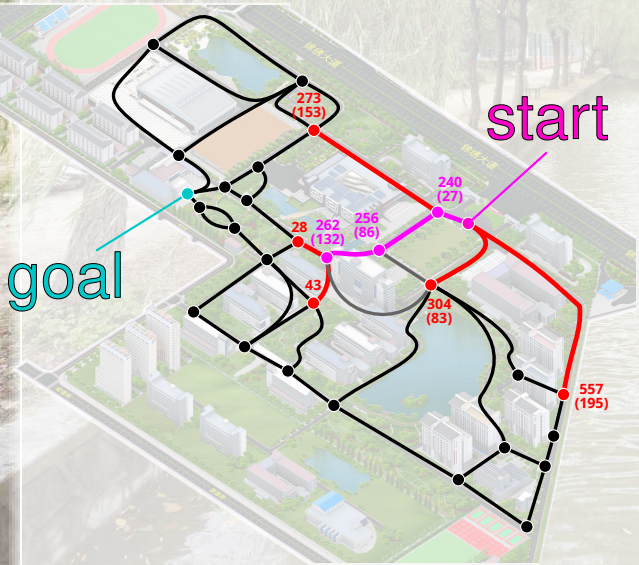

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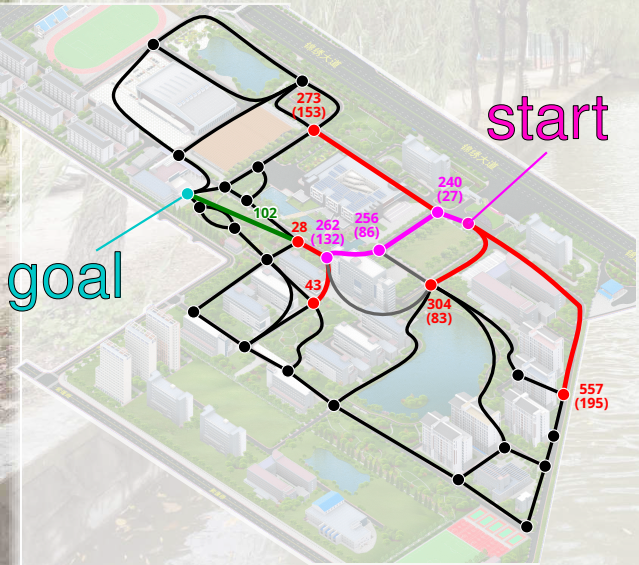

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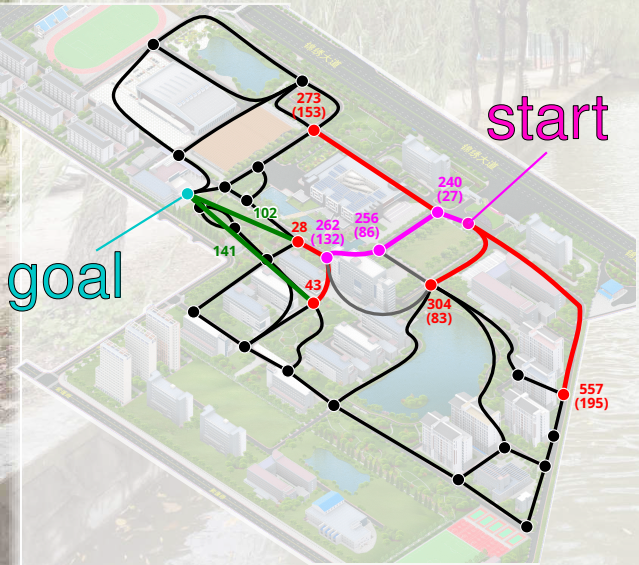

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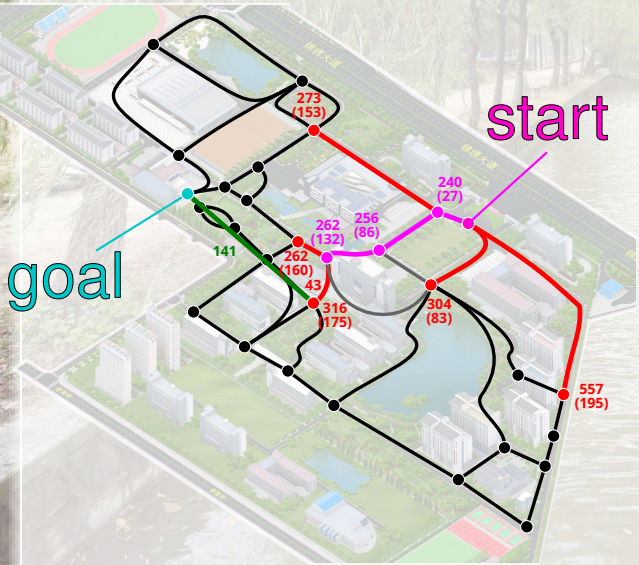

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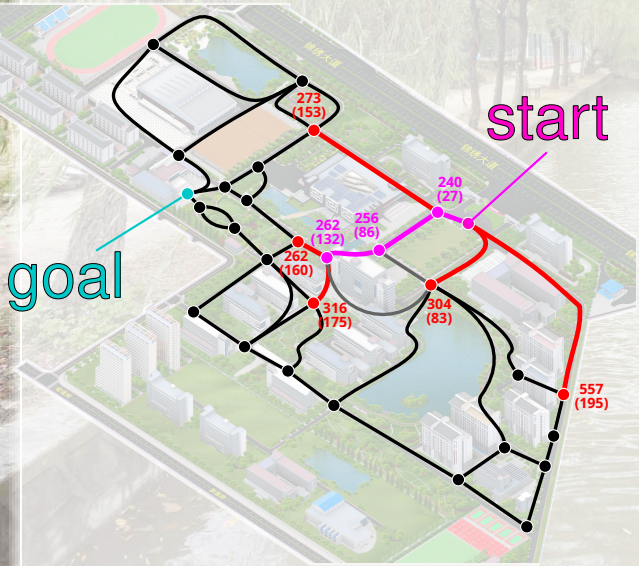

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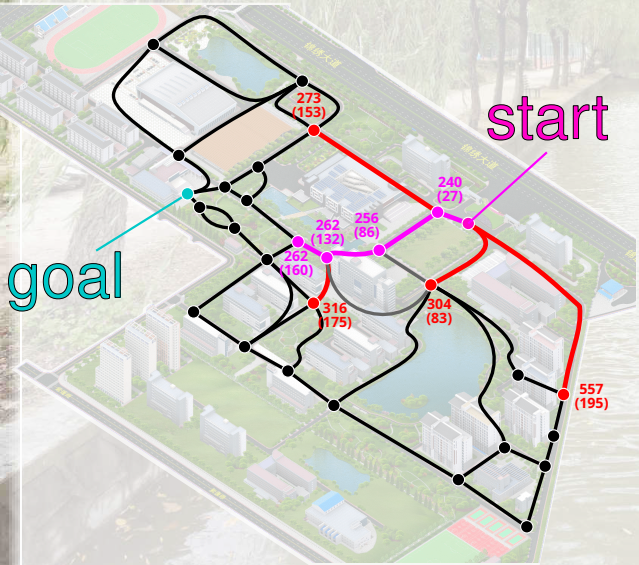

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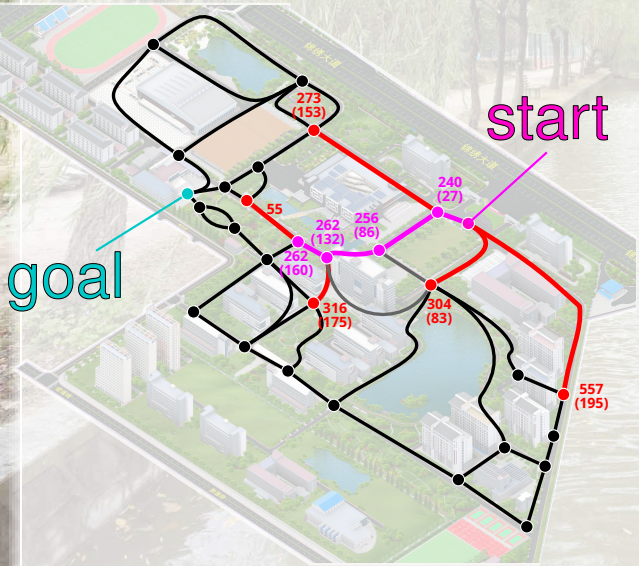

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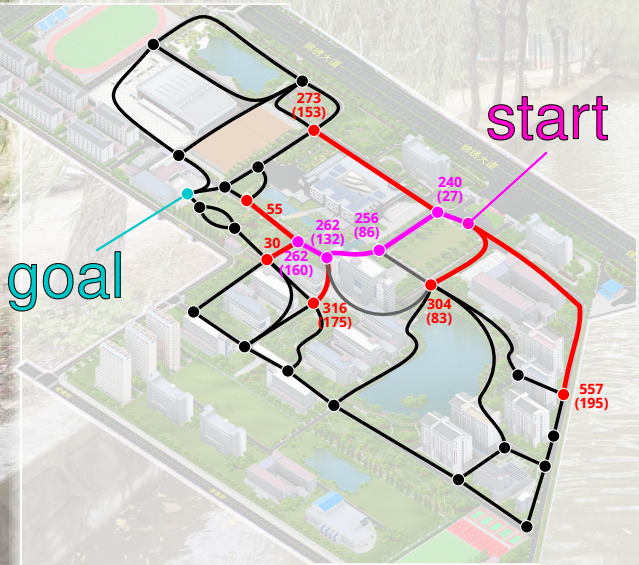

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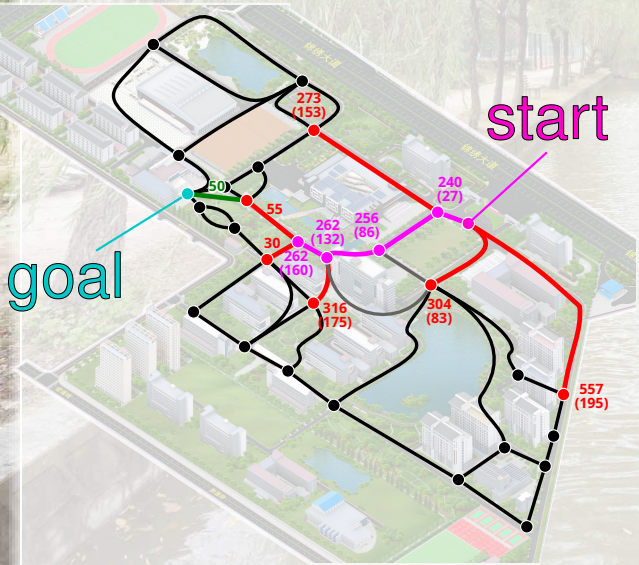

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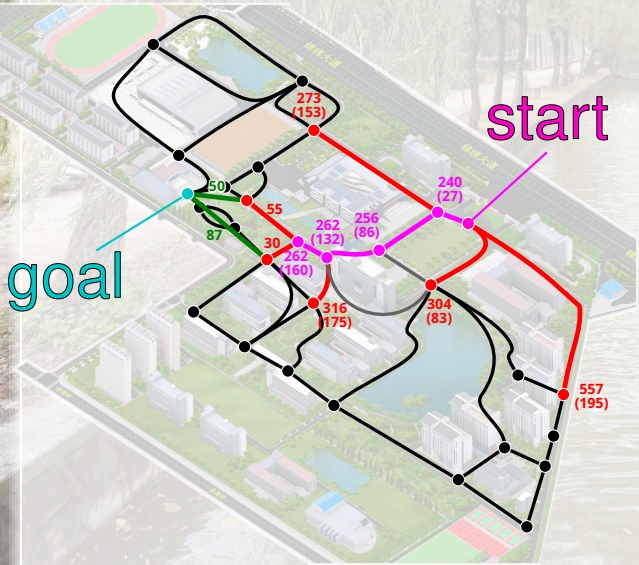

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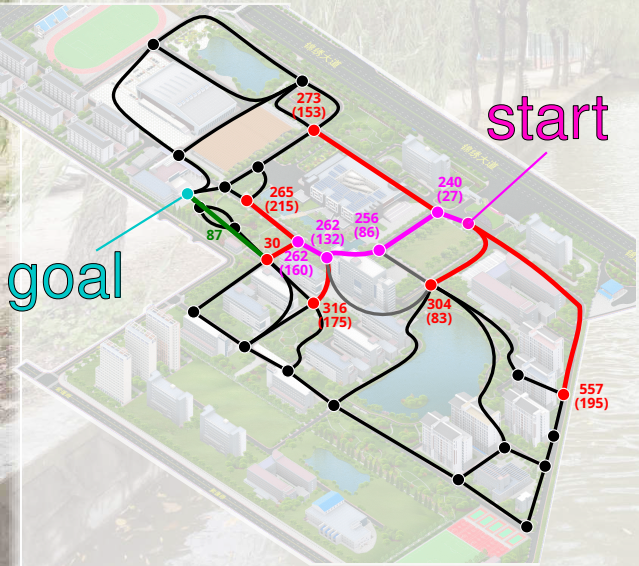

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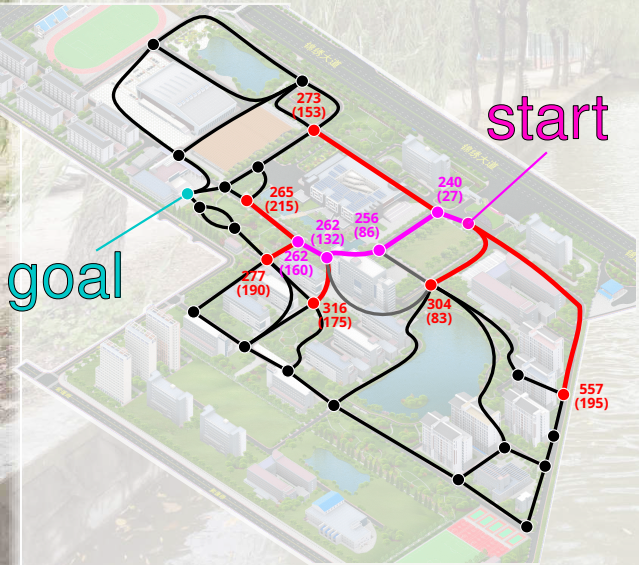

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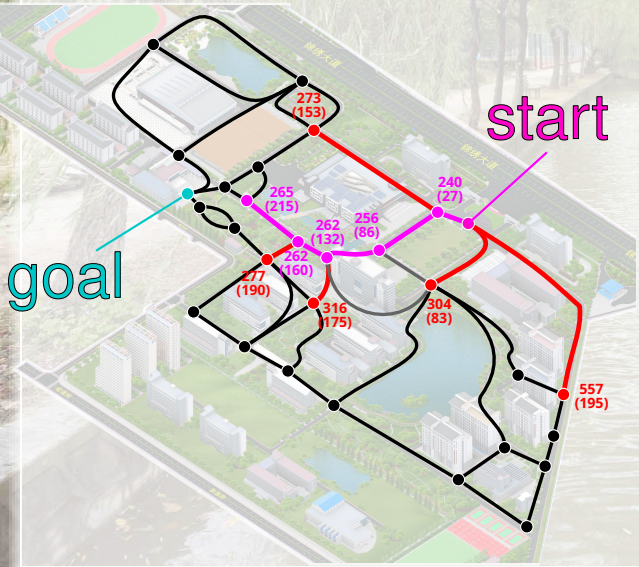

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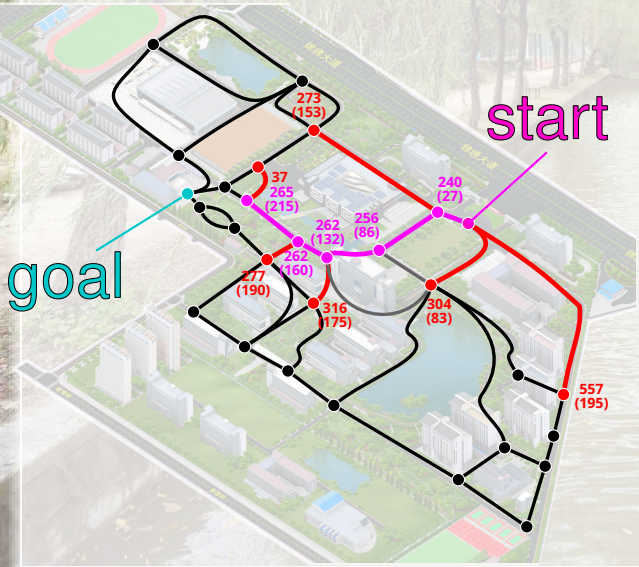

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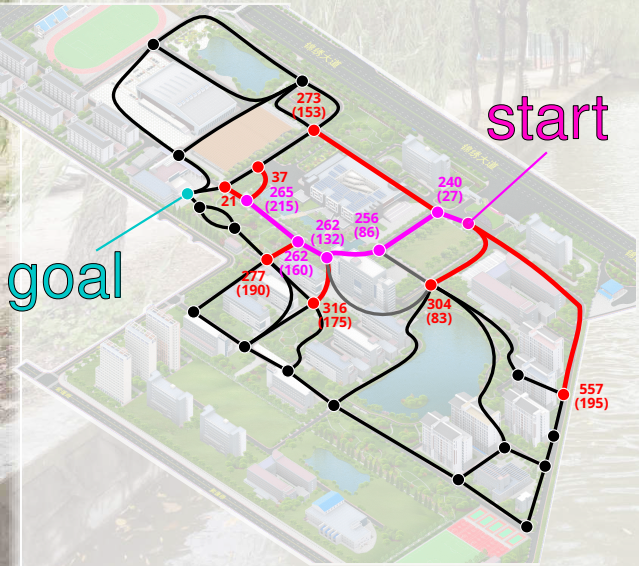

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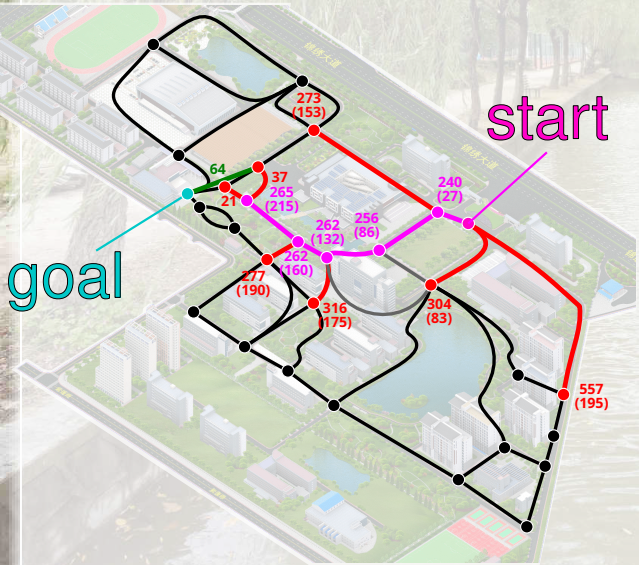

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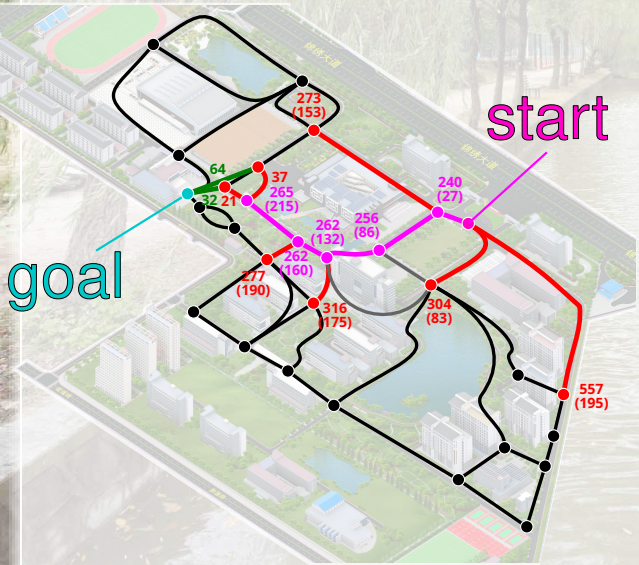

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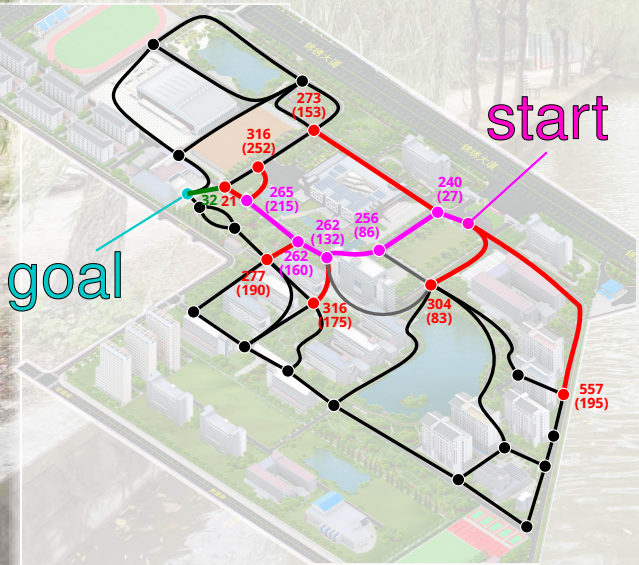

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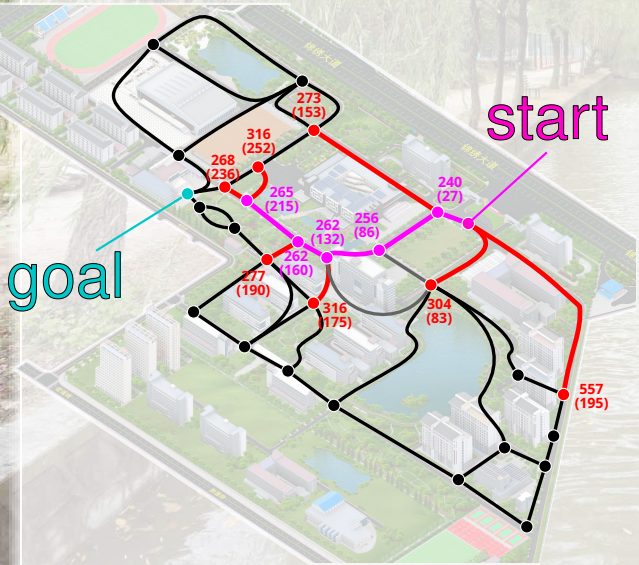

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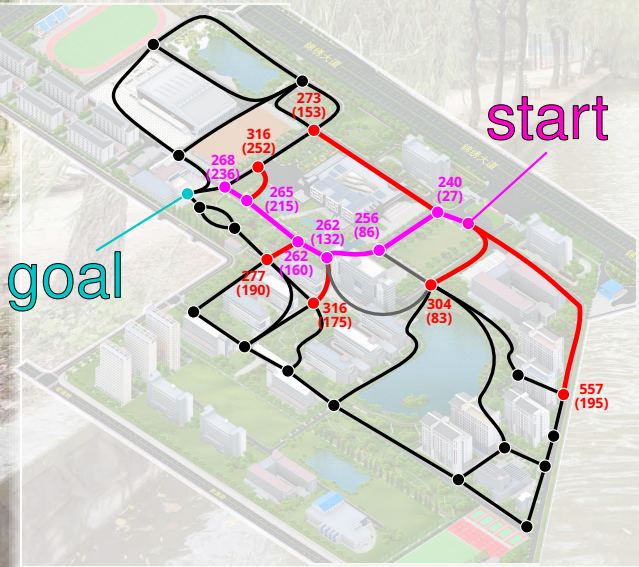

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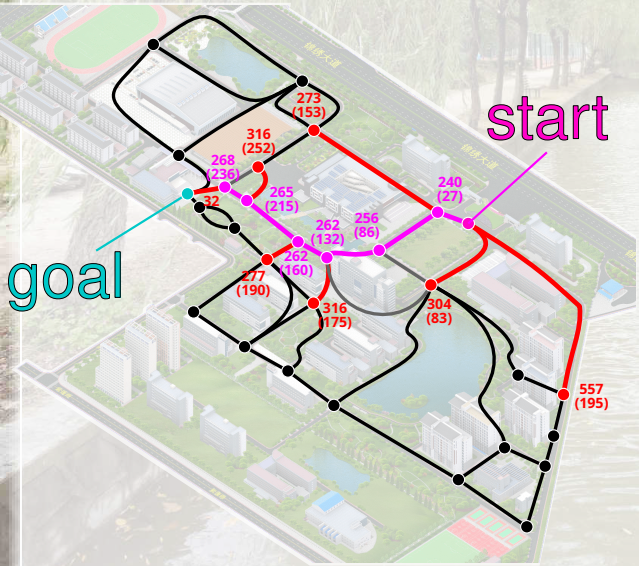

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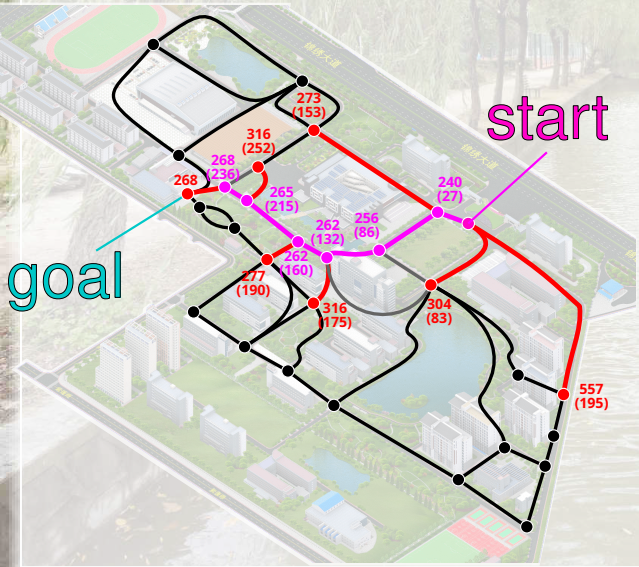

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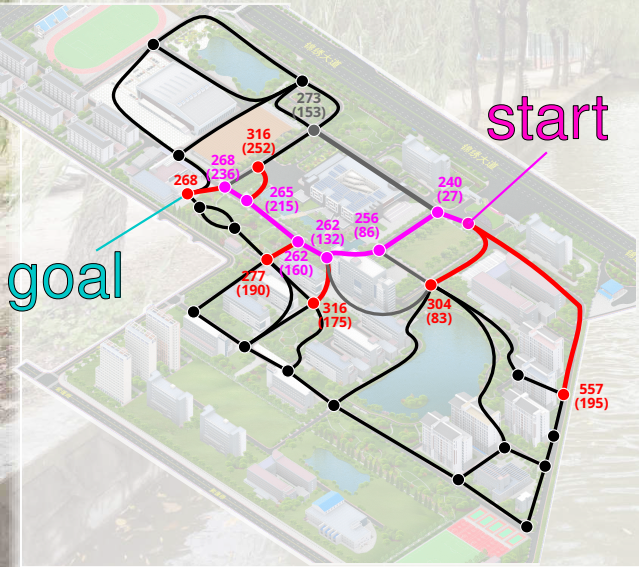

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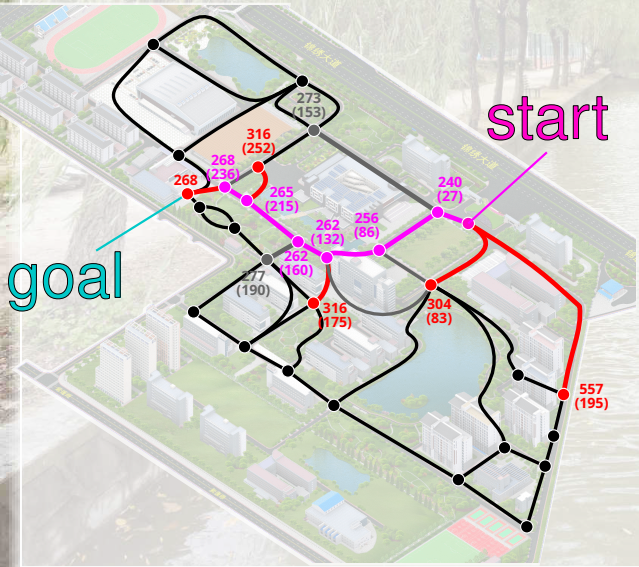

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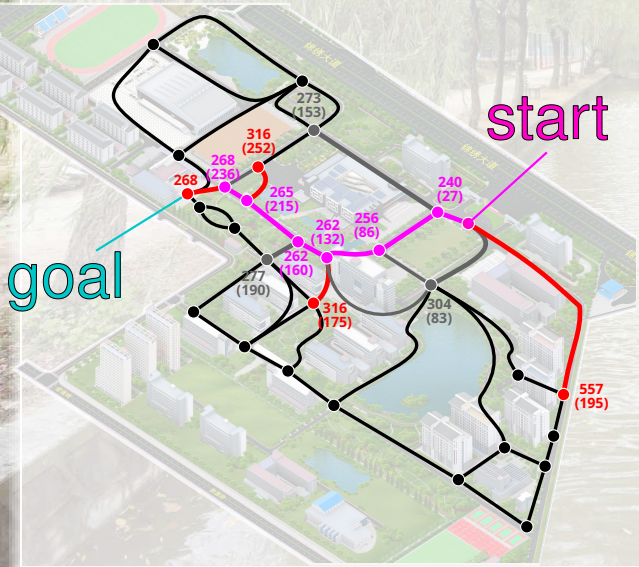

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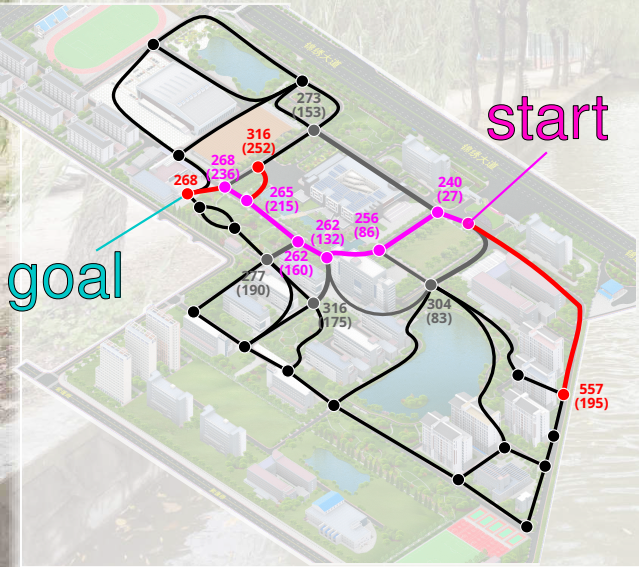

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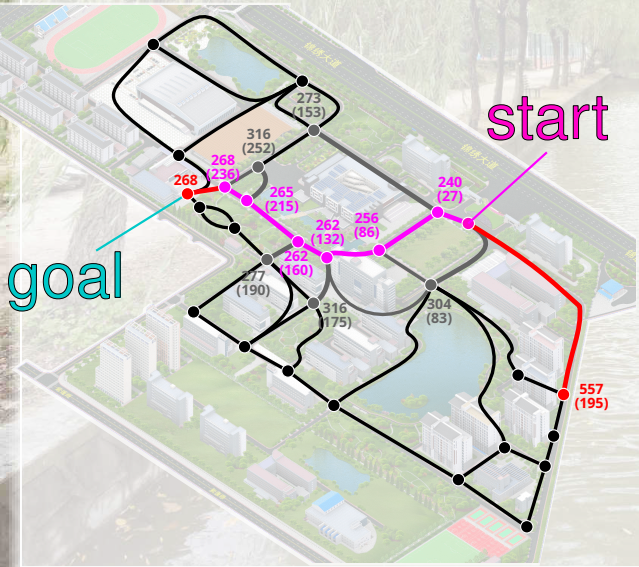

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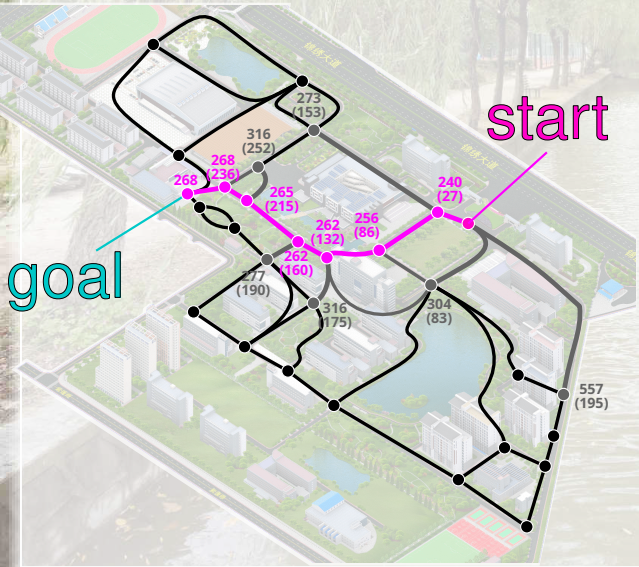

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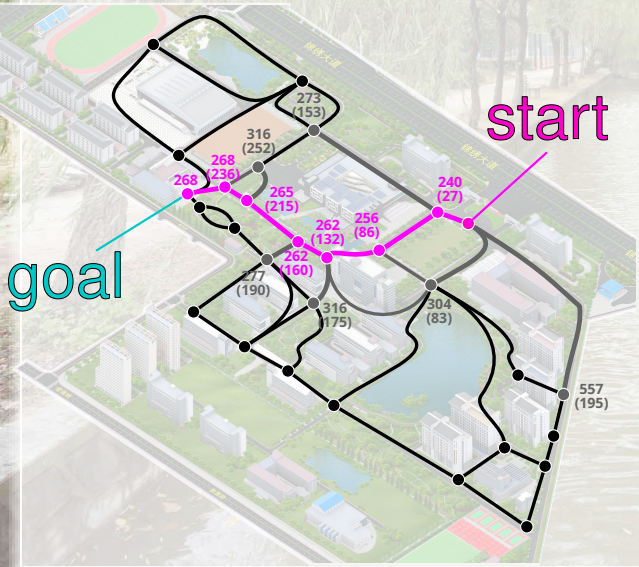

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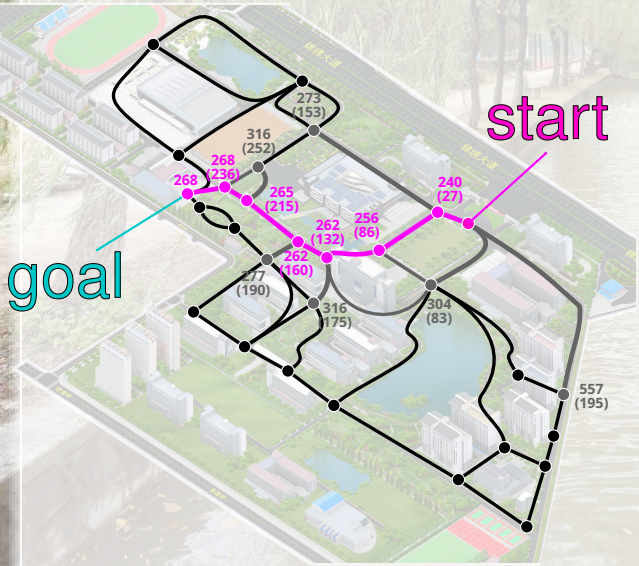
- We confirmed the shortest possible path and it takes $268s$.

Problems that algorithms can solve efficiently



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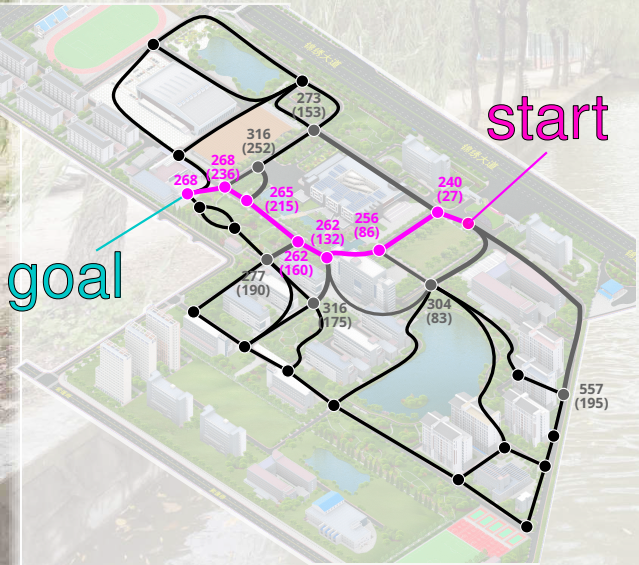

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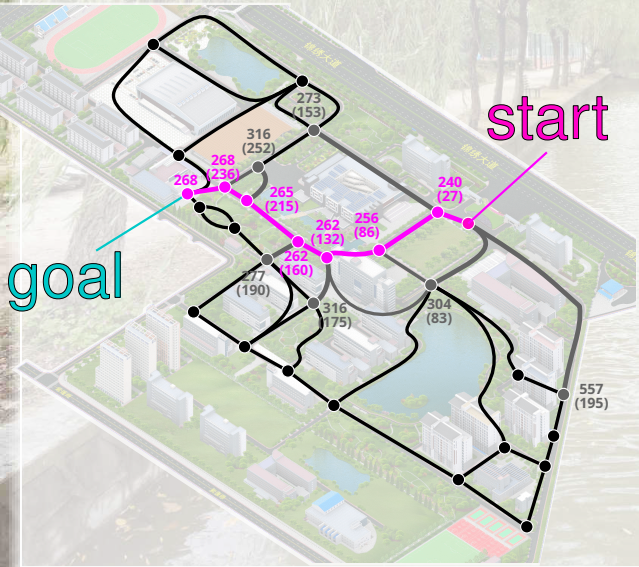

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- In the worst case, the A* Algorithm needs to "look" at all intersections (nodes) and streets (edges) once.
- Usually, it is quite efficient to find shortest paths on maps.
- If we look for shortest paths that do not visit any node twice, where the edge distances are not negative, where the number of choices per node is limited, and where the graph and path are not too big/long – then this algorithm is very efficient.

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- What about *hard* problems?
- What about problems where traditional Maths and algorithms do not really help us much?

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- Here we go.

Hard Problems (1)

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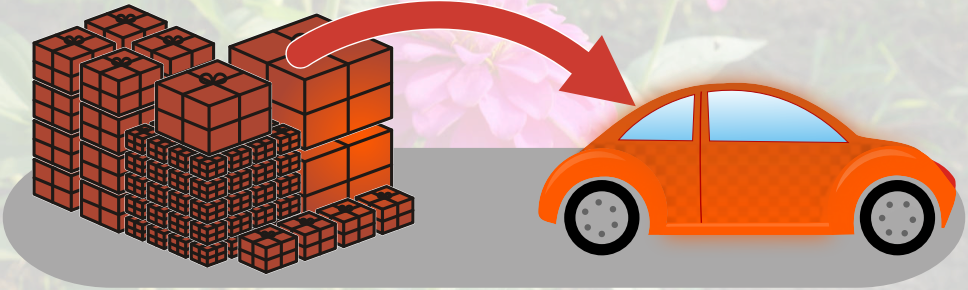
- A hard problem cannot be solved both exactly and efficiently.
- Let's look at two quick examples.

Bin Packing Problem

- My car can carry T kg of weight.



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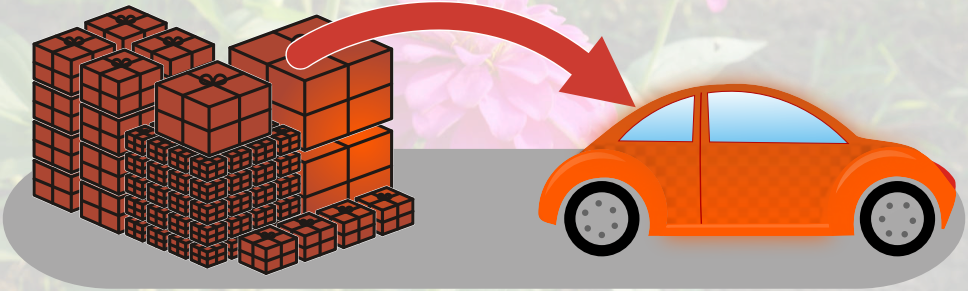
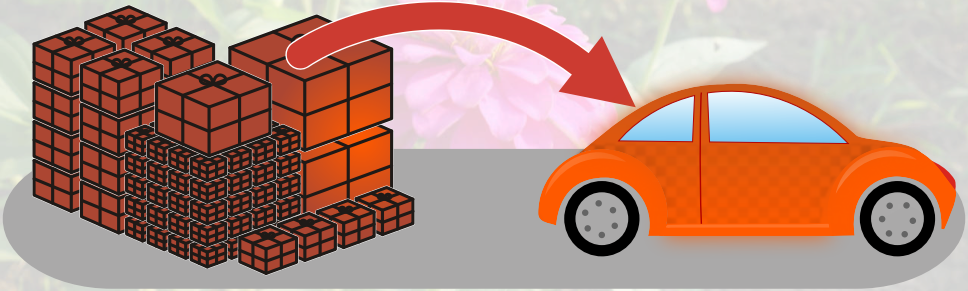


Bin Packing Problem

- My car can carry T kg of weight.
- I have n objects, each with weight w_i with i in 1 to n.
- How can I pack my car so that I can carry them from A to B with the fewest possible hauls?[259,260]

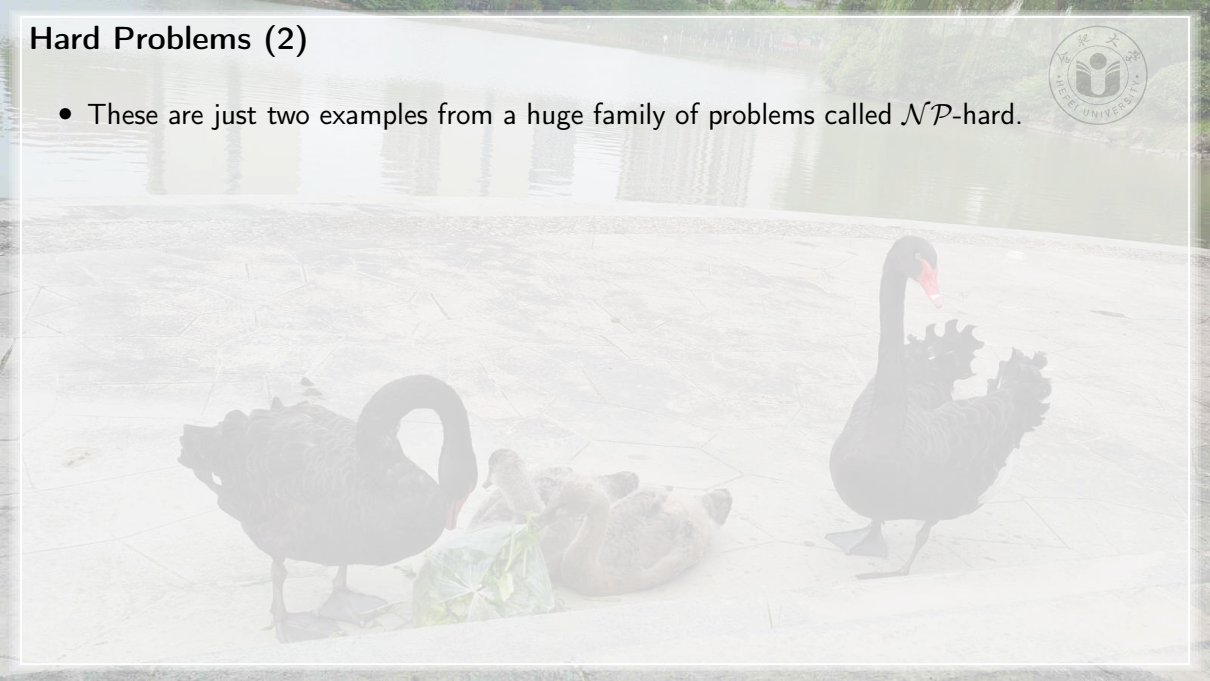


Traveling Salesperson Problem

- Find the shortest path that visits n locations and returns back to its origin.[5,144]

Hard Problems (2)

- These are just two examples from a huge family of problems called $\mathcal{NP}$-hard.



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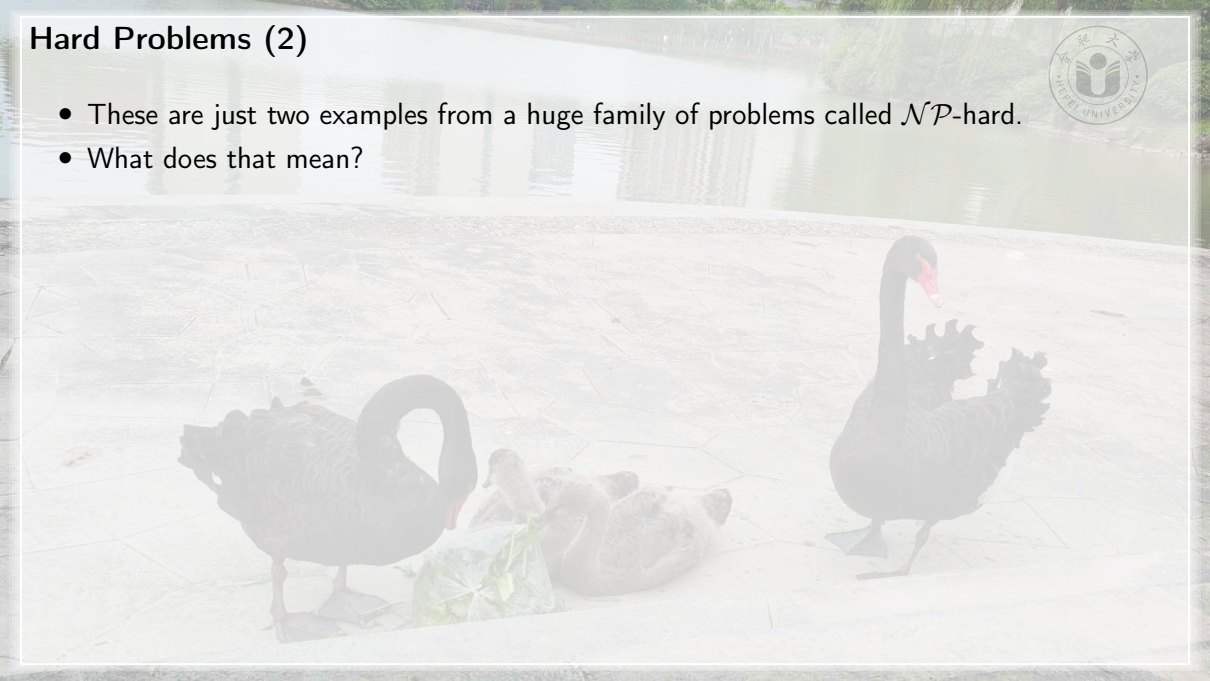

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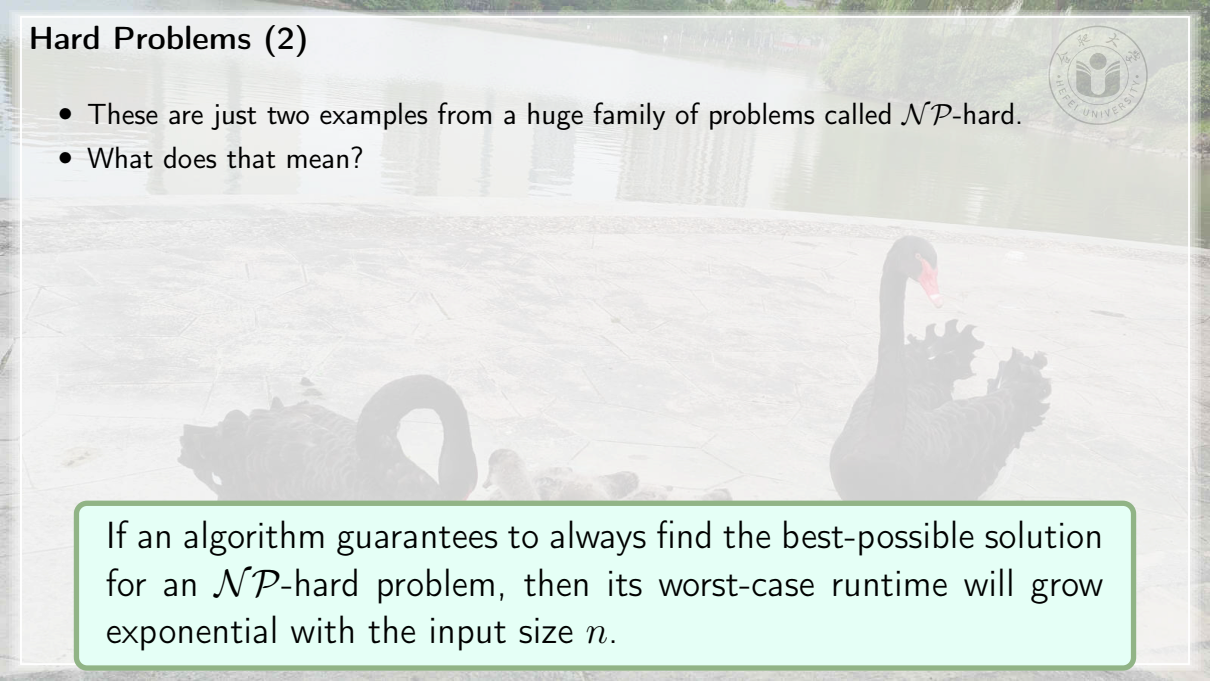

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If an algorithm guarantees to always find the best-possible solution for an $\mathcal{NP}$-hard problem, then its worst-case runtime will grow exponential with the input size n.

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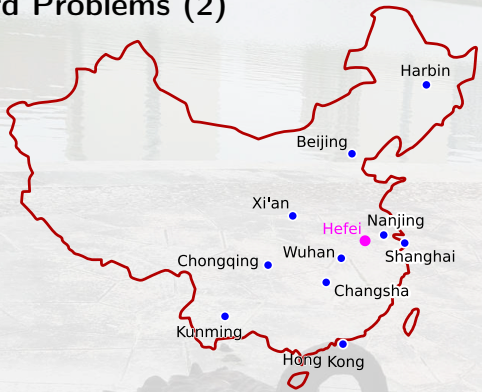

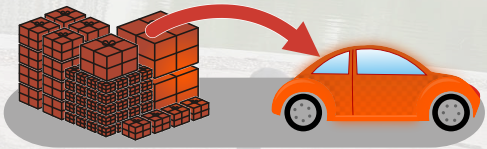

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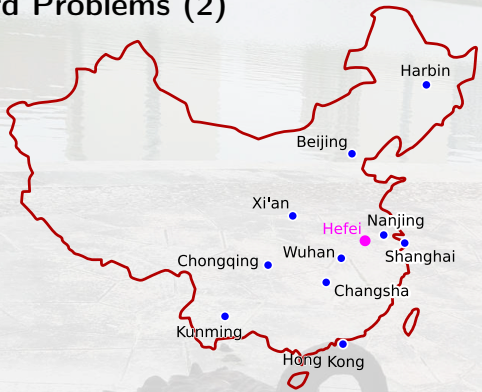

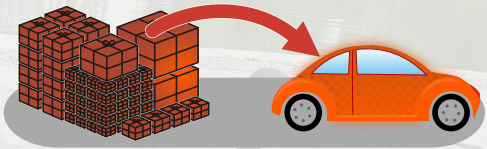

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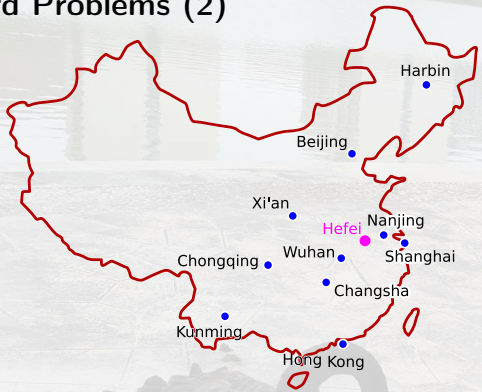
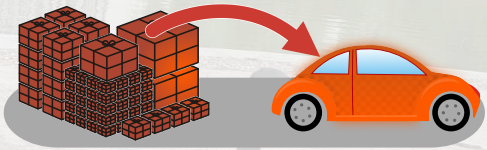


shortest round-trip tour



packing with fewest hauls

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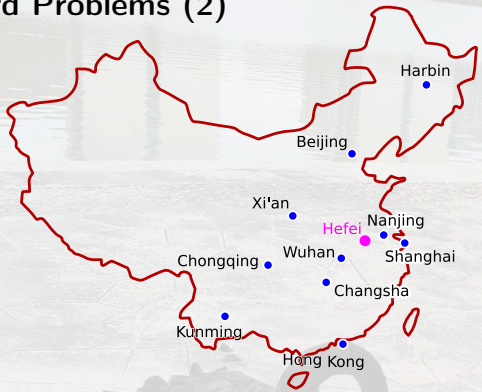
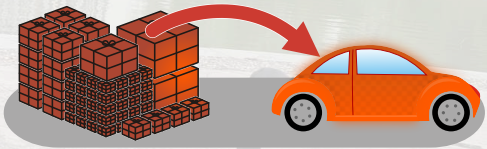
Hard Problems (2)





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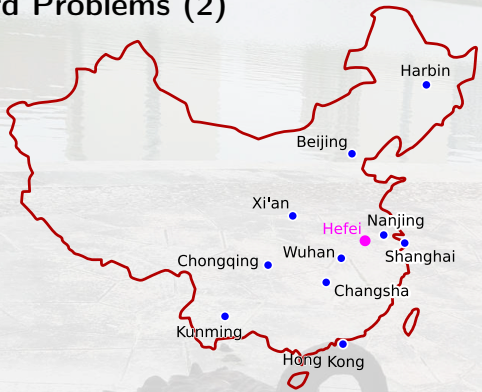
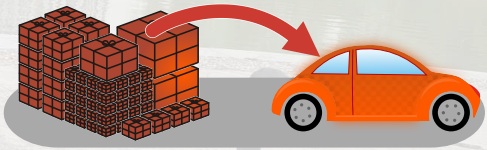


number n of cities to visit



number n of items to pack

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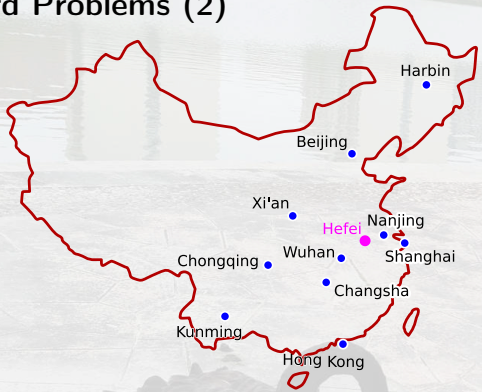

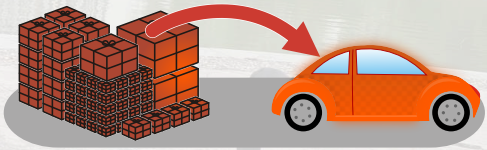

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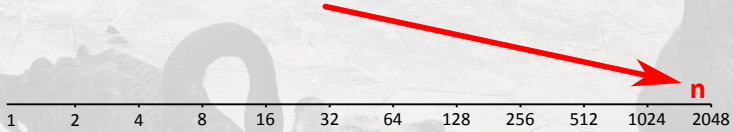
Hard Problems (2)

The problem scale n usually corresponding to the number of decision variables.



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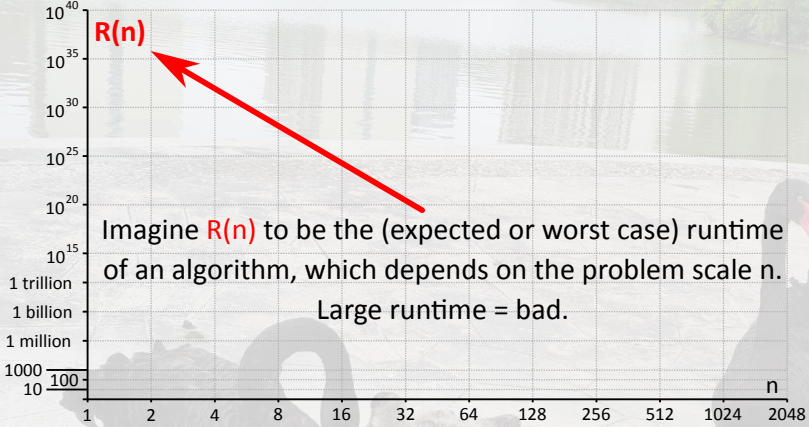

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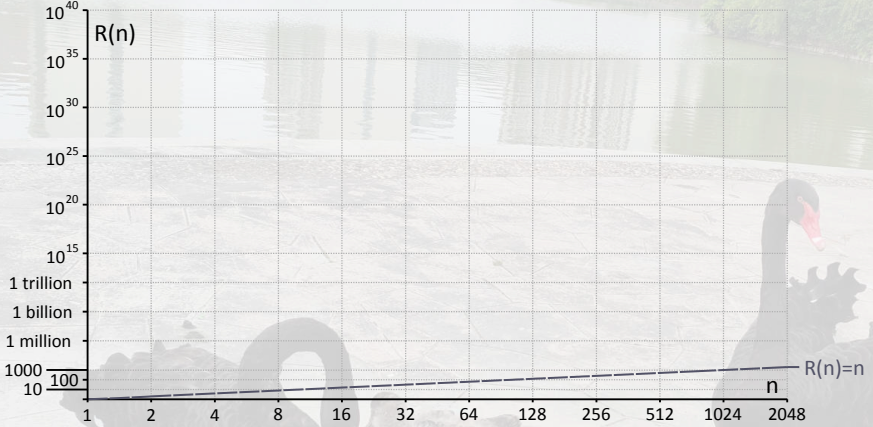

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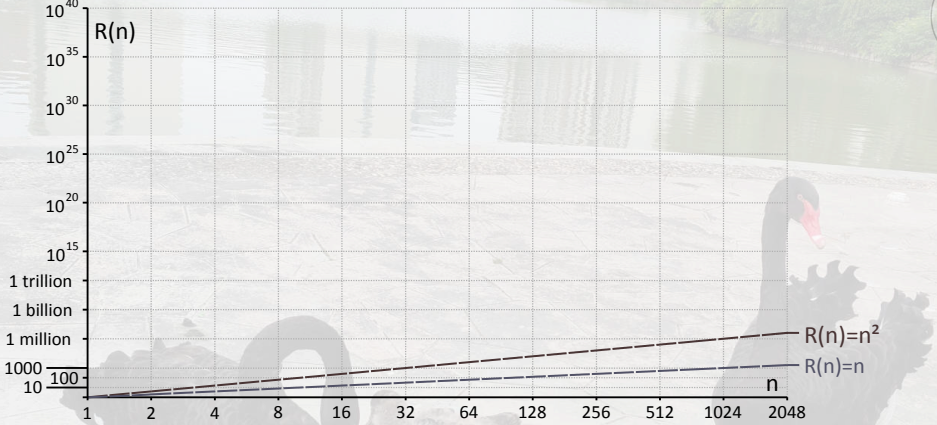

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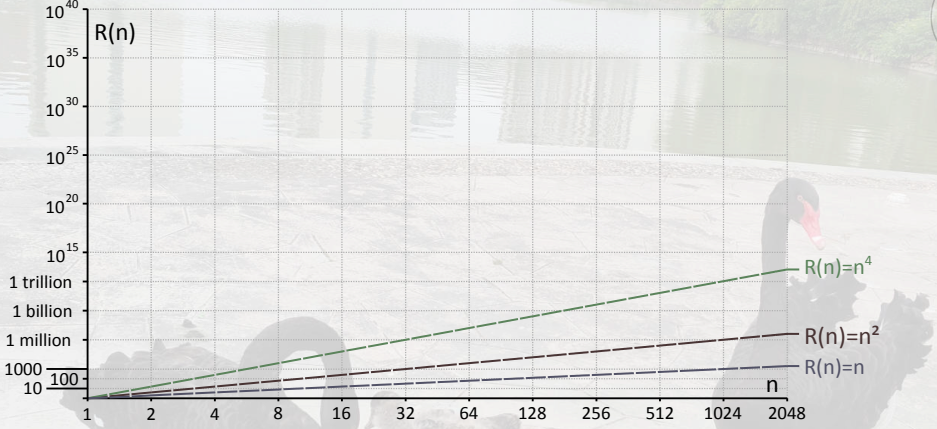

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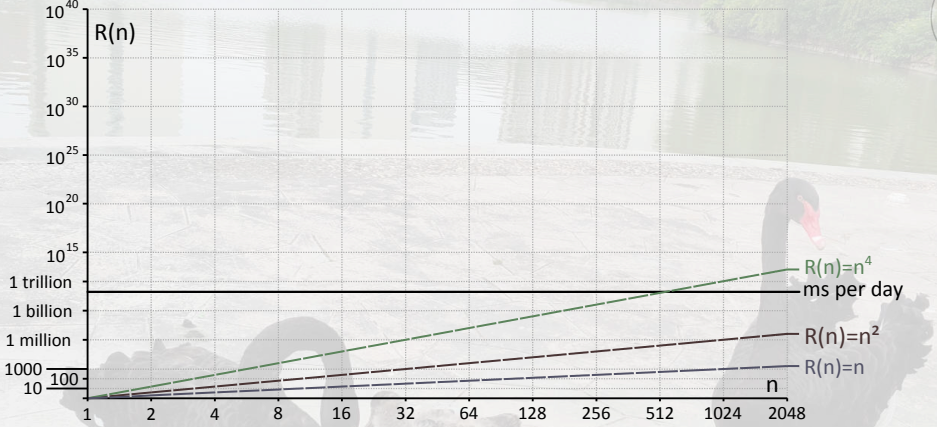

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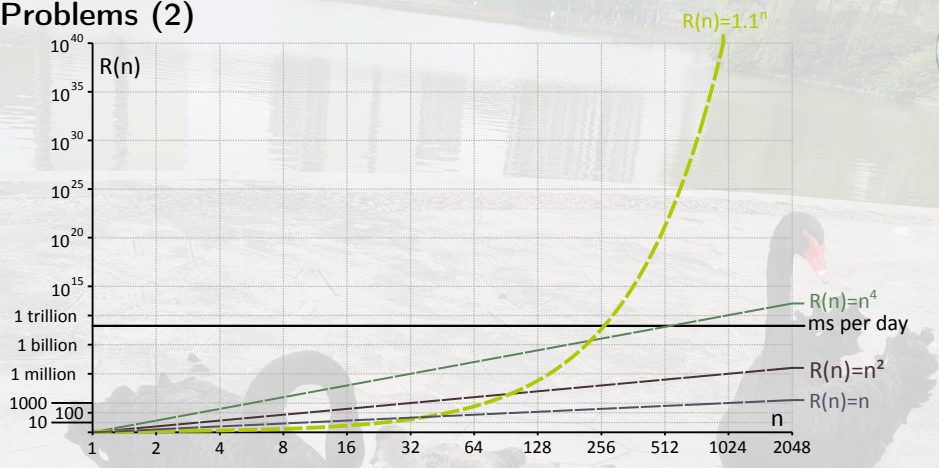

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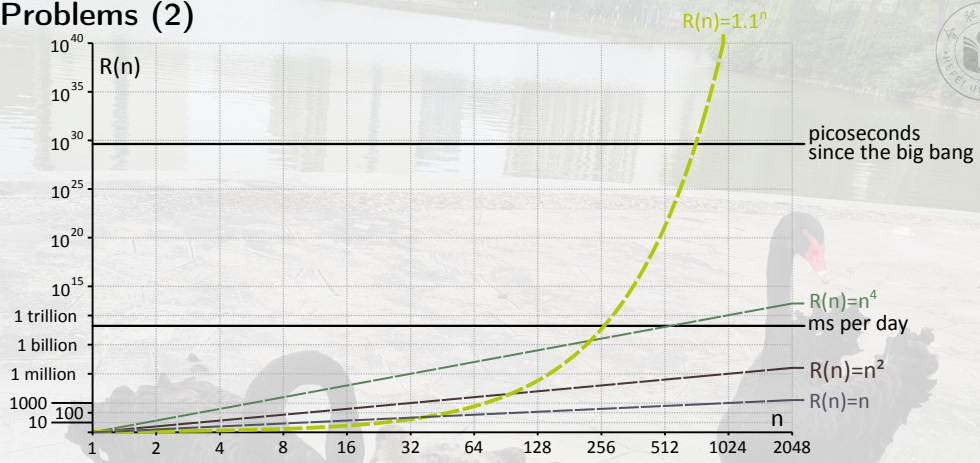

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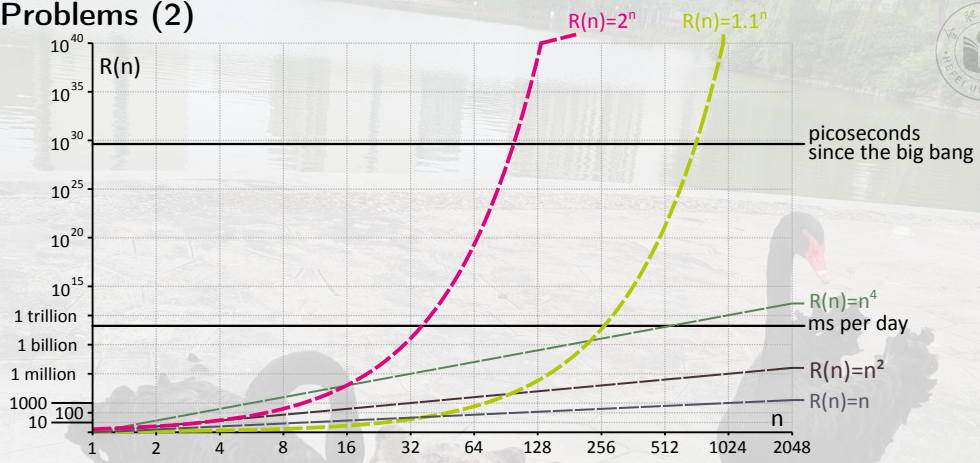

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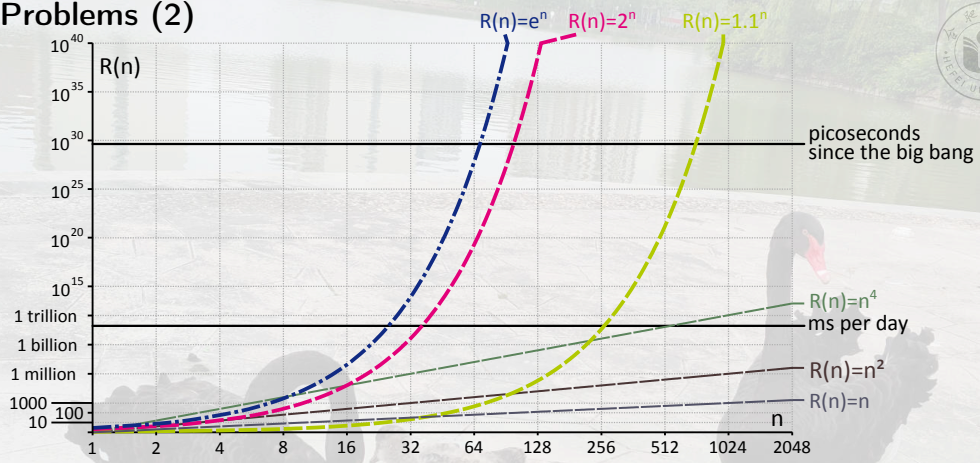

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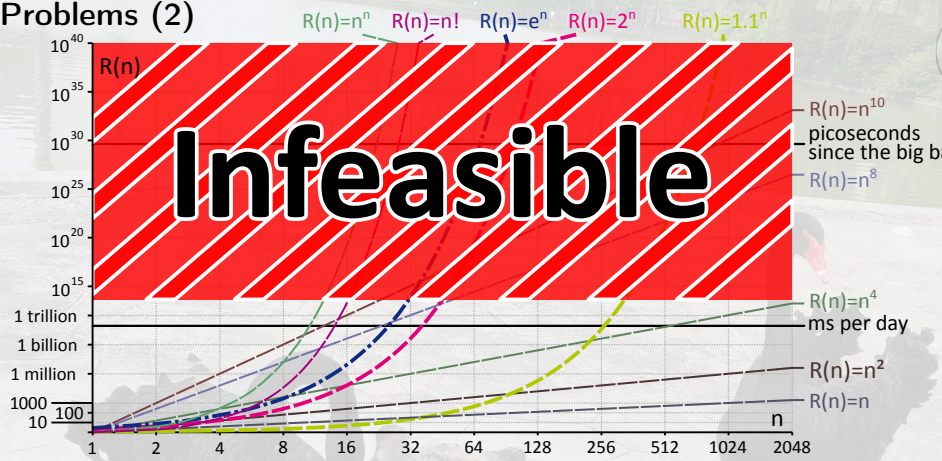

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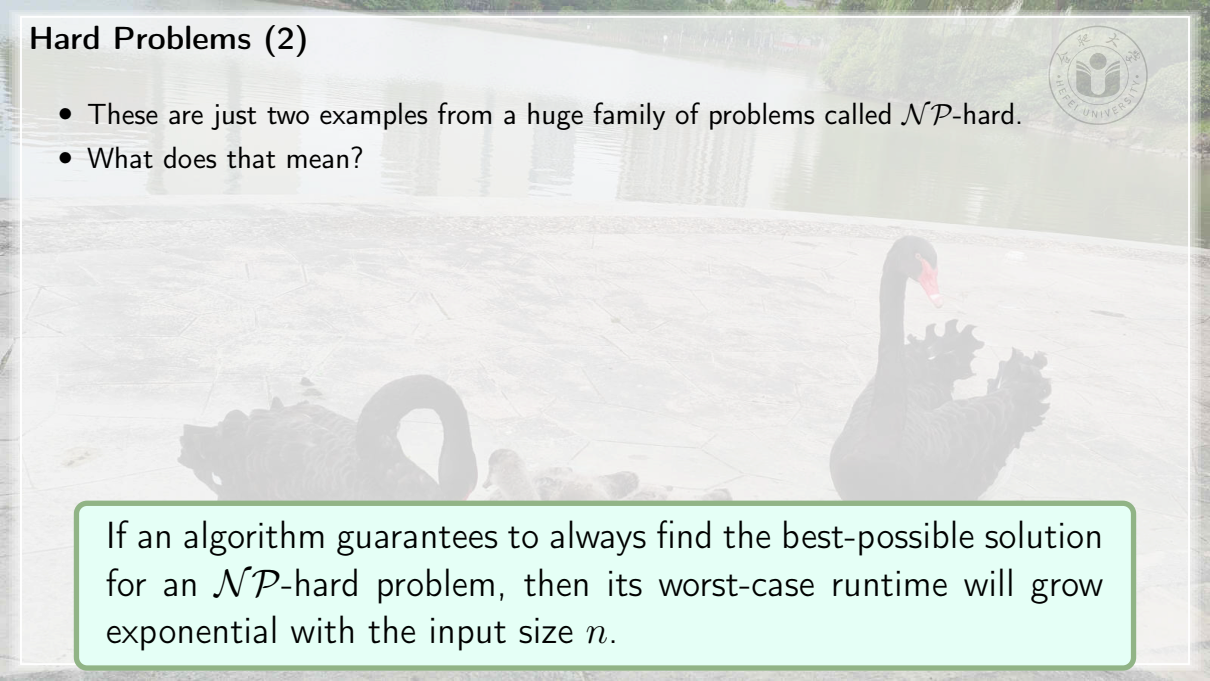

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- What does that mean?

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If we want to get the optimal solution,
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Metaheuristics

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Begin with a set $S_0 \subset \mathbb{X}$ of one or multiple randomly sampled solutions

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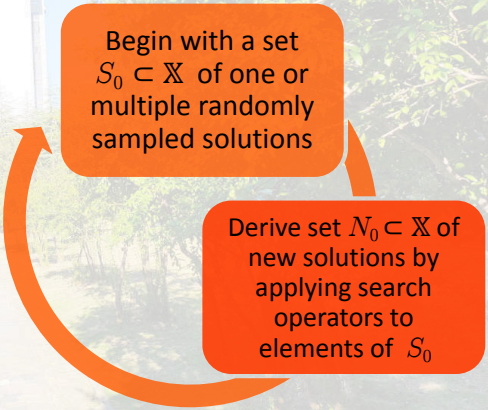

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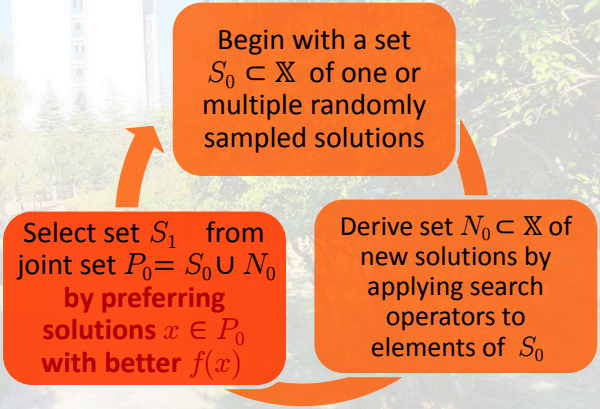

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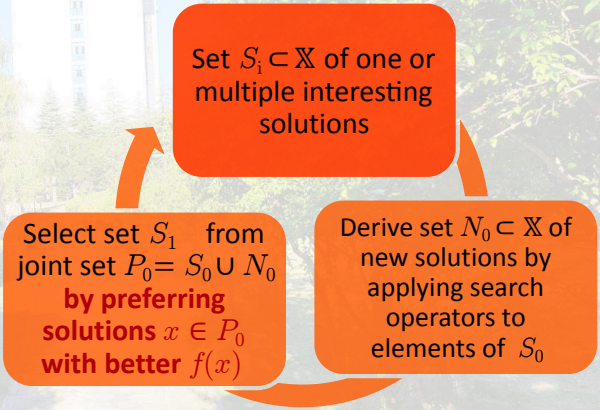

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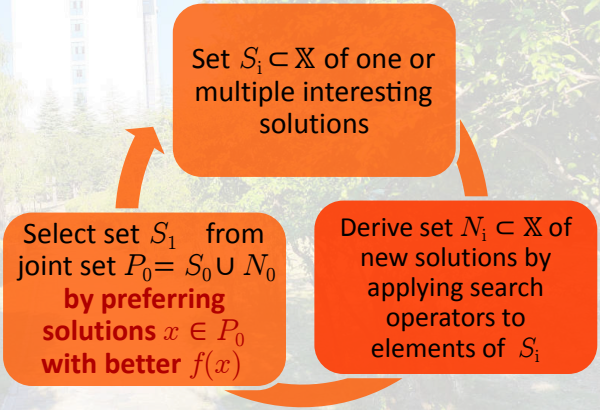

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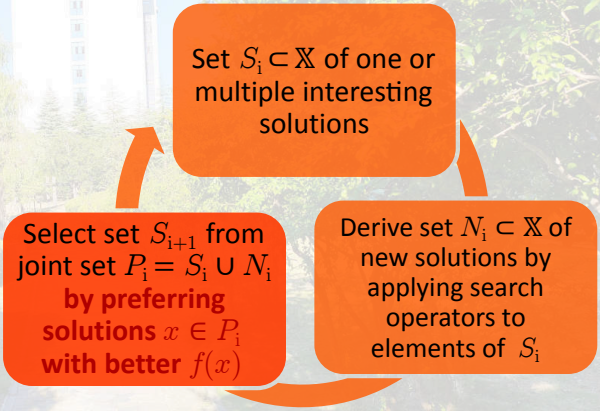

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- These are the usual main ingredients of metaheuristic optimization.

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- Randomized local search (RLS) and the $(1+1)$ evolutionary algorithm ($(1+1)$ EA) work according to this pattern[41,76].

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- SLS with $|S_i| = |N_i| = 1$ is the simplest realization of the metaheuristic idea.
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- Let's start with RLS/$(1+1)$ EA and modify it to become SA.

procedure SA($f : \mathbb{X} \mapsto \mathbb{R}$, T_0, ε)
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- $\mathfrak{R}_0^1$ be a random number uniformly distributed in $[0, 1)$.

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  while $\neg$ terminate **do**
    $x_n \leftarrow \texttt{move}(x_c)$; $y_n \leftarrow f(x_n)$;

    if $\mathfrak{R}_0^1 < e^{y_c - y_n}$ **then**
      $x_c \leftarrow x_n$; $y_c \leftarrow y_n$;

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  randomly sample x_c from $\mathbb{X}$; $y_c \leftarrow f(x_c)$;

  while $\neg$ terminate **do**
    $x_n \leftarrow$ `move`(x_c); $y_n \leftarrow f(x_n)$;

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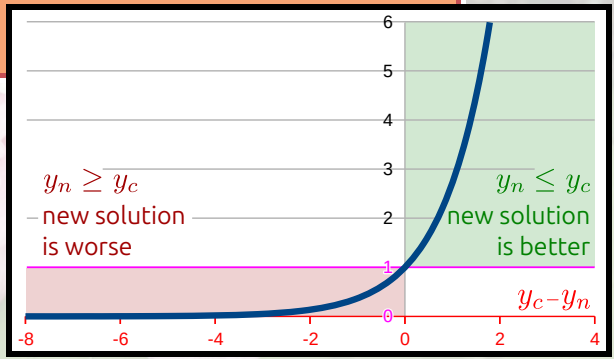

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    $T \leftarrow T_0$;
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  randomly sample x_c from $\mathbb{X}$; $y_c \leftarrow f(x_c)$;
  $x_B \leftarrow x_c$; $y_B \leftarrow y_c$; ▷ preserve best!
  $\tau \leftarrow 0$; ▷ τ is iteration counter
  while ¬ terminate **do**
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- So let's directly skip directly to the *research frontier* of this field.
- Let us therefore look a handful of discrete and combinatorial problems.
- (We ignore the continuous domain $\mathbb{X} = \mathbb{R}^n$.)

Discrete Optimization

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Black-Box Discrete Optimization Benchmarks

Black-Box Discrete Optimization Benchmarks: Introduction

- Problem: Search space $\mathbb{X}$ corresponds to the set $\{0,1\}^n$ of bit strings of length n and *nothing* is known about objective function $f : \{0,1\}^n \mapsto \mathbb{N}_0$.

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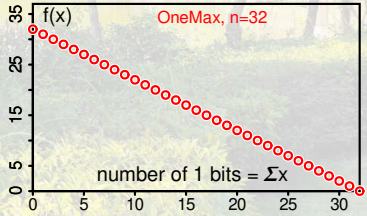
- The benchmarks are functions that scale with n and are usually computable in $\mathcal{O}(n)$ or $\mathcal{O}(n^2)$.
- Their optima are known (but the algorithms we apply to them don't know them).
- So what are such benchmarks?

Black-Box Discrete Optimization Benchmarks: Benchmarks

- OneMax[77,179]: maximize number of 1 bits, very easy, single optimum.



Black-Box Discrete Optimization Benchmarks: Benchmarks

- OneMax[77,179]
- Linear Harmonic Function (LinHarm)[70,76]: like OneMax, but bits have linearly increasing weights.

Black-Box Discrete Optimization Benchmarks: Benchmarks

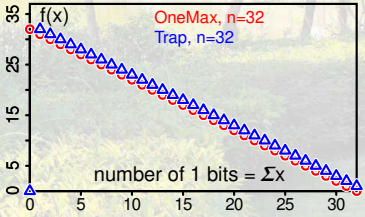
- OneMax[77,179], Linear Harmonic Function (LinHarm)[70,76]
- Binary Integer (BinInt)[222]: like LinHarm, but bits have exponentially increasing weights.

Black-Box Discrete Optimization Benchmarks: Benchmarks

- OneMax[77,179], Linear Harmonic Function (LinHarm)[70,76], Binary Integer (BinInt)[222]
- LeadingOnes[203,249]: number of leading 1 bits; contribution of variable x_i depends on all variables x_j with $j < i$.

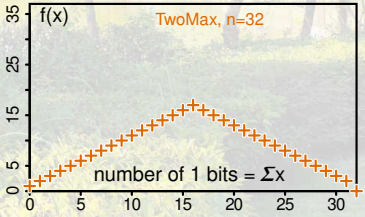
Black-Box Discrete Optimization Benchmarks: Benchmarks

- OneMax[77,179], Linear Harmonic Function (LinHarm)[70,76], Binary Integer (BinInt)[222], LeadingOnes[203,249]
- Trap[76,183]: OneMax, but optimum and worse solution swapped – extremely hard, local search needs exponential runtime.



Black-Box Discrete Optimization Benchmarks: Benchmarks

- OneMax[77,179], Linear Harmonic Function (LinHarm)[70,76], Binary Integer (BinInt)[222], LeadingOnes[203,249], Trap[76,183]
- TwoMax[92,228]: two optima, a global and a local one, of same size, local search needs exponential runtime with 50% probability.



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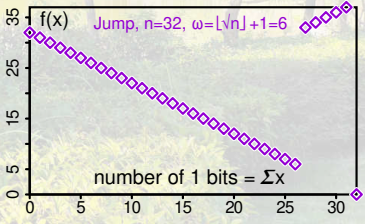
- Jump[76,92]: like OneMax, but last ω bit flips around optimum are deceptive = lead away from optimum.



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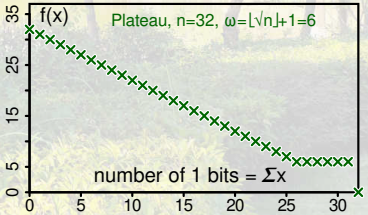
- Plateau[4]: like Jump, but last ω bit fips around optimum are neutral = have same objective value.



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- Ising Models[70,87,88]: variables should take on same values as neighbors in a given tology (ring, torus, ...); many interactions between variables.

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- N-Queens[70]: place N queens on a chessboard so that they cannot beat each other, many interactions between variables.

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- OneMax[77,179], Linear Harmonic Function (LinHarm)[70,76], Binary Integer (BinInt)[222], LeadingOnes[203,249], Trap[76,183], TwoMax[92,228], Jump[76,92], Plateau[4], Ising Models[70,87,88], N-Queens[70]
- W-Model[239,247]: allows you to tune and combine the above features.

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- OneMax[77,179], Linear Harmonic Function (LinHarm)[70,76], Binary Integer (BinInt)[222], LeadingOnes[203,249], Trap[76,183], TwoMax[92,228], Jump[76,92], Plateau[4], Ising Models[70,87,88], N-Queens[70], W-Model[239,247]
- If an algorithm performs well on all of these problems, then it can probably also solve real problems quite well...

Black-Box Discrete Optimization Benchmarks: Baselines

- So what are the baseline algorithms for black-box discrete optimization, i.e., the algorithms that you absolutely must beat and against which any new idea should be compared?

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- So what are the baseline algorithms for black-box discrete optimization, i.e., the algorithms that you absolutely must beat and against which any new idea should be compared?
  - randomized local search (RLS): unary operator flips 1 bit
  - $(1+1)$ EA: unary operator flips each bit with probability $1/n$

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- So what are the baseline algorithms for black-box discrete optimization, i.e., the algorithms that you absolutely must beat and against which any new idea should be compared?
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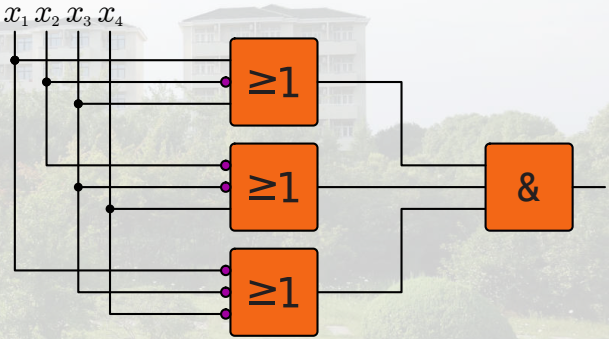
  - $(1+1)$ FEA by Weise (汤卫思), Wu (吴志泽), Li (李新路), and Chen (陈岩) [245, 246, 257], i.e., the $(1+1)$ EA with Frequency Fitness Assignment (FFA): the *only* algorithm achieving polynomial runtime on Trap, TwoMax, and Jump. Invented here.

Discrete Optimization

Maximum Satisfiability Problem

Maximum Satisfiability Problem: Introduction

- The goal of the Maximum Satisfiability (MaxSAT)[51,114] problem is to find a setting of n variables that makes a Boolean formula F become True. The variables appear directly or negated in m OR-clauses, whose results flow into one AND-clause.



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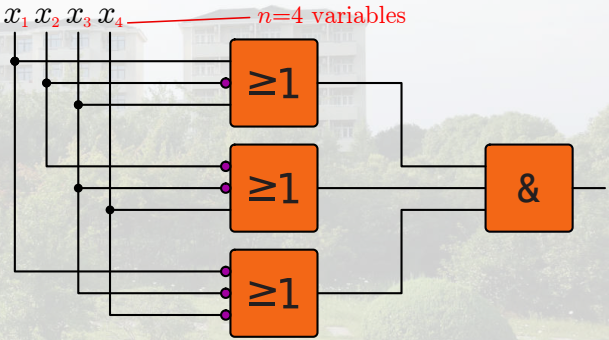

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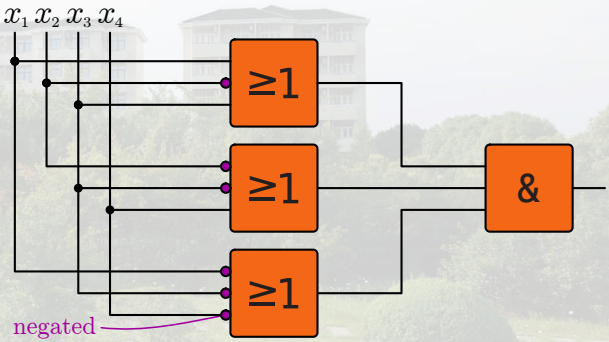

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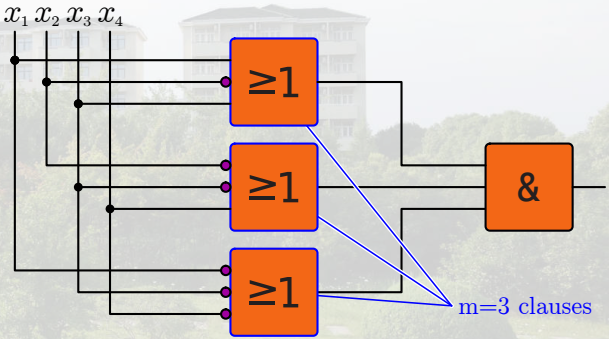

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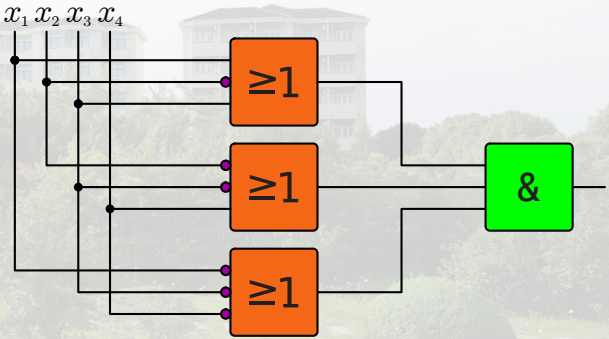

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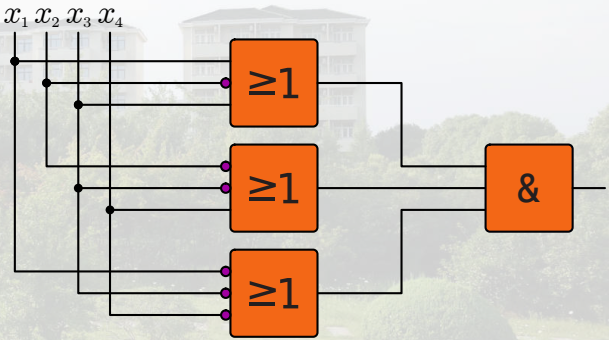

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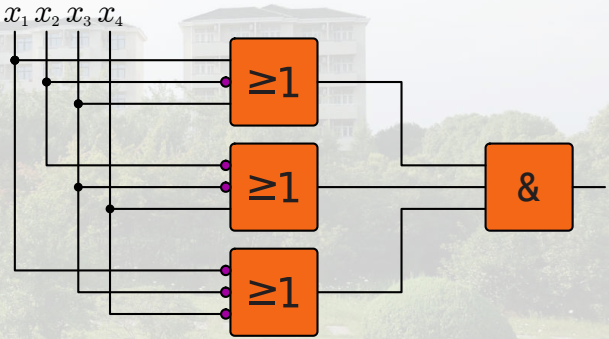

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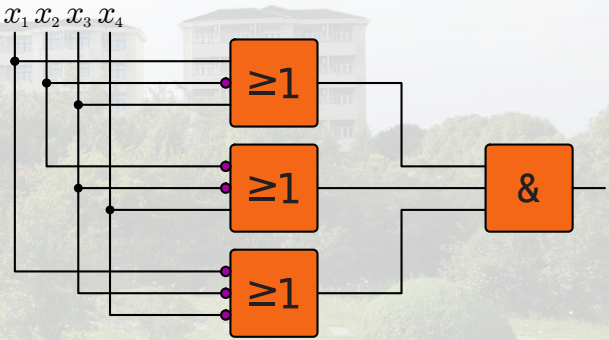

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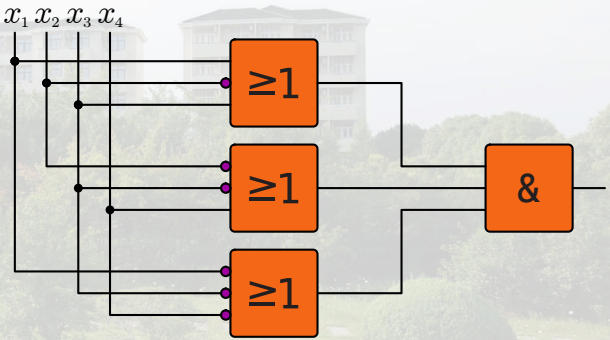
- MaxSAT is $\mathcal{NP}$-complete[51].



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- The MaxSAT Evaluation (MSE) offers a much more comprehensive and up-to-date set of benchmark instances[122] as well as solvers and results[229].

Maximum Satisfiability: Solver Types

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  - Incomplete Solvers: provide good solutions quickly, but may not guarantee optimality.
- Metaheuristics tend to be incomplete solvers and these are therefore more interesting for us.

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  - A good problem-specific baseline is WalkSAT by Selman, Kautz, and Cohen [206].

Maximum Satisfiability Problem: SOTA

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- The SOTA for heuristic incomplete solvers comprises SLS algorithms, often based on ideas dating back to WalkSAT, such as
  - Adaptive Memory-based Local Search (AMLS) by Lü and Hao [161] or
  - NuWLS by Chu, Cai, and Luo [49].

Discrete Optimization

Low Autocorrelation Binary Sequences

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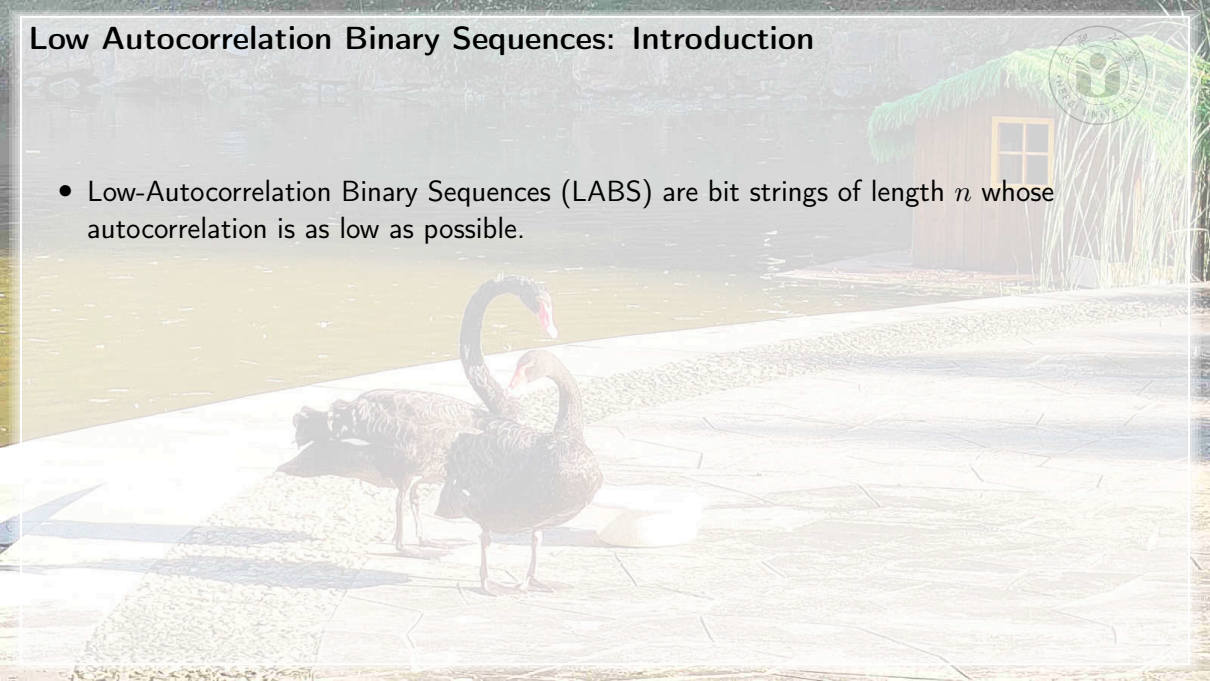

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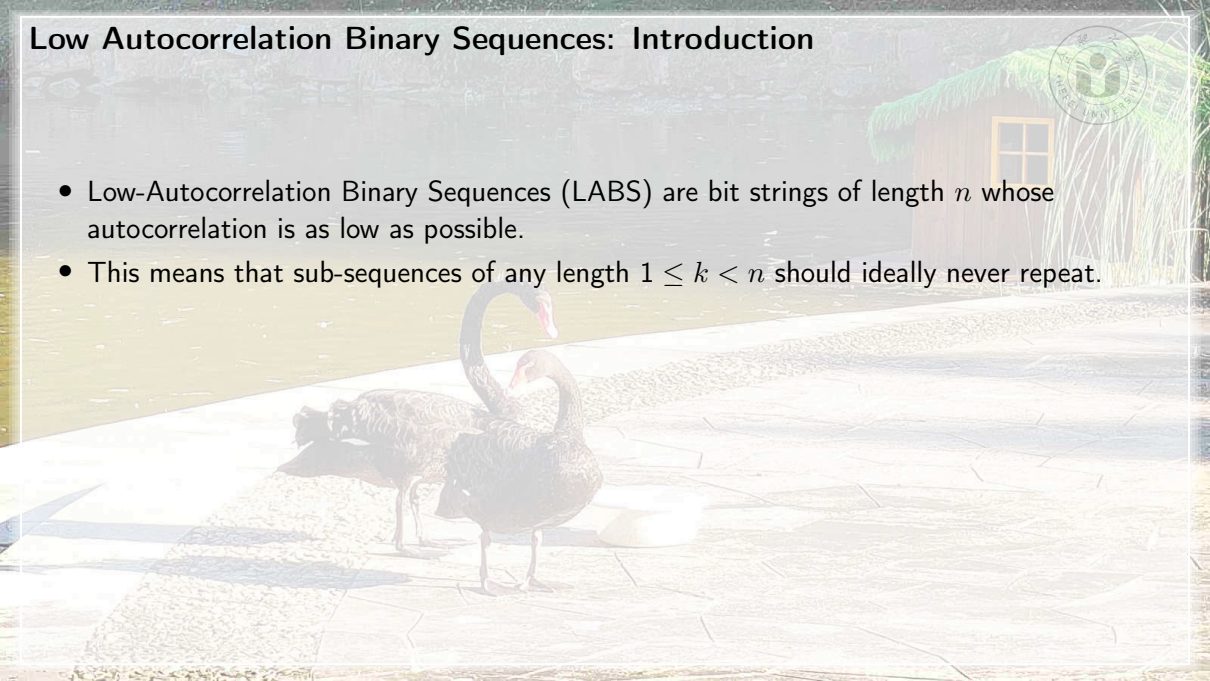

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- The LABS is $\mathcal{NP}$-hard[226].

Low Autocorrelation Binary Sequences: Benchmarks

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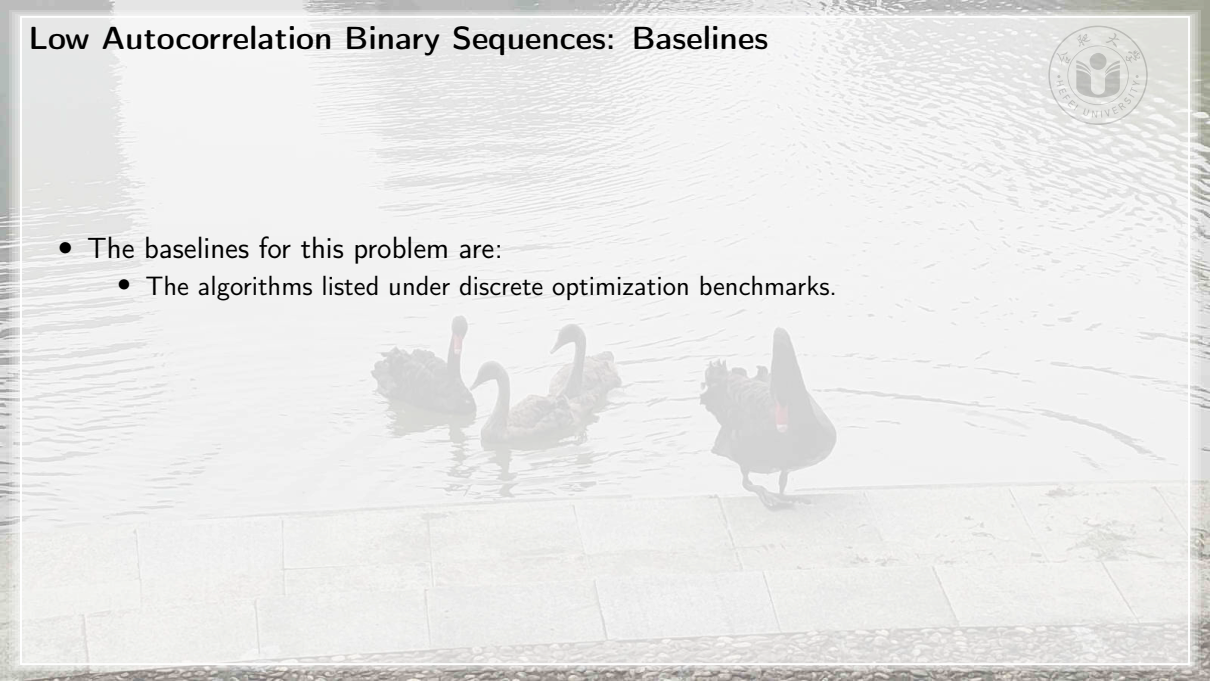

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(Multi-)Permutation Problems

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  - $(\mu+\lambda)$ evolutionary algorithm ($(\mu+\lambda)$ EA) with these operators and maybe Alternating Position Crossover (AP)[140,141].

(Multi-)Permutation Problems

Traveling Salesperson Problem

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  - The TSP Suite, which offers an integrated Java environment for benchmarking algorithms on the TSPLib instances[237].

Traveling Salesperson Problem: Baselines

- Besides permutation baselines, the baseline for the TSP is the Lin-Kernighan (LK) algorithm[152], which exchanges variable number of edges in a tour in each step.

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(Multi-)Permutation Problems
Quadratic Assignment Problem

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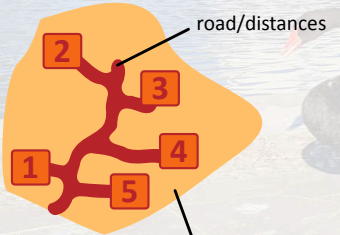

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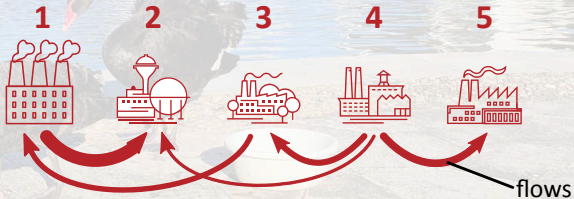

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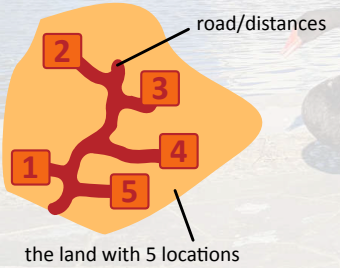

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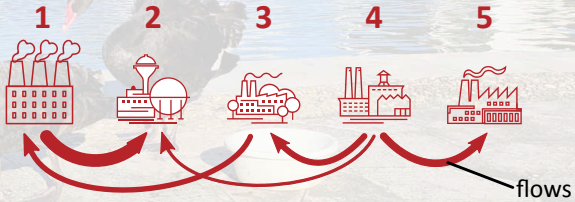

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- The QAP has applications such as building layout[56,81,137], keyboard layout[39], circuit design[82], wiring[211], and scheduling[210].
- The QAP is $\mathcal{NP}$-complete[75,204] and even among this problem class, is considered as particularly hard for metaheuristics[20,37,157,214].

Quadratic Assignment Problem: Benchmarks

- The typical benchmark for the QAP are the 136 instances of QAPLIB[36,38], 28 of which have not yet been solved to optimality[3] (at least by 2023).

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(Multi-)Permutation Problems

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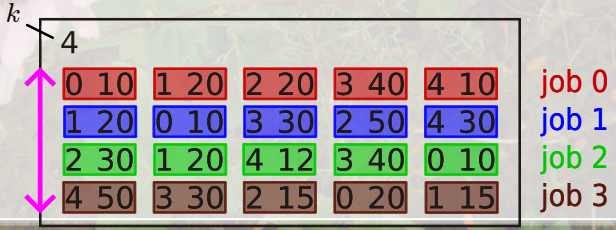

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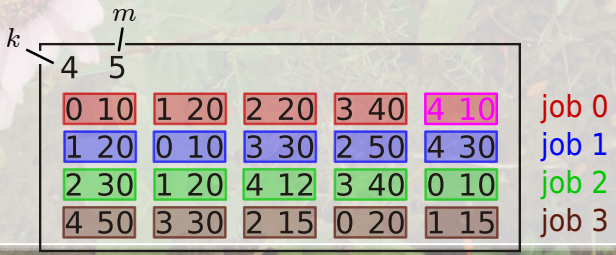

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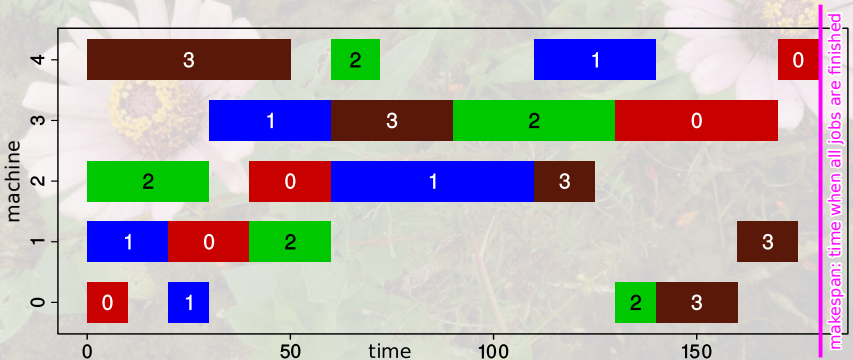

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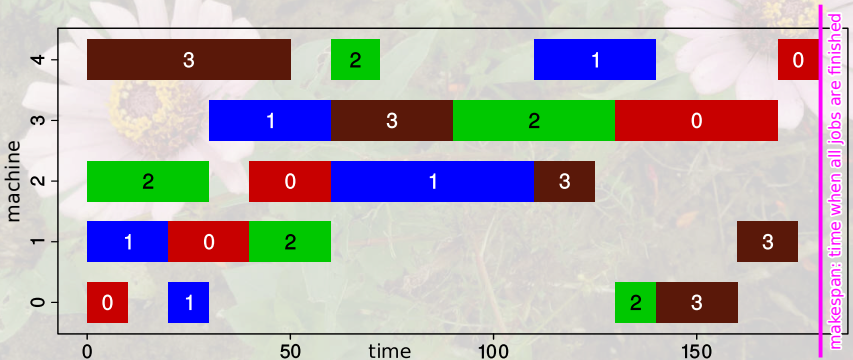

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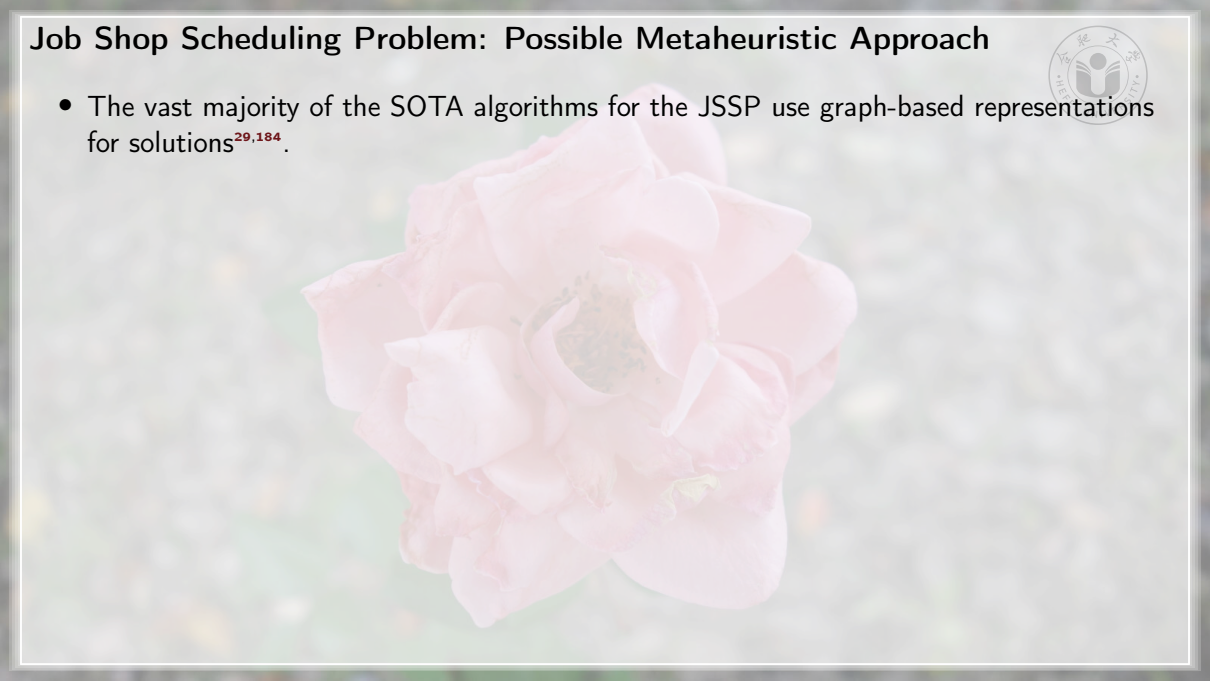

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  - Then we can use the algorithms and operators we already know...

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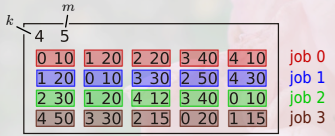

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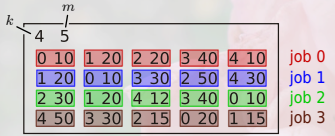

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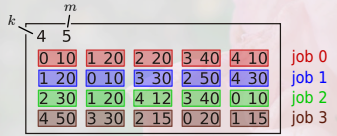

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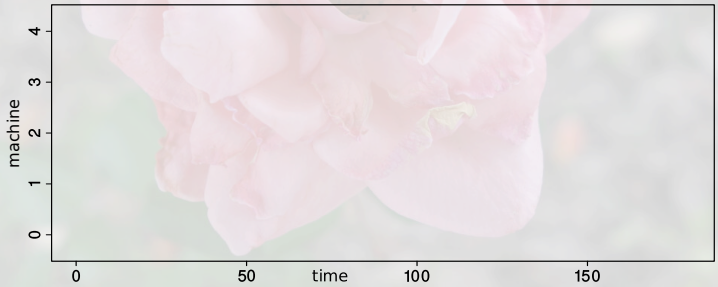

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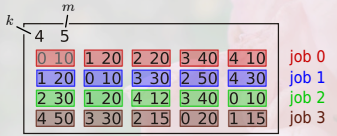

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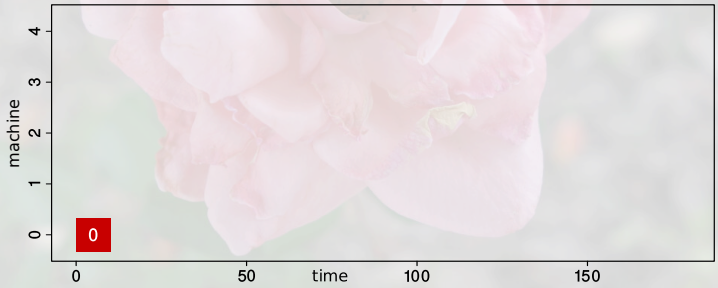

Job Shop Scheduling Problem: Possible Metaheuristic Approach

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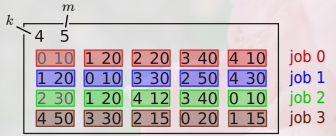

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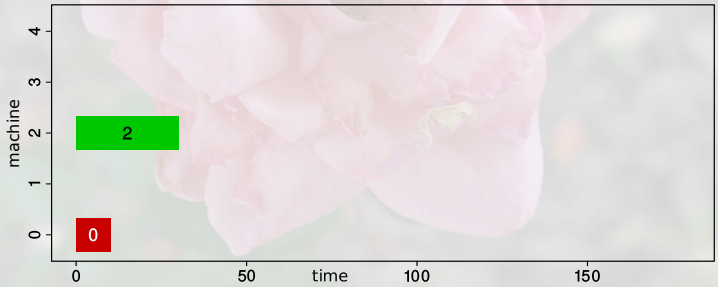

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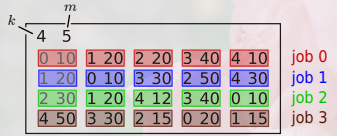

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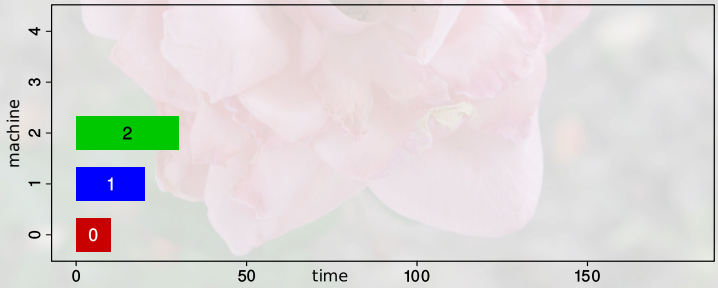

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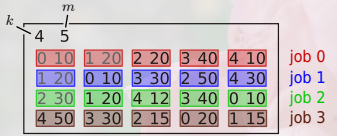

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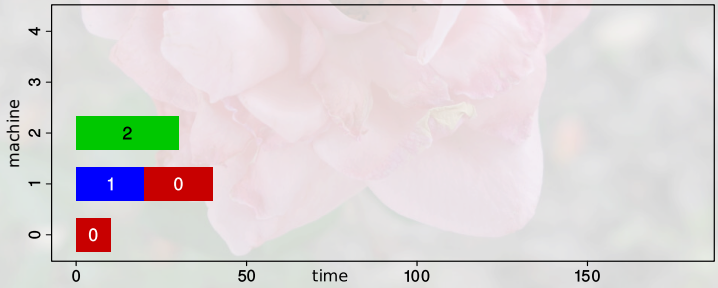

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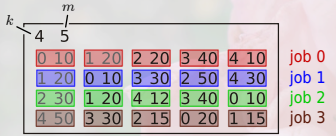

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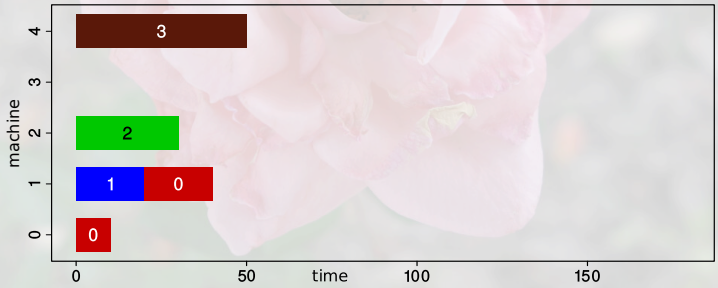

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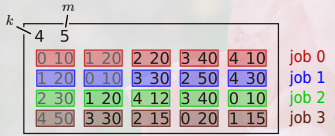

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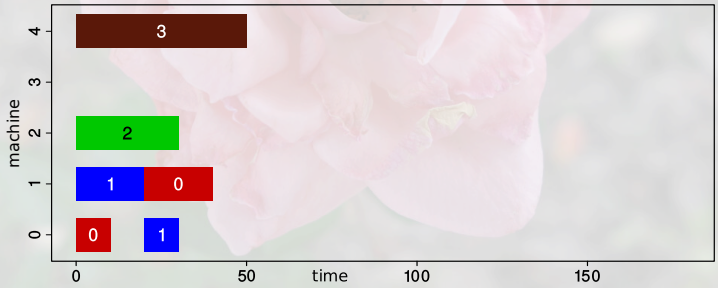

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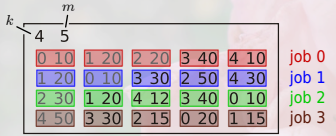

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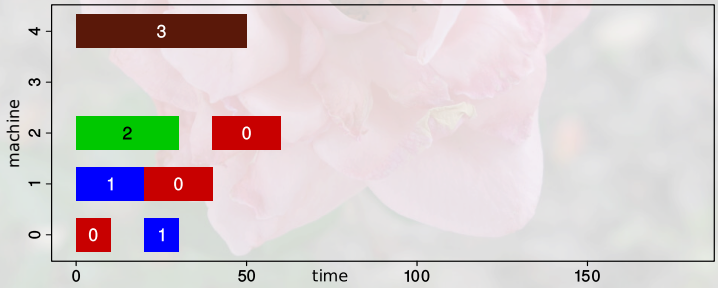

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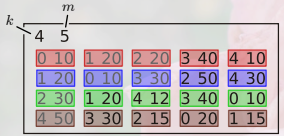

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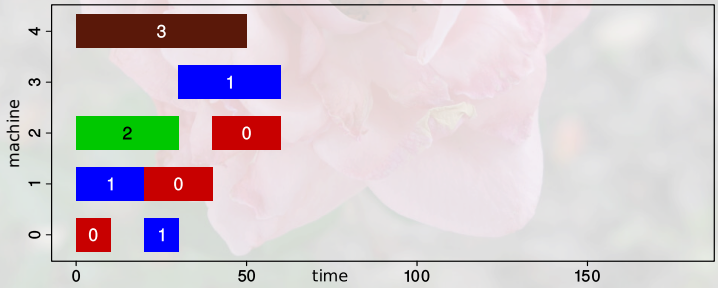

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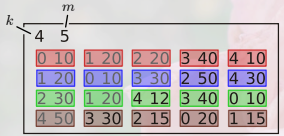

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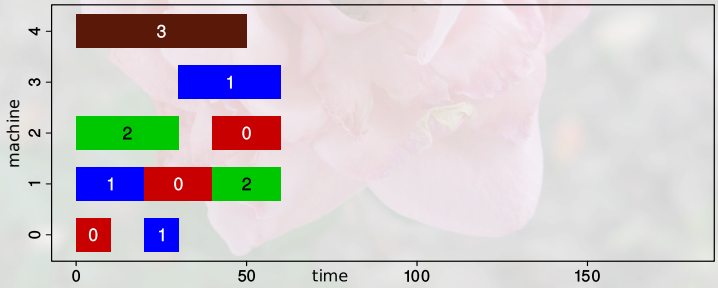

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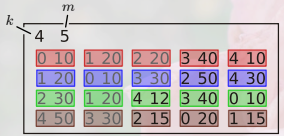

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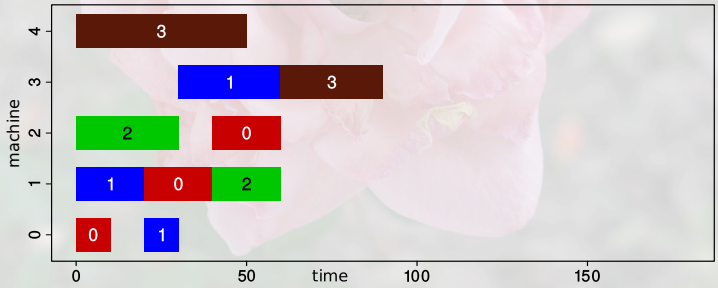

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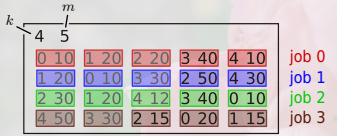

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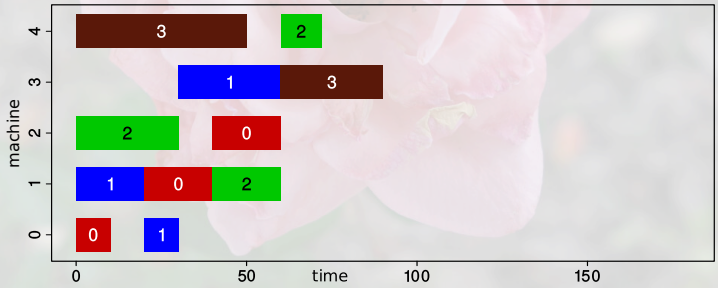

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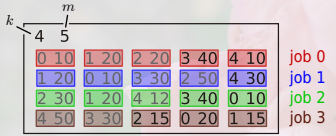

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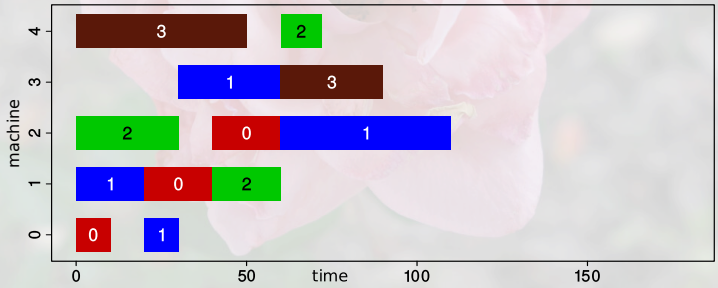

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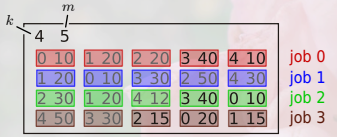

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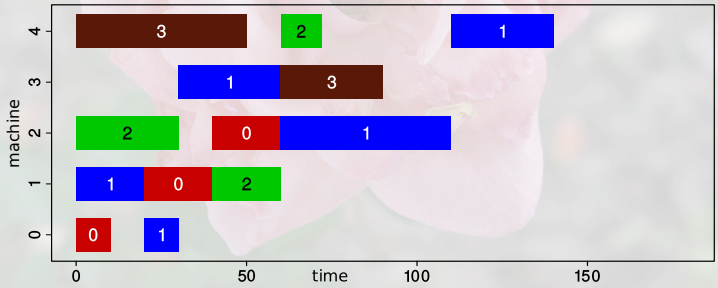

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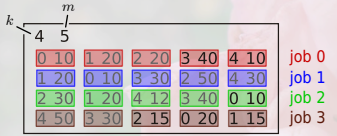

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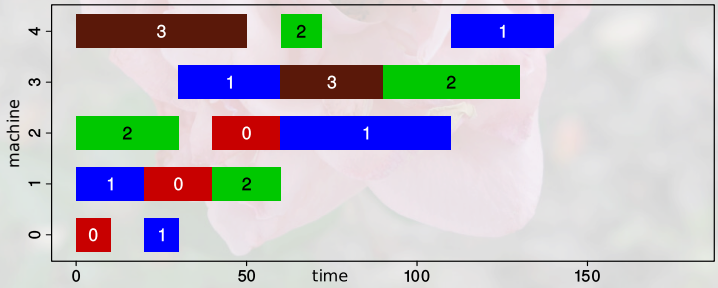

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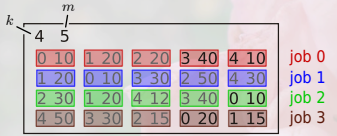

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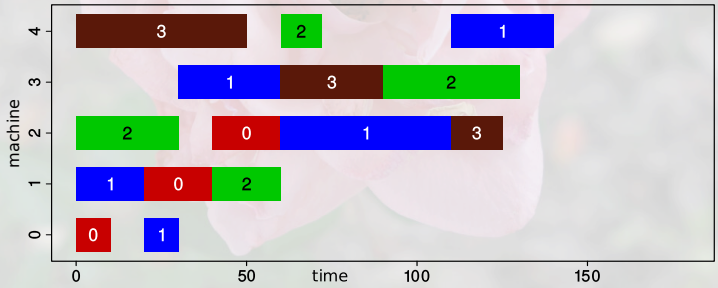

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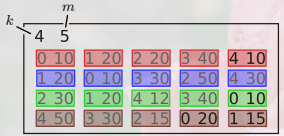

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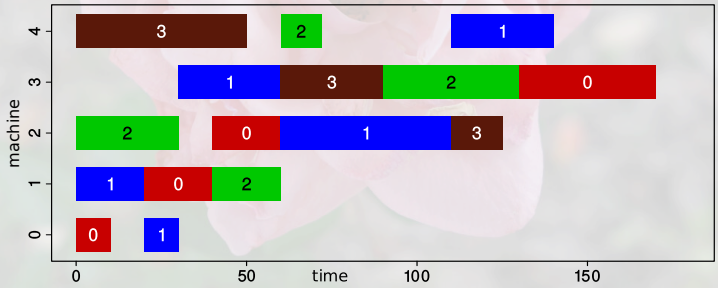

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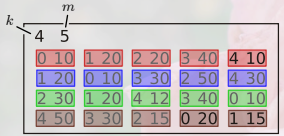

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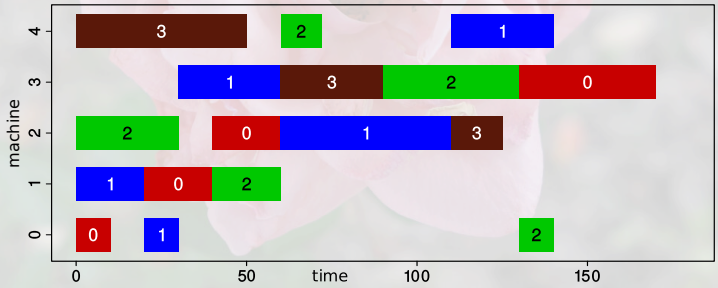

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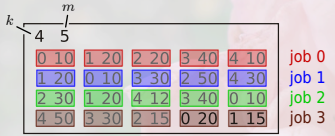

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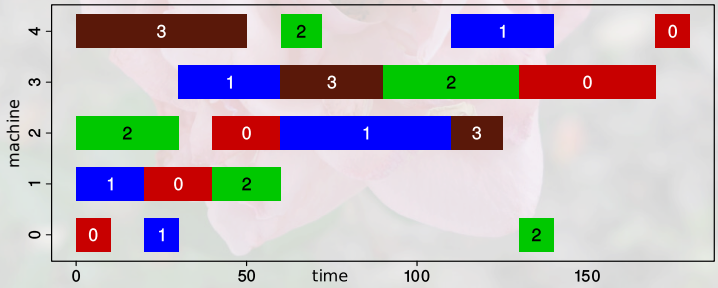

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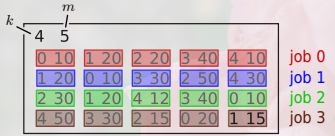

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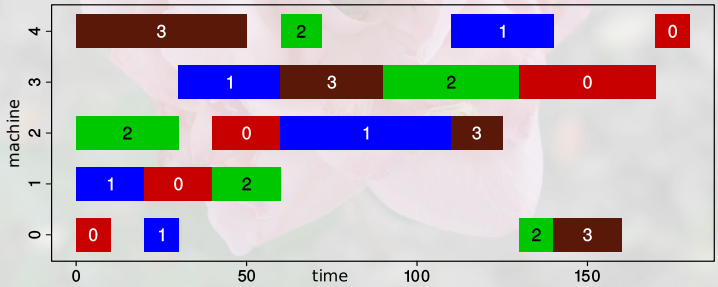

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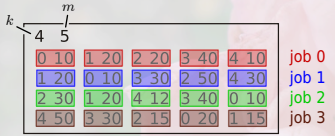

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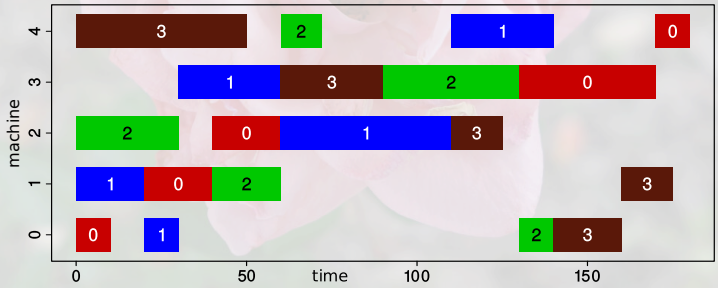

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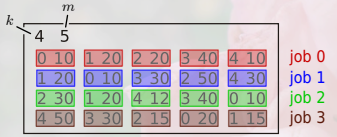

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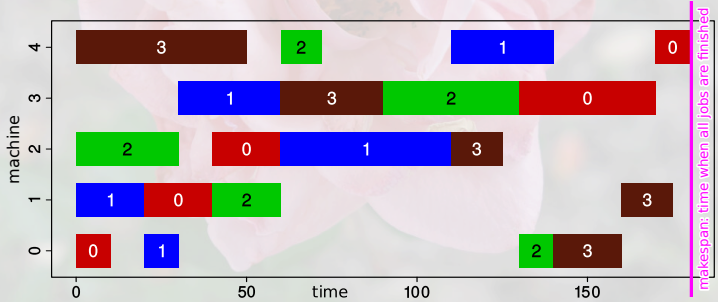

Job Shop Scheduling Problem: Benchmark

- The 8 most common sets of JSSP benchmark instances are *abz*[1], *dmu*[61], *ft*[89], *la*[145], *orb*[6], *swv*[212], *ta*[217], and *yn*[254].

Job Shop Scheduling Problem: Benchmark

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  - my own repository [234] at https://github.com/thomasWeise/jsspInstancesAndResults, where I collect all the above problem instances and many results from existing works.

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```
+++++++++++++++++++++++++++++++
Lawrence 15x10 instance (Table 7, instance 4); also called (setb4) or (B4)
15 10
 7  8 9 75 0 72 6 74 4 30 8 43 2 38 5 98 1 26 3 19
 6 19 8 73 3 43 0 23 1 85 4 39 5 13 9 26 2 67 7  9
 1 50 3 93 5 80 4  7 0 55 2 61 6 57 8 72 9 42 7 46
 1 68 7 43 4 99 6 60 5 68 0 91 8 11 3 96 9 11 2 72
 7 84 2 34 8 40 5  7 1 70 6 74 3 12 0 43 9 69 4 30
 8 60 0 49 4 59 5 72 9 63 1 69 7 99 6 45 3 27 2  9
 6 71 2 91 8 65 1 90 9 98 4  8 7 50 0 75 5 37 3 17
 8 62 7 90 5 98 3 31 2 91 4 38 9 72 1  9 0 72 6 49
 4 35 0 39 9 74 5 25 7 47 3 52 2 63 8 21 6 35 1 80
 9 58 0  5 3 50 8 52 1 88 6 20 2 68 5 24 4 53 7 57
 7 99 3 91 4 33 5 19 2 18 6 38 0 24 9 35 1 49 8  9
 0 68 3 60 2 77 7 10 8 60 5 15 9 72 1 18 6 90 4 18
 9 79 1 60 3 56 6 91 2 40 8 86 7 72 0 80 5 89 4 51
 4 10 2 92 5 23 6 46 8 40 7 72 3  6 1 23 0 95 9 34
```

15 jobs

10 machines

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```
+++++++++++++++++++++++++++++
Lawrence 15x10 instance (Table 7, instance 4); also called (setb4) or (84)
15 10                                  10 machines
15 jobs
 7  8  9 75  0 72  6 74  4 30  8 43  2 38  5 98  1 26  3 19
 6 19  8 73  3 43  0 23  1 85  4 39  5 13  9 26  2 67  7  9
 1 50  3 93  5 80  4  7  0 55  2 61  6 57  8 72  9 42  7 46
 1 68  7 43  4 99  6 60  5 68  0 91  8 11  3 96  9 11  2 72
 7 84  2 34  8 40  5  7  1 70  6 74  3 12  0 43  9 69  4 30
 8 60  0 49  4 59  5 72  9 63  1 49  7 99  6 45  3 27  2  9
 6 71  2 91  8 65  1 90  9 98  4  8  7 50  0 75  5 37  3 17
 8 62  7 90  5 98  3 31  2 91  4 38  9 72  1  9  0 72  6 40
 4 35  0 39  9 74  5 25  7 47  3 52  2 63  8 21  6 35  1 80
 9 58  0  5  3 50  8 52  1 88  6 20  2 68  5 24  4 53  7 57
 7 99  3 91  4 33  5 19  2 18  6 38  0 24  9 35  1 49  8  9
 0 68  3 60  2 77  7 10  8 60  5 15  9 72  1 18  6 90  4 18
 9 79  1 60  3 56  6 91  2 40  8 86  7 72  0 80  5 89  4 51
 4 10  2 92  5 23  6 46  8 40  7 72  3  6  1 23  0 95  9 34
 2 24  5 29  9 49  8 55  0 47  6 77  3 77  7  8  1 28  4 48
```

```
++++++++++++++++++++++++++++++
Storer, Wu, and Vaccari hard 50x10 instance (Table 2, instance 15)
50 10                                  10 machines
50 jobs
 2  93  4  40  0   1  3  77  1  77  5  16  9  74  8  11  6  51  7  92
 0  92  4  80  1  76  3  59  2  70  5  86  9  17  6  78  7  30  8  93
 1  44  2  92  3  96  4  77  0  53  9  10  7  49  5  84  8  59  6  14
 1  60  2  19  3  76  0  73  4  85  7  13  8  93  5  68  9  50  6  78
 2  20  0  24  3  41  1   2  4   4  9  44  7  79  8  81  5  16  6  39
 3  41  2  35  1  32  4  18  0  15  8  98  6  29  5  19  7  14  9  26
 1  59  0  45  4  53  3  44  2  98  5  84  6  23  7  45  8  39  9  89
 1  30  4  51  3  25  0  51  2  84  6  60  5  45  7  89  8  25  9  97
 0  47  3  18  2  40  4  62  1  58  5  36  7  93  8  77  9  90  6  15
 3  33  1  68  0  41  4  72  2  20  6  69  7  47  5  22  9  47  8  22
 2  28  1 100  4  20  0  35  3  26  5  24  9  41  6  42  7 100  8  32
 0  65  2  12  4  53  3  93  1  40  8  18  7  23  5  60  6  89  9  53
 0  58  1  60  4  97  3  31  2  50  9  85  5  64  7  38  6  85  8  35
 3  64  0  58  1  49  2  45  4   9  8  49  6  22  5  99  9  15  7   7
 0  10  4  85  3  72  2  37  1  77  5  70  7  45  9   8  6  83  8  57
 4  93  0  87  1  87  2  18  3   4  8  78  5  67  9  20  6  17  7  35
 4  72  0  56  3  57  2  15  1  45  6  41  5  40  9  85  8  32  7  81
 0  36  3  63  4  79  2  32  1   5  6  25  7  86  9  91  5  21  8  35
 2  83  4  29  0   9  1  38  3  73  7  50  9  99  5  18  8  29  6  41
 0 100  3  29  2  60  4  63  1  64  8  71  6  35  5  26  9   9  7  22
 1  81  0  60  3  62  4  48  2  68  7  28  5  69  8  92  6  79  9  10
 0  40  4  80  1  41  2  10  3  68  8  28  9  51  7  33  6  82  5  25
```

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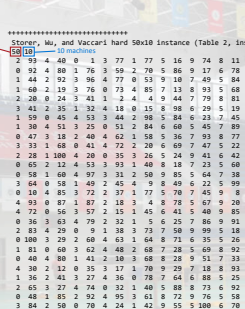

Job Shop Scheduling Problem: Baselines

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- Good classical algorithms to outperform are the Tabu Search (TS) algorithms by Nowicki and Smutnicki [184] and Pezzella and Merelli [192] or the local search (LS) by Balas and Vazacopoulos [11].

Job Shop Scheduling Problem: SOTA

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(Multi-)Permutation Problems

2D Bin Packing

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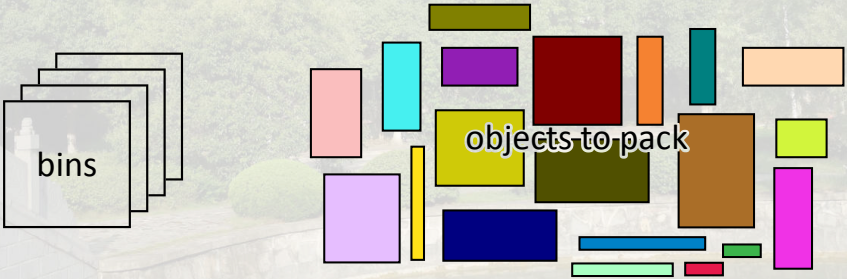

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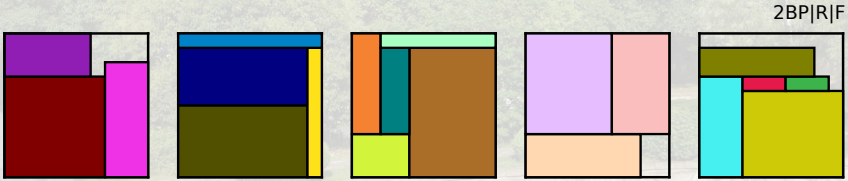

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- It has many practical applications, e.g., in cutting (wood, glas)[156], packing (transportation, warehouse)[156], newspaper page layout[213], as well as in scheduling and build formation in additive manufacturing[148,194].

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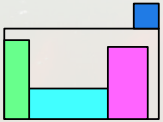
- The permutations are processed from beginning to end. The rectangle represented by the current number is placed on the top-right corner of the bin.



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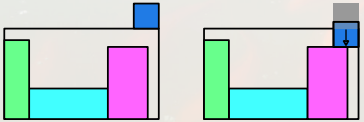
- As soon as the current rectangle can be moved down, it is moved down.



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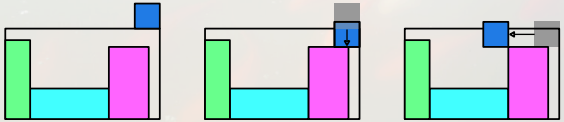
- Once it cannot be moved down anymore, it is moved to the left (until it could be moved down again).



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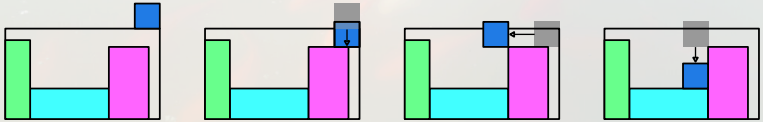
- The rectangle is moved down-left (with down prioritized) until it cannot be moved anymore.



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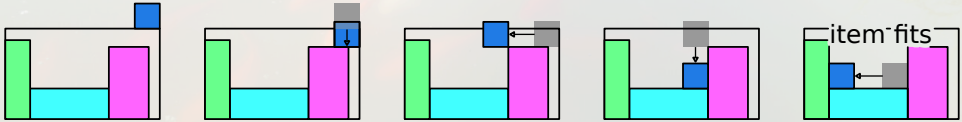
- Once it cannot be moved anymore, we check if it is completely inside the bin.



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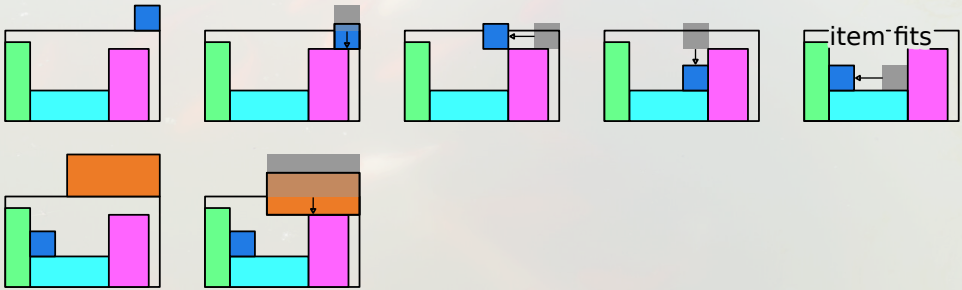

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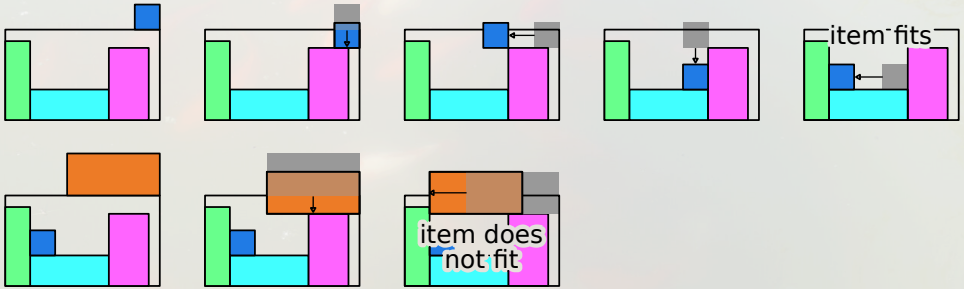

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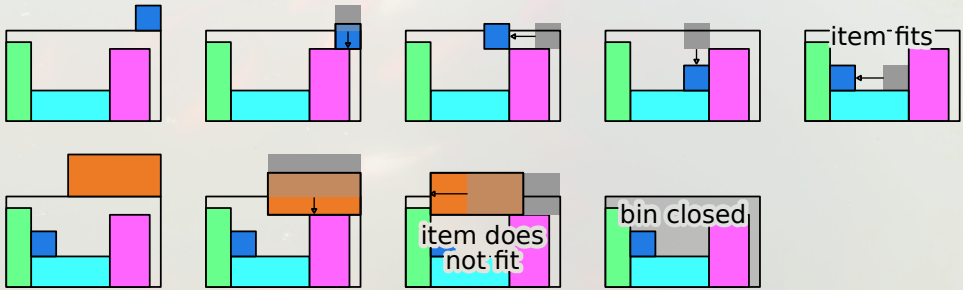
- If the rectangle does not fit into the bin, then we close the bin...



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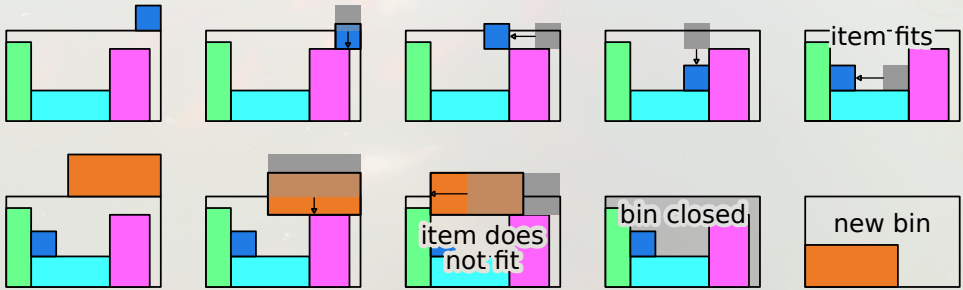

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- Either way: We can now use the same permutation search spaces and the operators and algorithms we already have developed for the TSP, QAP, and JSSP (maybe with that slight modification) to search for two-dimensional packings!

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- The almost-squares-in-almost-squares (Asqas) problem[227] asks us to put objects of sizes 1×2, 2×3, $\ldots$, $n \times (n+1)$ into a similarly almost-square bin without waste.

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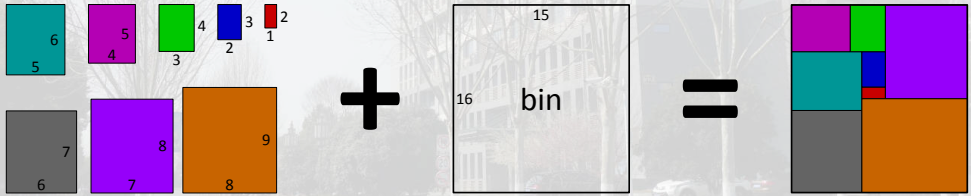
  - the *A* instance set[164] offers 43 instances with 13 to 809 items.
- The almost-squares-in-almost-squares (Asqas) problem[227] asks us to put objects of sizes 1×2, 2×3, $\ldots$, $n \times (n+1)$ into a similarly almost-square bin without waste. It has only 4 interesting instances, with bin sizes 5×4 for $n = 3$, 15×16 for $n = 8$, 55×56 for $n = 20$, and 119×120 for $n = 34$.



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  - Actually, the simple RLS by Zhao (赵睿), Wu (吴志泽), van den Berg, Thürer, Liang (梁天宇), Tan (檀明), and Weise (汤卫思) [259, 260] outperforms these methods on the 2BP|O|F and is quite competitive on the 2BP|R|F.

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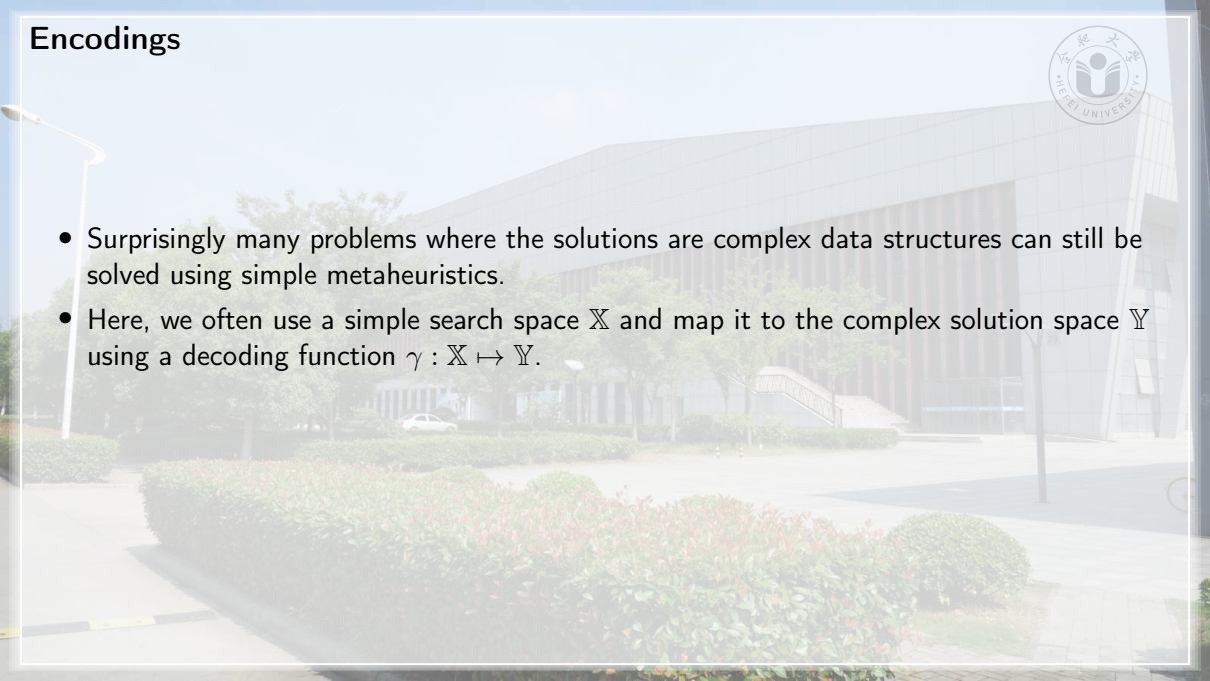
Encodings

- Surprisingly many problems where the solutions are complex data structures can still be solved using simple metaheuristics.

Encodings



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- Typical examples were the JSSP and the 2BP, where permutations are translated to Gantt charts and packings, respectively.

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- If you understand these concepts, then you can tackle basically *any* problem occuring in management, logistics, warehousing, planning, scheduling, packing, or even engineering.

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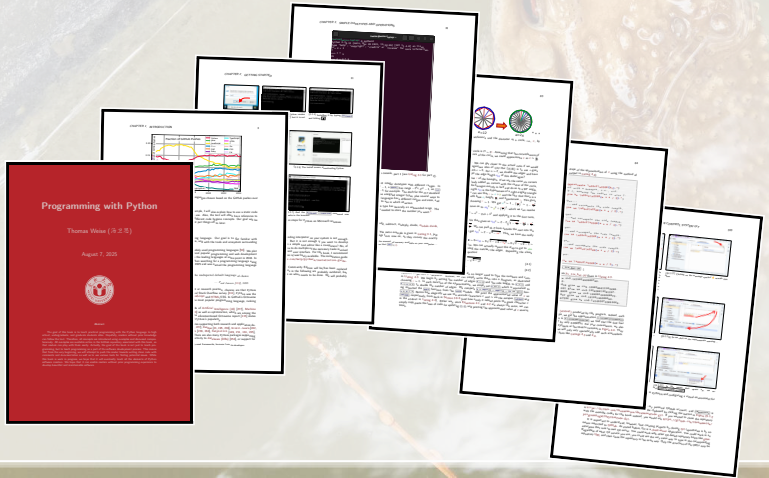
Programming with Python

We have a freely available course book on *Programming with Python* at `https://thomasweise.github.io/programmingWithPython`, with focus on practical software development using the Python ecosystem of tools[236].

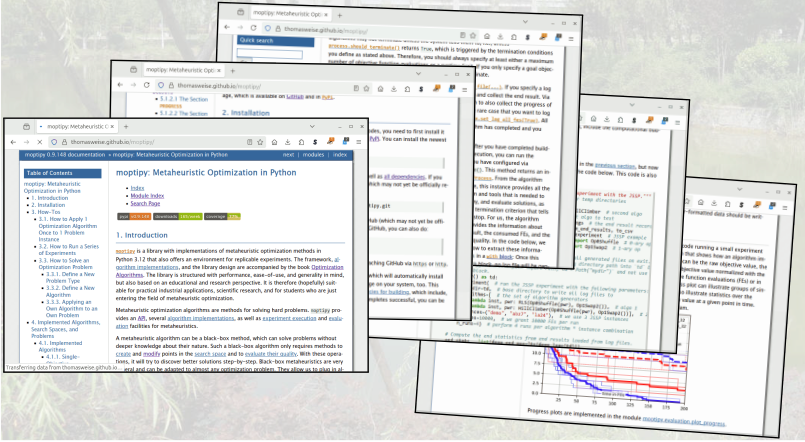


Databases

We have a freely available course book on *Databases* at `https://thomasweise.github.io/databases`, with actual practical examples using a real database management system (DBMS)[232].

Metaheuristic Optimization in Python: moptipy

We offer moptipy[244], a mature open source Python package for metaheuristic optimization, which implements several algorithms, can run self-documenting experiments in parallel and in a distributed fashion, and offers statistical evaluation tools.



谢谢您们！

Thank you!

Vielen Dank!

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Glossary I

(1 + 1) EA The $(1+1)$ EA is a local search algorithm that retains the best solution x_c discovered so far during the search[41,76]. In each step, it applies a unary search operator to this best-so-far solution x_c and derives a new solution x_n. If the new solution x_n is *better or equally good* when compared with x_c, i.e., not worse, then it replaces it, i.e., is stored as the new x_c. If the search space are bit strings of length n, then the $(1+1)$ EA uses a unary search operator that flips each bit independently with probability m/n, where usually $m = 1$. This operator is the main difference to randomized local search (RLS). The $(1+1)$ EA is a special case of the $(\mu+\lambda)$ EA where $\mu = \lambda = 1$.

(1 + 1) FEA The $(1+1)$ EA with FFA plugged in [245, 246, 257]. On the discrete bit-string domains, this algorithm has a very specific structure: Its unary search operator flips each bit with probability $1/n$, where n is the dimension of the search space, distinguishing it from the unary operator in FRLS which flips exactly one bit. On other domains, such as permutations, the term is used synonymously with FRLS.

(μ + λ) EA The $(\mu+\lambda)$ EA is an evolutionary algorithm (EA) where, in each generation, λ offspring solutions are generated from the current population of μ parent solutions. The offspring and parent populations are merged, yielding $\mu + \lambda$ solutions, from which then the best μ solutions are ratained to form the parent population of the next generation. If the search space is the bit strings of length n, then this algorithm usually applies a mutation operator flipping each bit independently with probability $1/n$.

BinInt The *Binary Integer Problem* is a benchmark problem for bit-string based black-box search[222,257]. It is very similar to OneMax: It tries to maximize the number of 1 bits in the bit string of a chosen dimension n. Different from OneMax, where each bit has the same weight in the objective function, the bits of BinInt have a exponentially increasing weight: The bit string is treated as a binary-encoded integer whose value should be maximized. The objective function $f(x)$ of the minimization variant of this problem can be defined as $f(x) = 2^n - 1 - \sum_{i=1}^{n} 2^{i-1} \cdot x_i$. The problem is unimodal, i.e., has only a single global optimum and no local optimum. It is fully separable, i.e., we can optimize each variable separately.

LinHarm The Linear Harmonic Function is a benchmark problem for bit-string based black-box search[70,76,257]. It is very similar to OneMax: It tries to maximize the number of 1 bits in the bit string of a chosen dimension n. Different from OneMax, where each bit has the same weight in the objective function, the bits of LinHarm have a linear increasing weight. The objective function $f(x)$ of the minimization variant of this problem can be defined as $f(x) = \frac{1}{2}n(n+1) - \sum_{i=1}^{n} i \cdot x_i$. The problem is unimodal, i.e., has only a single global optimum and no local optimum. It is fully separable, i.e., we can optimize each variable separately. The $(1+1)$ EA can solve this problem with an expected runtime in $\Theta(n \ln n)$[70,76].

Glossary II

TSP Suite is a open source benchmarking framework for the TSP written in Java[237]. It uses the instances of the TSPLib and offers both automated experiment execution as well as several metrics to measure runtime. It is available at https://thomasweise.github.io/tspSuite and has been used in many research works, such as[248]. It can be viewed as a predecessor of moptipy.

2BP In the two-dimensional Bin Packing Problem (BP), the goal is to pack rectangles into rectangular bins without overlap and such that their edges are parallel to the edges of the bins[119,154,156]. The 2BP is strongly $\mathcal{NP}$-hard[156]. The variants of this problem include the 2BP|O|F, 2BP|R|F, 2BP|O|G, and 2BP|R|G[155,156]. Different objective function for the 2BP|O|F and 2BP|R|F are presented in [259]. 2BPs have many practical applications, including cutting (wood, glas)[156], packing (transportation, warehouse)[156], and newspaper page layout[213].

2BP|O|F In the oriented 2BP with free cuts, the rectangles to pack have fixed orientations, i.e., cannot be rotated, but can otherwise be arranged freely[155,156,167,260]. Permutation-based search spaces can be used for this problem by using the IBL encoding. The SOTA for this problem comprises[26,31,50,102,189,260].

2BP|O|G In the oriented 2BP with guillotine cuts, the rectangles to pack have fixed orientations, i.e., cannot be rotated, and must be arranged such that they can be extracted by edge-to-edge cuts[155,156].

2BP|R|F In this variant of the 2BP with free cuts, the rectangles to pack can be rotated by 90° if need be and can freely be arranged[22,155,156,259,260]. Besides packing and cutting, this problem also occurs in scheduling and build formation in additive manufacturing[148,194]. The Asqas problem[227] is also a 2BP|R|F. Search spaces based on signed permutations can be used for this problem by using the IBL encoding. The SOTA for this problem comprises[26,50,102,259,260].

2BP|R|G In this variant of the 2BP with guillotine cuts, the rectangles to pack can be rotated by 90° if need be, but must be arranged such that they can be extracted by edge-to-edge cuts[155,156].

Glossary III

2DPackLib is a library of benchmark instances for the two-dimensional Bin Packing Problem (2BP)[120,121].The website of this benchmark set is https://site.unibo.it/operations-research/en/research/2dpacklib[120]. The benchmark set contains two-dimensional instances of the Strip Packing, BP, CSP, and Knapsack problems. The 2BP instances are from three groups: The 10 instances of type *beng*[22] have both bin and item dimensions drawn from uniform distributions, between 20 and 200 items, and the largest bin size is $(40, 25)$.The *class* instances[23,167] are divided into 10 classes based on the bin and item dimensions, the former of which ranges from $(10, 10)$ to $(300, 300)$. Each class is divided into five groups with $\{20, 40, 60, 80, 100\}$ items with 10 instances each. The set *A*[164] offers 43 instances with 13 to 809 items and bin sizes of either $(2750, 1220)$, $(2550, 2100)$, or $(2470, 2080)$.

A* Algorithm The A* Algorithm is an a greedy best-first-first search for finding the shortest path between two locations[110,190]. The algorithm iteratively constructs the shortest path. It maintains a list of current candidate paths. For choosing the next candidate path whose end node should be expanded, it computes a value $f(p)$ of these paths p. $f(p)$ is the sum of the cost $g(p)$ so far incurred by the path as well as a heuristic $h(p)$ predicting the cost from the current end of the path to the goal. $h(p)$ must never overestimate that cost.

ACO *Ant Colony Optimization* is a nature-inspired optimization method for combinatorial problems where solutions are generated by "ants" that move from node to node in a graph choosing edges based on (1) the simulated pheromone on the edges and (2) a per-edge heuristic value[72–74]. If an ant produced a good solution, "pheromone" is distributed over the edges it visited, making it more likely to be re-visited by other ants.

AMLS an *Adaptive Memory-based Local Search* algorithm, i.e., a SLS, for solving MaxSAT problems by Lü and Hao [161]. It inherits the diversification mechanism of WalkSAT.

AP *Alternating Position Crossover* is a binary operator for EAs using a permutation-based representation[140,141]. It simply creates an offspring by selecting alternately the next element of the first parent and the next element of the second parent, omitting the elements already present in the offspring. For example, if parent 1 is $(1, 2, 3, 4, 5, 6, 7, 8)$ and parent 2 is $(3, 7, 5, 1, 6, 8, 2, 4)$, AP yields $(1, 3, 2, 7, 5, 4, 6, 8)$ and exchanging the two parents would yield $(3, 7, 5, 1, 6, 8, 2, 4)$[140]. A generalized version of this operator will randomly choose a parent in each step, thus increasing the diversity of the results[253].

Glossary IV

Asqas An "almost-square" be a rectangle with size $i \times (i+1)$ with $i \in \mathbb{N}_{\mathbf{1}}$. The *almost-squares-in-almost-squares* problem[227] is a special case of the 2BP|R|F. Here, almost-square rectangles of sizes 1×2, 2×3, ..., $n \times (n+1)$ must be placed into an almost-square bin without waste. This requires that there exists a natural number $p \in \mathbb{N}_{\mathbf{1}}$ such that $p(p+1) = \sum_{i=\mathbf{1}}^{n} i(i+1)$[227]. There are exacty five Asqas instances[227], namely with bin sizes 1×2 for $n = 1$, $p = 1$, 5×4 for $n = 3$, $p = 4$, 15×16 for $n = 8$, $p = 15$, 55×56 for $n = 20$, $p = 55$, and 119×120 for $n = 34$, $p = 119$.

ATSP The general case for TSPs are *asymmetric Traveling Salesperson Problems*, where the $n \times n$ distance matrix A does not need to be symmetric. In other words, it is possible that $\exists a, b \in 1..n$ with $A[a, b] \neq A[b, a]$. Notice that while such index tuple *may* exist, this is not always or necessarily the case. In this case, one may consider the STSP as a special case of the ATSP, while ATSPs may be considered as synonymous to TSPs.

BP The *Bin Packing Problem*[79,205] is one of the classical problems from Operations Research (OR). The goal is to pack a set of objects into containers, using as few containers as possible. Different flavors of BPs can be distinguished, e.g., by the number of dimensions[60,154] that the objects and containers have (see, e.g., the 2BP), and/or whether the objects can be rotated[260], as well as other constraints and limitations. Both the one- and two-dimensional BP are strongly $\mathcal{NP}$-hard[156].

BRKGA The *Biased Random Key Genetic Algorithm* is a EA with a continuous search space $\mathbb{X} \subset [0, 1]^n$ and a decoding function γ translating this space to something else and biased, uniform, and elitist mating strategies[104,159].The *bias* comes from the fact that one of the parents used for the binary operation is biased to be of higher fitness than the other parent[104]. Often, γ is the Random Keys encoding[16,17] procedure translating $\mathbb{X}$ to permutations of the first n natural numbers[103]. Sometimes, it is actually a MA further refining the translation result with a LS[103]. Sometimes the permutations are then further translated to more complex data structures, such as Gantt charts for the JSSP[103], or packing plans for 2BP 3D packing[102].

C is a programming language, which is very successful in system programming situations[64,198,235].

Concorde is an exact solver for the TSP[5,52,53], which could even solve the largest instance of TSPLib (with 85 900 cities) to optimality.

CSP The *Cutting Stock Problem* is about cutting shapes of a larger shape while reducing waste[156,164]. It is closely related to the BP.

Glossary V

DB A *database* is an organized collection of structured information or data, typically stored electronically in a computer system. Databases are discussed in our book *Databases*[232].

DBMS A *database management system* is the software layer located between the user or application and the database (DB). The DBMS allows the user/application to create, read, write, update, delete, and otherwise manipulate the data in the DB[255].

EA An *evolutionary algorithm* is a metaheuristic optimization method that maintains a population of candidate solutions, which undergo selection (where better solutions are chosen with higher probability) and reproduction (where mutation and recombination create a new candidate solution from one or two existing ones, repectively)[9,233].

EAX *Edge assembly crossover* is a binary operator to be used inside a EA for the TSP[180–182]. It belongs to the SOTA on that field.

FFA *Frequency Fitness Assignment* (中文: 频率适应度分配) is a algorithm plugin for optimization methods applied to discrete or combinatorial problems with not-too-many different possible objective values[243,245,246]. It replaces the objective values in all comparisons with their absolute encounter frequency so far during the search. FFA has successfully been applied to the MaxSAT[245,246], the TSP[149,150], and the QAP[46,223], amongst others. Its properties have been intensely researched on discrete benchmark functions such as OneMax, LeadingOnes, Trap, TwoMax, Jump, and Plateau[245,246,257]. The most common realizations of FFA so far are the $(1+1)$ FEA and FRLS.

FSSP The *Flow Shop Scheduling Problem* is a variant of a JSSP where all products go through the same sequence of machines[106,130].

GA A *Genetic Algorithm* is an EA usually for bit-string based search spaces[9,57,100,170,176,233] (although many researchers use the terms EA and GA synonymously). It is a population-based metaheuristic applying a unary and a binary search operater as well as a selection scheme. The original GA is the Standard Genetic Algorithm (SGA), which today is considered outdated. Modern variants are the Greedy $(2+1)$ GA and the Self-Adjusting $(1+(\lambda,\lambda))$ Genetic Algorithm (Self-Adjusting $(1+(\lambda,\lambda))$ GA) as well as bit-string based $(\mu+\lambda)$ EAs.

Glossary VI

Gantt chart Gantt charts are two-dimensional diagrams illustrating the steps of a process[132,251]. Their horizontal axis represents the time, the vertical axis represents work packages, machines, or workers. In the diagram area, bars located at certain vertical locations indicate that the work package, machine, or worker is processed or works at the time range covered by the bar as projected to the horizontal axis.

GPX *Generalized Partition Crossover* (GPX) is a binary operator for EAs for the TSP[250]. It is *respectful*, because the generated tour inherits all edges common to both parents. It is *transmits alleles*, because it only uses edges that occur in at least one parent. It belongs to the SOTA on the TSP, especially if used in conjunction with LKH[225].

GRASP A *Greedy Randomized Adaptive Search Procedure* is a metaheuristic where each cycle consists of two main components[83,85,86]: First, a starting point is generated using a heuristic method (as opposed to being randomly sampled). Then, a LS is applied to refine the solution. Then, another cycle starts and proceeds in the same way. The overall best solution is retained as result. Many examples of applications of GRASPs are listed in [86]. GRASP in combination with VND has been applied to the 2BP|O|F in [189].

Greedy $(2+1)$ GA The Greedy $(2+1)$ GA[40,41,215,216] is an EA for bit-string based search spaces, i.e., a GA. It was shown to be twice as fast as any mutation-only EA on OneMax[215,216].

IBL The *improved bottom left* heuristic is a one-shot algorithm constructing a solution for 2BP[153]. For the 2BP|O|F problem variant, the input of the IBL algorithm is a permutation of the first n natural numbers, each number identifying one rectangle to be packed. For the 2BP|R|F problem flavor, the permutation elements are signed, where a negative sign indicates a rotation by 90° [153]. In[259,260] it is adapted to a multi-bin situation as follows: The algorithm iterates over the (potentially signed) permutation x and places the rectangles into the packing plan in the prescribed order. It first places the rectangle outside on top of the current bin b, with its bottom-right corner onto the top-right corner of the current bin b. It then moves the rectangle downwards and leftwards as far as possible, prioritizing the downward movement whenever possible. If the rectangle cannot be moved any further, it checks if it is completely contained in the bin b. If yes, it can remain there. Otherwise, a new bin is opened ($b \leftarrow b+1$) and the rectangle is placed at its bottom-left corner. Once a new bin is opened, this bin is used for all further insertions. IBL is suitable as a decoding function γ inside an EA[153]. Historically, it improves upon the BL algorithm[125]: That algorithm cannot represent all possible optimal packings using a permutation, but the IBL can[153].

Glossary VII

ILS An *iterated local search* is a variant of the LS which applies cycles of local search and perturbation[214]. First, an initial solution is generated and refined by the local search. Then the cycle begins: A perturbation, i.e., a large search move that likely destroys a good part of the solution, is applied. The resulting new solution is refined by local search. Then, an acceptance criterion decides whether we should use this or the previous solution to continue in the next cycle. ILS has successfully been applied to the QAP.

Ising Model The Ising models are benchmark problems for bit-string based black-box search[70,87,88,257]. Here, a neighborhood is defined around each bit. The bit needs to have the same value as all bits in the neighborhood structure T. Thus, we try to achieve that all bits have the same value. The objective function $f(x)$ of the minimization variant of this problem family can be defined as $f(x,T) = \sum_{(i,j)\in T} |x_i - x_j|$, where T contains the index pairs of neighboring bits and is defined differently for the Ising model instances Ising 1D and Ising 2D. The problems are highly epistatic, as all variables interact with some other variables.

Ising 1D The 1D is a benchmark problem for bit-string based black-box search[70,88,257]. The bits are arranged in a one-dimensional ring and we try to minimize the number of bits that have different values than their neighbors. Hence, the neighborhood structure T is defined as $T_{\mathbf{1}D} = \{(i+1, (i+1 \bmod n)+1) \mid i \in [0..n-1]\}$. The problems are highly epistatic, as all variables interact with some other variables. The $(1+1)$ EA has an expected runtime in $\mathcal{O}(n^{\mathbf{3}})$ on 1-Dimensional Ising Model (Ising 1D)[88].

Ising 2D The 2D is a benchmark problem for bit-string based black-box search[70,87,257]. The bits are arranged in a two-dimensional torus and we try to minimize the number of bits that have different values than their neighbors. Hence, the neighborhood structure $T = T_{\mathbf{2}D}$ is defined as the set of all $\{(\alpha+\beta N, \gamma+\delta N)\}$ pairs with $\alpha,\beta,\gamma,\delta \in 0..N-1$ and either ()$\gamma = (\alpha+1) \bmod N$ and $\delta = \beta$) or ($\gamma = \alpha$ and $\delta = (\beta+1) \bmod N$) with $n = N^{\mathbf{2}}$. The problems are highly epistatic, as all variables interact with some other variables. On a similar class of problems, the $(1+1)$ EA has an exponential expected runtime[87].

Java is another very successful programming language, with roots in the C family of languages[30,160].

Glossary VIII

JSSP The *Job Shop Scheduling Problem*[29,48,106,143] is one of the most prominent and well-studied scheduling tasks. In a JSSP instance, there are m machines and k jobs[217]. Each job must be processed once by each machine in a job-specific sequence and has a job-specific processing time on each machine. The goal is to find an assignment of jobs to machines that results in an overall shortest makespan, i.e., the schedule (Gantt chart) which can complete all the jobs in the shortest time. The JSSP is $\mathcal{NP}$-complete[44,142,143]. The 8 most common sets of JSSP benchmark instances are *abz*[1], *dmu*[61], *ft*[89], *la*[145], *orb*[6], *swv*[212], *ta*[217], and *yn*[254].Their data can be found (sometimes partially) in several repositories, such as the OR-Library[18,19], [208], and [234], the latter of which also contains many results from existing works. Encodings based on permutations with repetition are proposed in [27, 28, 95, 207] and used in [245]. The SOTA of the JSSP includes[103,113,188,191,209,231,258].

Jump The Jump problem is a benchmark problem for bit-string based black-box search[76,92,257]. Like the OneMax, it tries to maximize the number of 1 bits in a bit string of chosen length n. However, the solutions representing the last ω bits around the optimum to flip are deceptive, meaning that they are actually worse and getting worse than surrounding solutions. The objective function $f(x)$ of the minimization variant is $n - \|x\|_1$ if $\|x\|_1 = n$ or $\|x\|_1 \leq n - \omega$ hold, and $\omega + \|x\|_1$ otherwise, where $\|x\|_1 = \sum_{i=1}^{n} x_i$. This means that, the larger ω, the larger the deceptive basin around the optimum that an algorithm needs to cross. An $(1+1)$ EA will converge to the resulting local optimum and thus have an expected runtime of $\Theta(n^\omega + n \ln n)$[76].In experiments, the $(1+1)$ FEA achieves polynomial runtime on Jump, seemingly in $\mathcal{O}(n^2 \ln n)$[245,246,257].

LABS *Low-Autocorrelation Binary Sequences* are bit strings of length n whose autocorrelation is as low as possible[32,34,35,62,63,93,168,171,178,186,197]. This means that sub-sequences of any length $1 \leq k < n$ should ideally never repeat. The objective function $f : \{0,1\}^n \mapsto \mathbb{N}_0$ represents the energy of a bit string x and is subject to minimization. We can define it as $f(x) = \sum_{k=1}^{n-1} \left(\sum_{i=0}^{n-1-k} s_i \cdot s_{i+k}\right)^2$ with $s_j = 2x_j - 1$. Such sequences are needed for communication engineering[34,35,178], cryptography[62,178], as well as modulation pulses in radar and sonar ranging[186]. The LABS is $\mathcal{NP}$-hard[226]. The SOTA on LABS contains[32,35,62,63,178].

Glossary IX

LeadingOnes The LeadingOnes problem is a benchmark problem for bit-string based black-box search[203,249,257]. It tries to maximize the number of *leading* 1 bits in the bit string of a chosen dimension n. The objective function $f(x)$ of the minimization variant of this problem can be defined as $f(x) = n - \sum_{i=1}^{n} \prod_{j=1}^{i} x_j$. Different from OneMax, the contribution to the objective value of the bit at index i thus depends on the state of all the bits *before* this index. The problem is unimodal, i.e., has only a single global optimum and no local optimum. Different from OneMax, it is not separable, i.e., we can optimize each variable separately. The expected runtime of $(1+1)$ EA on LeadingOnes is in $\Theta(n^2)$[76]. The average runtime $(1+1)$ FEA on LeadingOnes seems to be slower by factor proportional to n, i.e., seems to be in $\mathcal{O}(n^3)$[245].

LK The *Lin-Kernighan* (LK) algorithm[152] is a SLS method for the TSP. The LK algorithm performs iterative k-opt moves, i.e., in each step, it constructs the two sets X and Y of k edges such that removing the edges in X from the current tour T and then adding the edges from Y to T results in a better tour.The LK algorithm is one of the most classical approaches to the TSP.

LKH The *Lin-Kernighan-Helsgaun* algorithm[53,111,112] is a SLS method for the TSP based on the Lin-Kernighan (LK) algorithm[152]. LKH-1 was proposed in [112] and LKH-2 in [111]. It holds the record in the *World TSP*[54] as of 2025.

LS A *local search* is an algorithm where a single "current" candidate solution is maintained and new solutions are selected from its neighborhood in each step. Most LSes are randomized, i.e., stochastic local search (SLS).

MA A *Memetic Algorithm* is an EA where each new solution is refined by a LS before it enters the population[233]. As LS, we can use, e.g., RLS, but TS[47,90] is maybe a better choise. Some SOTA algorithms in some domains may be classified as MAs, e.g., Lin-Kernighan Helsgaun (LKH) basically uses a population that is refined via a local search. MAs also worked well on the JSSP[47,113] and the TSP[90].

MaxSAT The goal of satisfiability problems is to find an assignment for n Boolean variables that make a given Boolean formula $F : \{0,1\}^n \mapsto \{0,1\}$ become true. In the *Maximum Satisfiability* problem (MaxSAT)[114], F is given in conjunctive normal form, i.e., the variables appear either directly or negated in m "or" clauses, which are all combined into one "and." The objective function $f(x)$, subject to minimization, computes the number of clauses which are false under the variable setting x. If $f(x) = 0$, then all clauses of F are true, which solves the problem. The MaxSAT problem is $\mathcal{NP}$-complete[51] and exhibits low epistasis but deceptiveness[200], making it hard for local searches. Typical benchmark instances can be found in SATLIB, which contains smaller-scale instances, and the up-to-date MSE, providing SOTA benchmarks, algorithms, and results.

Glossary X

metaheuristic A *metaheuristic* is an optimization algorithm which performs a cycle of trial and error during which candidate solutions are sampled from the solution space $\mathbb{Y}$ and evaluated using an objective function $f : \mathbb{Y} \mapsto \mathbb{R}$[98,233]. The results of the objective function are used to decide which solution is promising. The neighborhoods around promising solutions *or* similarly-structured solutions are then sampled in the next cycle. Prominent metaheuristics are SLSes, SA, TS, and EAs.

moptipy is the *Metaheuristic Optimization in Python* library[244]. It has been used in several different research works, including[46,149–151,223,253,259,260]. Algorithms implemented by moptipy include EAs, $(1+1)$ EA, RLS, SA, and the $(1+1)$ FEA, amongst others. Learn more at https://thomasweise.github.io/moptipy and https://thomasweise.github.io/moptipyapps.

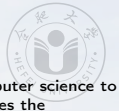
MSE The *MaxSAT Evaluation* is a regularly updated set of benchmark instances for the MaxSAT[122]. It also includes solver software and benchmark results[229].

N-Queens The N-Queens Problem is a benchmark problem for bit-string based black-box search[70,257]. The goal is to arrange N queens on an $N \times N$ chessboard so that none can attack the other. The search space has thus dimension $n = N^2$, with each bit standing for a field and encoding whether or not a queen is placed there. The objective function $f(x)$ of the minimization variant is $N - \left(\sum_{i=1}^{n} x_i\right) - N \sum_{\xi} \max\left\{0, Q_\xi(x) - 1\right\}$ where $Q_\xi(x)$ is the number of queens in the horizontal, vertical, or diagonale ξ. The problem is quite epistatic.

NuWLS a SLS algorithm by Chu, Cai, and Luo [49] for solving the (Weighted) Partial MaxSAT problem.

OneMax OneMax is the simplest possible benchmark problem for bit-string based black-box search[77,179,257]. The goal is to maximize the number of 1 bits in the bit string of a chosen dimension n. The objective function $f(x)$ of the minimization variant of this problem can be defined as $f(x) = n - \sum_{i=1}^{n} x_i$. The problem is unimodal, i.e., has only a single global optimum and no local optimum. It is fully separable, i.e., we can optimize each variable separately. The expected runtime of $(1+1)$ EA on OneMax is in $\Theta(n \ln n)$[66,179]. The average runtime $(1+1)$ FEA on OneMax seems to be slower by factor proportional to n, i.e., seems to be in $\mathcal{O}(n^2 \ln n)$[245].

Glossary XI



OR Operations Research (or Operational Research) is the application of sciences such as mathematics and computer science to the management and organization of systems, organizations, enterprises, factories, or projects. It encompasses the development and application of problem-solving methods and techniques (such as mathematical optimization, simulation, queueing theory and other stochastic models) with the goal to improve decision-making and efficiency[58].

OR-Library is a library of benchmark instances for various OR problems[18,19] located at https://people.brunel.ac.uk/~mastjjb/jeb/info.html. It contains instances of the JSSP, 1D- and 2D-Packing, Knapsack, periodic TSPs, amongst others.

Plateau The Plateau problem is a benchmark problem for bit-string based black-box search[4,257]. It is structured exactly like Jump, but instead of being deceptive, the last ω bits to flip around the optimum are neutral. The objective function $f(x)$ of the minimization variant is $n - \|x\|_1$ if $\|x\|_1 = n$ or $\|x\|_1 \leq n - \omega$ hold, and ω otherwise, where $\|x\|_1 = \sum_{i=1}^{n} x_i$. This means that, the larger ω, the larger the neutral plane around the optimum that an algorithm needs to cross. An $(1+1)$ EA has an expected runtime of $\Theta(n^\omega)$ on plateaus[4]. The average runtime of the $(1+1)$ FEA on Plateaus is slower by a factor seemingly proportional in n.

PostgreSQL An open source object-relational DBMS[84,185,196,221]. See https://postgresql.org for more information.

Python The Python programming language[117,146,163,236], i.e., what you will learn about in our book[236]. Learn more at https://python.org.

QAP The *Quadratic Assignment Problem* is an optimization problem where the goal is to assign a set of n facilities to a set of n locations[37,46,136,157]. Such an assignment can be represented as a permutation x of the first n natural numbers, where x_i specifies the location where facility i should be placed. For each QAP instance, a distance matrix A is given, where $A[a,b]$ specifies the distance from location a to location b, as well as a flow matrix B, where $B[i,j]$ is the amount of material flowing (e.g., daily) from facility i to facility j. The objective function $f(x)$ then rates a permutation x as $f(x) = \sum_{i=1}^{n} \sum_{j=1}^{n} A[x_i, x_j] * B[i,j]$. The QAP is $\mathcal{NP}$-complete[75,204] and even among this problem class, is considered as particularly hard for metaheuristics[20,37,157,214]. The QAP has applications such as building layout[56,81,137], keyboard layout[39], circuit design[82], wiring[211], and scheduling[210]. Typical benchmark instances can be found in QAPLIB. SOTA algorithms for the QAP include[20,90,169,175,219,220].

Glossary XII

QAPLIB is a library of 136 benchmark instances for the Quadratic Assignment Problem (QAP)[36,38], 28 of which have not yet been solved to optimality[3] (at least by 2023).

Random Keys encoding is a decoding procedure $\gamma : \mathbb{X} \mapsto \mathbb{Y}$ that maps a continuous search space $\mathbb{X} \subset [0,1]^n$ which is a subset of the n-dimensional real numbers to the space $\mathbb{Y}$ of permutations of the first n natural numbers[16,17]. For example, on $n = 5$, the vector $x = (0.49, 0.91, 0.33, 0.75, 0.51)$ is translated to the permutation $y = (3, 1, 5, 4, 2)$, because $x_{\mathbf{3}} = 0.33$ has the smallest value, putting 3 at the beginning, $x_{\mathbf{1}} = 0.49$ the second-smallest, placing 1 second, ..., and $x_{\mathbf{2}} = 0.91$ being the largest value meaning that the last number in y will be 2[17]. The advantage of the encoding is that we can search in $\mathbb{X}$ applying arbitrary mathematical operators, crossover, and mutation procedures, but will always get correct permutations. The disadvantage is that $|\mathbb{X}| \gg |\mathbb{Y}|$ and that there actually is no meaning underpinning moves in $\mathbb{X}$ in relation to their projected result in $\mathbb{Y}$.

RLS Randomized local search retains the best solution x_c discovered so far during the search and, in each step, it applies a unary search operator to this best-so-far solution x_c and derives a new solution x_n. If the new solution x_n is *better or equally good* when compared with x_c, i.e., not worse, then it replaces it, i.e., is stored as the new x_c. If the search space are bit strings of length n, then RLS uses a unary search operator that flips exactly one bit. This operator is the main difference to $(1 + 1)$ EA.

FRLS The RLS with FFA plugged in [257]. On the discrete bit-string domains, this algorithm has a very specific structure: Its unary search operator flips exactly one bit, distinguishing it from the unary operator in the $(1 + 1)$ FEA, which flips each bit with probability $1/n$. On other domains, such as permutations, we use the term synonymously with $(1 + 1)$ FEA.

RTS *Reactive Tabu Search* is a variant of TS where the length of the tabu list is adapted to the problem and to the current evolution of the search, and an escape strategy for diversifying the search is used when that is not sufficient[12–15]. It has successfully been applied to a wide range of problems, from the QAP[13,14], to Knapsack problems[12,14], the N-K landscape[12] to the training of neural networks[15].

SA *Simulated Annealing* is a SLS that sometimes accepts a worse solution[42,124,131,193,233]. The probability to do so decreases over time and with the difference in objective values, i.e.,is the lower the worse the new solution is. SA performed well on a variety of problems, including the QAP[173,224] and the JSSP[188].

Glossary XIII

SATLIB is a library of benchmark instances for the Maximum Satisfiability (MaxSAT) available at https://www.cs.ubc.ca/~hoos/SATLIB/index-ubc.html[115,116].

Self-Adjusting $(1 + (\lambda, \lambda))$ GA The Self-Adjusting $(1 + (\lambda, \lambda))$ GA is a theory-inspired EA for bit string-based search spaces[41]. It has linear runtime on the OneMax problem[65].

SGA The Standard Genetic Algorithm[9,57,100,170,176,233] was the first EA. It maintains a population of solutions and applies mutation and crossover to generate offspring solutions. It uses fitness proportionate selection to choose which solutions should "survive" into the next generation, which today is considered a very bad design choice, see, e.g., [249].

SLS *Stochastic local search* is a local search (LS), one of the simplest metaheuristic search paradigms[114]: A single "current" candidate solution is maintained and new solutions are *stochastically* sampled from its neighborhood, i.e., in a way that involves random numbers. If these solutions are deemed interesting based on their objective value, they replace the "current". The cycle continues. Basic examples are RLS and $(1 + 1)$ EA. More complex examples that add mechanisms to escape local optima on top of the SLS paradigm are TS and SA.

SOTA State-of-the-Art, i.e., the best-performing methods for a field or problem type

STSP A *symmetric Traveling Salesperson Problem* is a special case of the TSP, where the $n \times n$ distance matrix A is symmetric, i.e., $A[a, b] = A[b, a] \forall a, b \in 1..n$.

Trap The Trap problem is a benchmark problem for bit-string based black-box search[76,183,257]. It is structured exactly like OneMax, however the worst and the best solution are swapped. This makes the problem highly deceptive, as following the "gradient" in the search space leads the search away from the global optimum. For bit strings of length n, the objective function $f(x)$ of the minimization variant is 0 if $\|x\|_1 = 0$ and $n - \|x\|_1 + 1$ otherwise, where $\|x\|_1 = \sum_{i=1}^{n} x_i$. An $(1 + 1)$ EA will converge to the resulting local optimum and thus have an expected runtime of $\Theta(n^n)$[76]. In experiments, the $(1 + 1)$ FEA achieves polynomial runtime on Trap, seemingly in $\mathcal{O}(n^2 \ln n)$[245,246,257].

Glossary XIV

TS *Tabu Search* is one of the most successful SLSes[96,97,99,126]. In order to avoid returning to previous solutions, it declares certain solutions or search moves as "tabu." For example, if a bit in a discrete problem has been flipped by the search operator, it may be forbidden to flip the bit again for some time, thus preventing the algorithm from getting trapped in a cycle. This rule may be overridden if a new best-so-far solution is discovered that outperforms all previous solutions, in which case it would be accepted (this is the so-called aspiration criterion). TS has performed well on the QAP[13,14,172,175,218,219] and on the JSSP[184,191,192,209,258].

TSP In an instance of the *Traveling Salesperson Problem*, also known as *Traveling Salesman Problem*, a set of n cities or locations as well as the distances between them are given[5,108,144,149,237,248]. The goal is to find the shortest round-trip tour that starts at one city, visits all the other cities one time each, and returns to the origin. The TSP is one of the most well-known $\mathcal{NP}$-hard combinatorial optimization problems[94,108,138]. The TSP has many applications[5], including in transportation, logistics, (circuit) plotting and drilling problems[107], telescope aiming, and genome sequencing. Solutions are often encoded as permutations x of the first n natural numbers, where each city gets a number and the order of numbers in the permutation corresponds to the order of cities in the tour. The distances between the n cities are usually given in a $n \times n$ distance matrix A, where $A[a, b]$ is the distance from city a to city b with $a, b \in 1..n$. Then, the objective function subject to minimization is $f(x) = A[x_n, x_\mathbf{1}] + \sum_{i=\mathbf{1}}^{n-\mathbf{1}} A[x_i, x_{i+\mathbf{1}}]$. Typical benchmark instances can be found in TSPLib[201,202], the 8th DIMACS challenge[127–129], and the *World TSP* repository[54].

TSPLib is a library of benchmark instances for the Traveling Salesperson Problem (TSP)[201,202]. Its original URL was available until 2025 and then became defunct. The repository https://github.com/shredderzwj/TSPLIB contains the data. Additionally, you may also check https://github.com/thomasWeise/tspSuite, which contains all the TSPLib instances as well as Java code to load them. The TSPLib contains 110 instances of the symmetric Traveling Salesperson Problems (STSPs) with scales n ranging from 14 to 85 900, but 93 instances have less than 2000 nodes. There also are 19 asymmetric Traveling Salesperson Problems (ATSPs) with scales $n \in 17..443$.

Glossary XV

TTP The *Traveling Tournament Problem* is the combinatorial optimization problem of both efficiently and fairly organizing a tournament of n teams that play against each other in a pairwise fashion[80,253]. The efficient part boils down to arranging the games such that the total travel length is short, which is somewhat similar to the classical TSP. Initially, each team is at its home location. On each day, a team needs to travel if its scheduled game is not at its present location. On the last day, each team may need to travel back home unless their last game is a home game. The total travel length sums up the lengths of all travels over all teams. The fair part is represented in several constraints, such as doubleRoundRobin, compactness, maxStreak, and noRepeat. The TTP is $\mathcal{NP}$-hard[230].

TwoMax The TwoMax problem is a benchmark problem for bit-string based black-box search[92,228,257]. It has a global and a local optimum, both of about the same size, at opposite ends of the search space. For bit strings of length n, the objective function $f(x)$ of the minimization variant is 0 if $\|x\|_1 = n$ and $1 + n - \max\{\|x\|_1, n - \|x\|_1\}$ otherwise, where $\|x\|_1 = \sum_{i=1}^{n} x_i$. An $(1+1)$ EA has a 50/50 chance to converge to the local or global optimum and thus has an expected runtime in $\Theta(n^n)$[91,92]. In experiments, the $(1+1)$ FEA achieves polynomial runtime seemingly in $\mathcal{O}(n^2 \ln n)$ on TwoMax[245,246,257].

URI A *Uniform Resource Identifier* is an identifier for an abstract or physical resource in the internet[256]. It can be a URL, a name, or both. URIs are supersets of URLs. The connection strings of the PostgreSQL DBMS are examples of Uniform Resource Identifiers (URIs).

URL A *Uniform Resource Locator* identifies a resource in the World Wide Web (WWW) and a way to obtain it by describing a network access mechanism. The most notable example of URLs is the text you write into web browsers to visit websites[25]. URLs are subsets of URIs.

VND *Variable Neighborhood Descend*; see VNS. VND in combination with GRASP has been applied to the 2BP|O|F in [189].

Glossary XVI

VNS *Variable Neighborhood Search* is a metaheuristic framework combining a systematic change of neighborhood with LS[109,177]. A *neighborhood* structure in LS defines which solutions can be reached from a given solution within one search step. Usually, one uses a single unary operator and, hence, has a single neighborhood structure. However, for many problems, we can easily design many operators. For permutations, for example, one operator may swap two elements, another one may reverse a section of the permutation, and yet another one, while yet another one may rotate a subsequence one step to the left or right[237]. In the VNS framework, a multiple neighborhoods are investigated in a systematic fashion. Assume that we have a list of neighborhood structures. Variable Neighborhood Descend (VND) works in a deterministic fashion. It finds the best solution that can be reached in one step from the current solution under the current neighborhood. If this is better than the current solution, we accept it and continue with the first neighborhood. Otherwise, we continue with the next neighborhood in the list. In Basic VNS, instead of picking the best move from the current neighborhood, the first improving move is used. Also, at the beginning of each cycle, a random point from the current neighborhood around the current solution is chosen as starting point.

W-Model The W-Model is a tunable benchmark function for bit-string based black-box search[239,247].It allows you to adjust the amount of ruggedness, deceptiveness, epistasis, and uniform neutrality in the search space.

WalkSAT a SLS algorithm by Selman, Kautz, and Cohen [206] for solving the MaxSAT problem.

WWW World Wide Web[24,59]

$i!$ The factorial $a!$ of a natural number $a \in \mathbb{N}_1$ is the product of all positive natural numbers less than or equal to a, i.e., $a! = 1 * 2 * 3 * 4 * \cdots * (a-1) * a$[43,78,162]. In Python, the module `math` provides the function `factorial` for computing the factorial.

$\circ^\star$ the superscript $\star$ symbol denotes that $\circ$ is a global optimum, i.e., there is no better solution than $\circ$.

$i..j$ with $i, j \in \mathbb{Z}$ and $i \leq j$ is the set that contains all integer numbers in the inclusive range from i to j. It always holds that $i..j \subset \mathbb{Z}$. For example, 5..9 is equivalent to $\{5, 6, 7, 8, 9\}$

Glossary XVII

A the $n \times n$ distance matrix. $A[a, b]$ indicates the distance between locations a and b with $a, b \in 1..n$. A distance matrix can be symmetric, in which case $A[a, b] = A[b, a]\ \forall a, b \in 1..n$. Distance matrices represent the problem instance data in TSPs and part of the instance data of QAPs.

B the $n \times n$ flow matrix. $B[a, b]$ is the flow of goods between locations a and b with $a, b \in 1..n$. Flow matrices are one half of the problem instance data of QAPs.

$f(y)$ The *objective function* $f : \mathbb{Y} \mapsto \mathbb{R}$ can compute a cost or rating for each candidate solution y in the solution space $\mathbb{Y}$. The smaller this value, the better the solution, i.e., in the context of this work, objective functions are subject to minimization. Objective functions are not necessarily continuous and it is also quite common that they have integer or natural numbers as results, e.g., we often have $f : \mathbb{Y} \mapsto \mathbb{N}_0$ instead of $f : \mathbb{Y} \mapsto \mathbb{R}$.

$\gamma(x)$ The decoding function $\gamma : \mathbb{X} \mapsto \mathbb{Y}$ translates point from the search space $\mathbb{X}$ to one point y in the solution space $\mathbb{Y}$. It is a very common case that $\mathbb{X} = \mathbb{Y}$ and that the decoding function γ is the identity mapping. Examples for encodings that translate from a simple search space $\mathbb{X}$ to a complex solution data structure $\mathbb{Y}$ are the IBL algorithm that translates (potentially signed) permutations to two-dimensional packings on the 2BP, the permutation-based encoding of Gantt charts for the JSSP from[27,28,95,207,238], or the permutation-based encoding of game plans for the Traveling Tournament Problem (TTP) from[253]. Encodings can also translate between types of search spaces. The Random Keys encoding, for example, translates from $[0, 1]^n$ to permutations of the first n natural numbers..

k the number of jobs in an instance of the JSSP.

m the number of machines in an instance of the JSSP.

Glossary XVIII

n We usually use the value $n \in \mathbb{N}_0$ to denote the *scale*, i.e., the size of the problem to solve. This notion is very general in computer science and not limited to optimization. For example, if we sort a list or search in a list, the scale n is the length of the list, i.e., the number of elements in it. In a TSP, this may be the number of cities. In a discrete benchmark problem such as OneMax or Trap, this may be the number of bits over which the problem is defined. In a QAP, this may be the number of facilities and locations. In a MaxSAT problem, this may be the number of variables that are input to the Boolean formula that we want to become true. There are also problems where the scale is not defined straightforwardly like this. For example, in a JSSP, we have a number of machines and a number of jobs, which together define the size of the problem. But even then, the scale n relates to the size of the solution space $\mathbb{Y}$, and, hence, also of the search space $\mathbb{X}$, which usually grow exponentially with n. And this is already the main issue: The larger n, the (exponentially) larger the number of possible solutions and thus, the larger the runtime of algorithms. This is known as the "curse of dimensionality"[21,45]. Scales n are also often used in conjunction with the $\mathcal{O}$-notation to denote algorithm complexity. For example, comparison-based sorting a list of n items takes $\Theta(n \log n)$ steps[135] (because they need $\lceil \log_2 n! \rceil$ comparisons in the worst case).

$\mathbb{N}_0$ the set of the natural numbers *including* 0, i.e., 0, 1, 2, 3, and so on. It holds that $\mathbb{N}_0 \subset \mathbb{Z}$ and that $\mathbb{N}_0 = \mathbb{N}_1 \cup \{0\}$.

$\mathbb{N}_1$ the set of the natural numbers *excluding* 0, i.e., 1, 2, 3, 4, and so on. It holds that $\mathbb{N}_1 \subset \mathbb{Z}$ and that $\mathbb{N}_1 = \mathbb{N}_0 \setminus \{0\}$.

$\mathcal{NP}$ is the class of computational problems that can be solved in polynomial time by a non-deterministic machine and can be verified in polynomial time by a deterministic machine (such as a normal computer)[105].

$\mathcal{NP}$-complete A decision problem is $\mathcal{NP}$-complete if it is in $\mathcal{NP}$ and all problems in $\mathcal{NP}$ are reducible to it in polynomial time[105,199]. A problem is $\mathcal{NP}$-complete if it is $\mathcal{NP}$-hard and if it is in $\mathcal{NP}$.

$\mathcal{NP}$-hard A problem is $\mathcal{NP}$-hard if all problems in $\mathcal{NP}$ are reducible to it in polynomial time[105]. Algorithms that guarantee to find the correct solutions of $\mathcal{NP}$-hard problems[44,51,143] have a worst-case runtime that grows exponentially with the problem scale.

$\Omega(g(x))$ If $f(x) = \Omega(g(x))$, then there exist positive numbers $x_0 \in \mathbb{R}^+$ and $c \in \mathbb{R}^+$ such that $f(x) \geq c * g(x) \geq 0$ for all $x \geq x_0$[55,133,134]. In other words, $\Omega(g(x))$ describes a lower bound for function growth.

$\mathcal{O}(g(x))$ If $f(x) = \mathcal{O}(g(x))$, then there exist positive numbers $x_0 \in \mathbb{R}^+$ and $c \in \mathbb{R}^+$ such that $0 \leq f(x) \leq c * g(x)$ for all $x \geq x_0$[8,55,133,134,139]. In other words, $\mathcal{O}(g(x))$ describes an upper bound for function growth.

Glossary XIX

$\Theta(g(x))$ If $f(x) = \Theta(g(x))$, then $f(x) = \mathcal{O}(g(x))$ and $f(x) = \Omega(g(x))$[133,134], meaning that there exist positive constants $c_1, c_2, x_0 \in \mathbb{R}^+$ such that $0 \leq c_1 g(x) \leq f(x) \leq c_2 g(x)$ for all $x \geq x_0$[55]. In other words, $\Theta(g(x))$ describes an exact order of function growth.

$\mathrm{o}(g(x))$ If $f(x) = \mathrm{o}(g(x))$, then there exist positive numbers $x_0 \in \mathbb{R}^+$ such that $0 \leq f(x) \leq c * g(x)$ for all $x \geq x_0$ *all* $c \in \mathbb{R}^+$[55]. Different from $\mathcal{O}(g(x))$, $\mathrm{o}(g(x))$ holds for *all* constants $c > 0$, whereas $\mathcal{O}(g(x))$ implies the existence of *some* constant $c > 0$. This basically means that $f(x)$ becomes insignificant compared to $g(x)$ as x_0 grows toward infinity. It is often written that $\lim_{x_0 \to \infty} \frac{f(x)}{g(x)} = 0$. In other words, $\mathrm{o}(g(x))$ describes an upper bound for function growth that is not asymptotically tight, i.e., an upper bound that grows faster than $f(x)$.

$\mathbb{R}$ the set of the real numbers.

$\mathbb{R}^+$ the set of the positive real numbers, i.e., $\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\}$.

$\omega(g(x))$ If $f(x) = \omega(g(x))$, then for every positive constant $c > 0$, there exists a number $x_0 \in \mathbb{R}^+$ such that $0 \leq c * g(x) < f(x)$ for all $x \geq x_0$[55]. In other words, $f(x) = \omega(g(x))$ if and only if $g(x) = \mathrm{o}(f(x))$. In other words, ω describes a lower bound that is not asymptotically tight.

$\mathbb{X}$ The search space $\mathbb{X}$ is the space through which an optimization algorithm navigates in order to find good solutions. The search operators of metaheuristics are used to sample points $x \in \mathbb{X}$, for example. On many benchmark problems, the objective functions f directly evaluate the elements $x \in \mathbb{X}$, i.e., are defined as $f : \mathbb{X} \mapsto \mathbb{R}$. However, very often these elements x are not the same data structures that the user is interested in and/or that the objective functions evaluate: It is common that the search space $\mathbb{X}$ is a simple and plain data structure that can be tackled with standard operations or mathematical tools, such as real vectors $\mathbb{R}^n$, permutations, or bit strings, whereas the solution space $\mathbb{Y}$ can be something more complicated. Then, the user only cares about the solutions $y \in \mathbb{Y}$ and not about the search space $\mathbb{X}$. A decoding $\gamma : \mathbb{X} \mapsto \mathbb{Y}$ is employed to translate the $x \in \mathbb{X}$ to $y = \gamma(x)$ with $y \in \mathbb{Y}$.

x a point in the search space $\mathbb{X}$.

Glossary XX

$\mathbb{Y}$ the set of candidate solutions of an optimization problem is called the *solution space* $\mathbb{Y}$. The solution space corresponds to the data structure holding the candidate solutions y. These are the elements that are passed to the user after the optimization process is completed. They are also the parameters of the objective function(s) $f : \mathbb{Y} \mapsto \mathbb{R}$.

y a candidate solution $y \in \mathbb{Y}$ is one possible solution to an optimization problem. After the optimization process is completed, one or multiple such solutions are passed to the user.

$\mathbb{Z}$ the set of the integers numbers including positive and negative numbers and 0, i.e., . . . , -3, -2, -1, 0, 1, 2, 3, . . . , and so on. It holds that $\mathbb{Z} \subset \mathbb{R}$.